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Three stage based reinforcement learning for combining multiple metaheuristic algorithms

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ABSTRACT

Combined of metaheuristic algorithms can effectively improve the performance of algorithms by utilizing the characteristics of different metaheuristic algorithms, and the key is how to combine multiple metaheuristic algorithms. Reinforcement learning is one of the effective methods for combining multiple metaheuristic algorithms. However, designing a competitive reinforcement learning approach to achieve efficient collaboration among metaheuristic algorithms is a highly challenging task. Therefore, this study proposes a three-stage reinforcement learning for combining multiple metaheuristic algorithms (TSRL-CMM). TSRL-CMM is divided into three stages: the exploration stage, the stage with both exploration and exploitation, and the exploitation stage. On this basis, an adaptive action selection strategy and a reward function are designed. The proposed action selection strategy can adaptively select appropriate metaheuristic algorithms based on the state of the population, achieving a balance between exploration and exploitation. The proposed reward function can effectively guide the population to transition to the expected state based on the iteration stage and state transitions. To verify the effectiveness of TSRL-CMM, we evaluated it using the CEC2017 test suite, nine real-world engineering design problems and six power system optimization problems. TSRL-CMM was compared with 10 state-of-the-art metaheuristic algorithms, and experimental results showed that TSRL-CMM performed better than the compared algorithms in both artificial and real-world problems. Furthermore, TSRL-CMM was specifically compared with three CEC winner algorithms on the CEC 2017 benchmark test suite. The experimental results show that the proposed algorithm is highly competitive. The source code can be obtained from https://github.com/xtongliu/TSRL-CMM-code.

1. Introduction

For large-scale complex engineering optimization problems, such as production scheduling, power system optimization, portfolio optimization, and biomedical image enhancement [1], metaheuristic algorithms [2] are among the effective solution methods. Metaheuristic algorithms are a type of optimization algorithm designed based on evolutionary principles, social behavior rules, and natural phenomena. They search the problem space using specific search operators and find the optimal (or near-optimal) solution to the problem through continuous iteration. Metaheuristic algorithms have attracted much attention from researchers due to their problem-independence, simplicity of implementation, and excellent performance [3].

To date, there are approximately over 1000 metaheuristic algorithms [4]. Each metaheuristic algorithm has its own characteristics, mainly reflected in the search operators and parameter settings. Among them, the African Vulture Optimization Algorithm (AVOA) [5], Whale Optimization Algorithm (WOA) [6], Marine Predators Algorithm (MPA) [7], and Particle Swarm Optimization (PSO) [8] are competitive metaheuristic algorithms, each with its own advantages in the optimization process. AVOA generates new solutions by simulating the foraging behavior of vultures, utilizing various foraging strategies (such as static foraging and dynamic foraging) to help the algorithm balance between global and local search, thereby avoiding local optima. AVOA is particularly suitable for handling complex multimodal optimization problems [9]. WOA generates new solutions by simulating the

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bubble-net hunting strategy and spiral updating mechanism of whales. The bubble-net hunting mechanism provides excellent local search capability, while the spiral updating aids in global search. WOA's unique search capabilities enable it to perform exceptionally well in various continuous optimization problems [10]. MPA generates new solutions by simulating various predation strategies of marine predators. By employing different predation strategies, such as chasing and ambushing, MPA enhances the breadth of the search and the robustness of the algorithm, making it particularly adept at handling highly complex and dynamically changing optimization problems [11]. PSO algorithm generates new solutions by learning from individual historical best and global historical best position, primarily involving position and velocity update formulas. PSO has been demonstrated good convergence speed in high-dimensional search spaces [12].

According to the no free lunch (NFL) theorem, no single algorithm can perform well on all problems [13]. Therefore, combining multiple metaheuristic algorithms to leverage the advantages of different metaheuristic algorithms to enhance performance has attracted considerable attention from researchers [14]. Most combination methods are constructed based on human expertise. Common methods include selecting one of the metaheuristic algorithms based on a specific probability [15], or executing a particular metaheuristic algorithm at a certain iteration stage [16]. Combining metaheuristics based on empirical methods is very limited and challenging to generalize. In addition to combining multiple metaheuristic algorithms to leverage their respective advantages, parameter tuning [17] also plays a crucial role in optimization performance. Each metaheuristic algorithm has internal control parameters, such as population size, learning rate, or mutation rate, which significantly affect its search efficiency. Manually setting optimal parameters is often challenging, as different problems require different configurations to achieve the best performance. To address this issue, researchers have proposed various adaptive parameter tuning strategies. For example, the Success-History Based Parameter Adaptation for Differential Evolution (SHADE) [18] method dynamically adjusts parameters based on past performance, while reinforcement learning has been applied to online parameter adaptation [19], enabling agents to optimize parameters in real time based on feedback. Additionally, statistical tuning methods, such as F-Race [20], iteratively compare different parameter configurations and eliminate suboptimal ones to improve optimization efficiency. While this study primarily focuses on reinforcement learning-based metaheuristic algorithm combinations rather than parameter tuning, integrating adaptive parameter control mechanisms could further enhance the efficiency of algorithm combinations.

Reinforcement learning (RL) can dynamically adjust strategies based on feedback from the environment and the search process, adaptively selecting the most appropriate algorithm or operation [21]. Additionally, RL has a mechanism to balance exploration and exploitation, making it less likely to get trapped in local optima [22]. The combination of multiple metaheuristic algorithms based on reinforcement learning (CMM-RL) models the selection of metaheuristic algorithms as a reinforcement learning problem. During the iterative process, CMM-RL selects the most appropriate metaheuristic algorithm from multiple metaheuristic algorithms based on the state of the population to search the solution space, achieving efficient collaboration among various metaheuristic algorithms [23]. The definition of states, the selection of actions, and the setting of rewards are key factors influencing CMM-RL [24,25]. However, existing CMM-RL methods still exhibit certain limitations in these key factors, which constrain their optimization efficiency and adaptability. First, the definition of states in most CMM-RL approaches is often overly simplistic, relying on basic statistical metrics such as population diversity or best fitness, which may fail to fully capture the underlying search dynamics. Second, the selection of actions in traditional CMM-RL methods frequently employs fixed exploration-exploitation mechanisms, such as a static ϵ-Greedy strategy, limiting adaptability at different stages of the optimization process. Third, the setting of rewards in CMM-RL is typically based on short-term fitness improvements, lacking a long-term perspective to effectively guide algorithm selection. These limitations reduce the ability of CMM-RL to dynamically adjust search strategies in response to evolving optimization landscapes.

To address the above issues, we propose a three-stage reinforcement learning for combining multiple metaheuristic algorithms (TSRL-CMM). First, TSRL-CMM divides the iterative process into three stages and defines eight states for the population. On this basis, we propose an adaptive action selection strategy and design a reward function based on state transitions. In the experiments, we evaluated the performance of TSRL-CMM using the CEC 2017 benchmark suite, eight constrained engineering design problems, and parameter estimation problems for photovoltaic modules. TSRL-CMM was compared with 10 state-of-the-art metaheuristic algorithms. The experimental results indicate that TSRL-CMM outperforms the compared algorithms in both artificial and real-world problems. The main contributions of this paper are as follows:

(1) To improve efficient collaboration among metaheuristic algorithms, this paper proposes a three-stage reinforcement learning approach (TSRL). The three stages are the exploration stage ($Stage_e$), a stage combining both exploration and exploitation ($Stage_{eu}$), and the exploitation stage ($Stage_u$). We use AVOA, WOA, MPA, and PSO as candidate metaheuristic algorithms, utilizing TSRL to select the most appropriate metaheuristic algorithm to search the solution space during the iterative process. First, based on the population's diversity, convergence, and fitness metrics, eight states were constructed for the population. On this basis, we designed an adaptive action selection strategy and a reward function to achieve efficient collaboration among the four metaheuristic algorithms.

(2) To effectively balance the exploration and exploitation in TSRL-CMM, this paper proposes an adaptive ϵ-Greedy action selection strategy (AEAS). For the three stages, AEAS constructs three different methods for setting ϵ. In $Stage_e$, the algorithm focuses on exploration. Therefore, we set the value of ϵ based on the diversity metric and keep ϵ at a relatively high level. In $Stage_{eu}$, the algorithm simultaneously possesses a certain degree of exploration and exploitation. We also set the value of ϵ based on the diversity metric, but in this stage, the value of ϵ is lower than in the first stage. In $Stage_u$, the algorithm should converge quickly. Therefore, we set the value of ϵ based on the convergence metric and maintain ϵ at a relatively low level.

(3) This paper proposes a reward function based on three stages and state transitions. We have defined eight states that can reflect the algorithm's performance, taking into account that the same state may reflect different algorithm's performance at different stages. We have returned different rewards for different state transitions. Additionally, in different stages, we return different rewards for the same state transitions. This reward function helps guide the population towards expected state transitions at different stages.

(4) We validated the proposed method using the CEC 2017 test suite and nine real-world optimization problems. To verify the performance of AEAS, we compared it with five existing action selection strategies. The experimental results demonstrate that AEAS outperforms the five compared action selection strategies. To verify the performance of TSRL-CMM, we compared it with 10 state-of-the-art algorithms. The experimental results demonstrate that TSRL-CMM significantly outperforms these 10 comparison algorithms. Additionally, we conducted sensitivity analysis on TSRL-CMM's parameters. The results demonstrate that TSRL-CMM is insensitive to parameter variations, thus confirming its robustness.

The remaining parts of this paper are arranged as follows. Section 2 introduces various combinations of metaheuristic algorithms and related work on reinforcement learning. Section 3 first describes the overall framework of TSRL-CMM, then introduces the proposed adaptive action selection strategy and reward function, and finally analyzes the algorithm's complexity. Section 4 verifies the performance

of TSRL-CMM and AEAS on the CEC 2017 test suite, and conducts sensitivity analysis on TSRL-CMM's parameters. Section 5 verifies the performance of TSRL-CMM in solving real-world problems. Finally, Section 6 summarizes the paper and proposes suggestions for future work.

2. Related work

2.1. Problem definition

This study primarily addresses single-objective optimization problems [26]. A typical single-objective optimization problem can be described by Eq. (1), which mainly includes the objective function, inequality constraints, equality constraints, and the definition of decision variables.

$$\min_{x\in\Re^D} f(x) \quad \text{subject to} \begin{cases} h_i(x)=0, i=1,2,\ldots,p \\ g_j(x)\le 0, j=1,2,\ldots,q \\ l_d \le x_d \le u_d, d=1,2,\ldots,D \end{cases} \tag{1}$$

where $x=(x_1,x_2,\ldots,x_D)$ represents a solution containing D decision variables, $f(x)$ is the objective function, $h_i(x)=0$ denotes the ith equality constraint, p is the number of equality constraints, $g_j(x)\le 0$ denotes the jth inequality constraint, q is the number of inequality constraints, and l_d and u_d are the lower and upper bounds of the dth decision variable, respectively.

2.2. Related work on metaheuristic algorithms

Metaheuristic algorithms can be broadly classified into swarm intelligence-based, evolutionary computation-based and physics-based algorithms.

Swarm intelligence-based metaheuristic algorithms draw inspiration from the collective behavior of biological groups. PSO [8] guides particles to move towards the best-known positions, making it highly effective in high-dimensional continuous optimization problems. Artificial Bee Colony (ABC) [27] simulates the foraging behavior of bees and has been successfully applied to combinatorial optimization tasks such as vehicle routing. WOA [6] models the bubble-net hunting strategy of whales and has shown great promise in high-dimensional problems, including image processing and medical diagnosis. Recently, Improved Grey Wolf Optimizer (IGWO) [28] has been proposed to enhance convergence speed and maintain population diversity, proving effective in solving engineering optimization problems. Furthermore, Discrete Improved Grey Wolf Optimizer [29] has demonstrated strong performance in community detection tasks, highlighting the applicability of swarm intelligence algorithms beyond traditional continuous domains.

Evolutionary computation-based metaheuristic algorithms are inspired by natural selection and genetic evolution. Genetic Algorithm (GA) [30] applies selection, crossover, and mutation operations to iteratively improve candidate solutions, making it widely used in combinatorial optimization problems such as job scheduling. Differential Evolution (DE) [15] employs mutation and recombination mechanisms to efficiently explore search spaces, particularly excelling in engineering design and energy scheduling problems. Additionally, Evolutionary Strategy (ES) [31] has been successfully applied to unconstrained optimization, including robotic path planning and spacecraft trajectory optimization. In recent advancements, Multi-Trial Vector-based Differential Evolution (MTDE) [32] has been introduced to improve search efficiency in engineering design problems.

Physics-based metaheuristic algorithms are inspired by natural physical principles such as thermal dynamics and gravitational forces. Simulated Annealing (SA) [33] follows the concept of gradual temperature reduction to escape local optima and is commonly applied in scheduling problems and the traveling salesman problem. Gravitational Search Algorithm (GSA) [34] simulates the attraction of masses and has demonstrated effectiveness in signal processing and feature selection tasks. Multi-Trial Vector-based Sine Cosine Algorithm (MTV-SCA) [35] combines multiple search strategies to enhance convergence performance in real-world applications These physics-based algorithms provide unique search dynamics that complement traditional population-based methods.

2.3. Related work on the combination of multiple metaheuristic algorithms

In the field of solving single-objective optimization problems, Parouha et al. [36] designed an advanced hybrid algorithm based on a multi-population approach (haDEPSO) to address optimization problems. This algorithm divides the population into two sub-populations, each using the advanced differential evolution algorithm (aDE) and the advanced particle swarm optimization algorithm (aPSO) respectively to search the problem space. By leveraging the search characteristics of the two different algorithms, haDEPSO performs better than compared algorithms. Dhiman [37] developed a combination optimization algorithm (SSC) based on the Sine Cosine Algorithm (SCA), the Spotted Hyena Optimizer (SHO), and the Chimp Optimization Algorithm (ChoA). In SSC, the exploration and exploitation capabilities of the algorithm are enhanced by incorporating SCA and SHO into the position update equations of ChoA. Duque et al. [38] proposed a dynamic parameter adaptation method for solving the Quadratic Assignment Problem (QAP), called DPA-QAP. DPA-QAP consists of two main components: worker nodes and a master node. During the iterations, the worker nodes respectively run three different metaheuristic algorithms: Robust Tabu Search (RoTS), Extremal Optimization (EO), and Multi-Start Local Search (MLS). The master node adjusts the algorithm parameters based on feedback from the worker nodes. Kalayci et al. [39] proposed an efficient combination metaheuristic algorithm for solving portfolio optimization problems with cardinality constraints. In each iteration, the Continuous Ant Colony Optimization (CACO) is used to generate new solutions, the GA optimizes the solutions through elitism, crossover, and mutation, and the Artificial Bee Colony (ABC) algorithm improves the solutions through local search and probabilistic selection. Salgotra et al. [40] proposed a novel multi-hybrid algorithm (MHA) based on the Naked Mole-Rat Algorithm (NMRA). The algorithm divides the iterations into four stages. In the first stage, the basic NMRA is used for the search. In the second stage, a random probability is used to select between exploration and exploitation strategies within the Whale Optimization Algorithm (WOA). In the third stage, MPA is employed to transition from exploration to exploitation, controlling individual movements with Brownian motion and Lévy flight. In the fourth stage, the Moth–Flame Optimization (MFO) is utilized for local search. Additionally, to address stagnation and prevent the algorithm from getting trapped in local optima, new solutions are generated using the mean of random solutions from the Grey Wolf Optimizer (GWO) and the Cauchy mutation from Cuckoo Search (CS).

In the field of solving multi-objective optimization problems, Tian et al. [41] proposed a reinforcement learning-based operator selection method. This method considers the solutions and their weight vectors as states and the candidate operators as actions, including GA, Non-dominated Sorting Genetic Algorithm II (NSGA-II), and DE. A deep neural network is used to learn a policy that estimates the Q-values of each action given a state, selecting the most appropriate operator based on these Q-values. Chegari et al. [42] developed a novel and efficient multi-objective optimization algorithm based on Building Performance Optimization (BPO) techniques. In each iteration, this method uses a Multi-layer Feedforward Neural Network (MFNN) to select the most appropriate algorithm from three metaheuristic algorithms to search the problem space. These three metaheuristic algorithms are NSGA-II, Multi-Objective Particle Swarm Optimization (MOPSO), and Multi-Objective Genetic Algorithm (MOGA). Ming et al. [43] proposed an online operator selection framework assisted by deep reinforcement learning, where the population's convergence, diversity, and feasibility

are considered as states. Two candidate operators are considered as actions, including GA and DE; and the improvements in the population's states are considered as the reward. By using a Q-network to learn a policy that estimates the Q-values of all actions, the framework adaptively selects the optimal operator based on the current state. Farag et al. [44] combined NSGA-II with SCA to form a new combination algorithm: Non-dominated Sorting Sine Cosine Genetic Algorithm (NS-SCGA), to solve multi-objective optimization problems. NS-SCGA divides the parent population into two equal subpopulations, with the two subpopulations being updated by NSGA-II and SCA, respectively. Finally, the two subpopulations are merged to generate a new population. Peng et al. [45] proposed a method for combining multiple metaheuristic algorithms to determine the optimal sizing of a Hybrid Renewable Energy System (HRES). This algorithm introduces Chaos Search (CS) and Simulated Annealing (SA) into the Harmony Search (HS) algorithm. First, a new solution is generated using an improved SA, and then a feasible random integer is generated through a chaotic sequence, which is used to adjust the position of the new solution. Zhang et al. [46] proposed a novel combination optimization algorithm for optimal sizing of hybrid solar and wind energy systems. The algorithm divides the iteration process into three stages: the initial search stage, the local optimization stage, and the fine-tuning stage. The CS algorithm is used in the initial search stage, the HS algorithm is used in the local optimization stage, and the SA algorithm is used in the fine-tuning stage. Askarzadeh et al. [47] proposed a SA algorithm based on discrete chaotic HS to optimize hybrid photovoltaic/wind energy systems. Specifically, the discrete SA is used to generate new solutions, while discrete HS generates new candidate solutions. These solutions are then combined with random integers generated through chaotic sequences to increase solution diversity.

2.4. Related work on ε-Greedy strategy in reinforcement learning

The balance between exploration and exploitation in the action selection process of reinforcement learning has a significant impact on the performance of the algorithm [48]. The ε-Greedy strategy is an action selection method that balances exploration and exploitation. In each iteration, a random number r_ε within the range [0,1] is generated. When $r_\varepsilon < \varepsilon$, the agent randomly selects an action to explore the solution space. When $r_\varepsilon \geq \varepsilon$, the agent selects the action with the highest Q-value to exploit the solution space. In the traditional Q-learning algorithm, ε is usually a fixed value ($\varepsilon < 1$) [49], which limits the adaptability of Q-learning. Zhang et al. [50] proposed a linearly decreasing ε adjustment method, where ε linearly decreases from 0.99 to 0.05 as the number of iterations increases. This method helps prevent the algorithm from getting trapped in local optima. Pei et al. [51] introduced an important mechanism from the SA algorithm (the cooling schedule) to modify the ε-Greedy strategy, establishing a nonlinear cooling schedule for the probability ε. Andrea Ortiz et al. [52] proposed a nonlinear decreasing ε adaptive method, where ε is set to $e^{-0.1*t}$, with t being the number of iterations. Mojgan Fayyazi et al. [22] proposed a time-varying search factor that divides the ε-Greedy strategy into two phases, designing an adaptive exploration factor for the probability ε in each phase, enabling ε to change adaptively. This method effectively balances exploration and exploitation.

3. Proposed algorithm

3.1. Motivation

The objective of this study is to utilize reinforcement learning for combining multiple metaheuristic algorithms. We select PSO, AVOA, MPA, and WOA as candidate metaheuristic algorithms based on their complementarity in search strategies and their wide applications in the field of metaheuristic optimization. First, these four algorithms show a good balance between exploration and exploitation. For example, PSO has strong convergence ability and is good at local exploitation, while AVOA and MPA emphasize global exploration and help to jump out of local optimal solutions. WOA combines the spiral search mechanism and shows strong refined optimization ability in local search. Second, these algorithms are diverse in search strategies, covering different mechanisms such as particle swarm collaboration, predator behavior simulation, and adaptive search strategy. Finally, to verify their complementarity, we conduct a quantitative analysis in Section 4.3. Experimental results show that these four algorithms each show unique advantages in the CEC 2017 benchmark test, and can complement each other in different problems and dimensions, thereby providing TSRL-CMM with stronger search capabilities. This analysis provides a theoretical basis and experimental support for algorithm selection. During the iteration process, reinforcement learning is used to adaptively select the optimal metaheuristic algorithm. This approach addresses the issue of traditional combination methods that rely on human experience and cannot adaptively select the optimal metaheuristic algorithm based on the state of the algorithm. In constructing a combination method of multiple metaheuristic algorithms based on reinforcement learning, the action selection method and the reward function are crucial [53]. The action selection method must balance between exploration and exploitation; neglecting exploration can cause the algorithm to get trapped in local optima, while neglecting exploitation can make it difficult for the algorithm to converge.

The ε-Greedy strategy is a widely used action selection method, where the value of ε has a crucial impact on exploration and exploitation [54]. A small ε value makes the algorithm focus on exploitation, while a large ε value makes the algorithm focus on exploration. Since the exploration and exploitation capabilities required by the algorithm differ at various stages of the iteration process, the early iterations should focus more on exploration, while the later iterations should focus more on exploitation. Additionally, the value of ε should be associated with the state of the environment. For example, when the diversity of the algorithm is poor, a larger ε value is needed to enable the algorithm to explore more of the solution space. Similarly, when the convergence of the algorithm is poor, a smaller ε value is needed to enable the algorithm to exploit the optimal actions and accelerate its convergence. Therefore, this paper proposes an adaptive adjustment method for ε, considering both the iteration state and the state of the environment. Firstly, we divided the iteration process into three stages, each with its own adaptive adjustment method for ε. In constructing the adaptive adjustment method for ε, we considered the diversity and convergence of the algorithm. When the diversity of the algorithm decreases, we increase the value of ε, and vice versa. Similarly, when the convergence of the algorithm is poor, we decrease the value of ε, and vice versa.

The reward function is another important factor affecting reinforcement learning [55]. The reward function is often problem-specific and needs to be constructed optimally for different problems. Therefore, this study constructs a corresponding reward function for the combination of multiple metaheuristic algorithms. The reward function should be related to the state of the environment. When an action transitions the agent to a better state, it should return a positive reward. Conversely, when an action transitions the agent to a worse state, it should return a negative reward. In the iteration process of metaheuristic algorithms, good and bad states are dynamically changing. For example, in the later stages of iteration, high diversity and poor convergence are considered a bad state, but in the early stages of iteration, this can be considered a good state. Therefore, when constructing the reward function, the iteration state should be taken into account. To this end, this paper constructs a reward function that considers both state transitions and the iteration state.

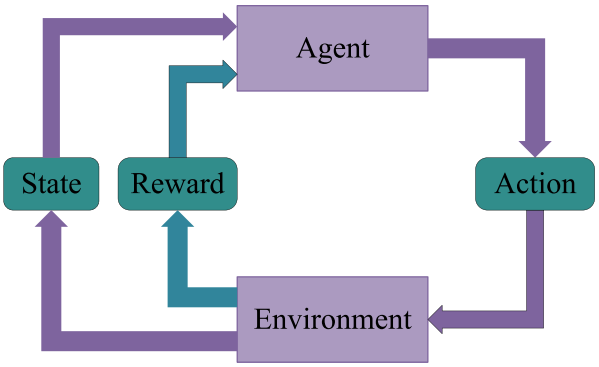


Fig. 1. Q-learning framework.

3.2. Overall framework of TSRL-CMM

Metaheuristic algorithms are a class of general optimization algorithms that are suitable for various optimization problems, but different metaheuristic algorithms have their own respective characteristics [56]. Therefore, multiple metaheuristic algorithms can be combined to enhance algorithm performance by leveraging the advantages of each metaheuristic algorithm. The most critical aspect of combining multiple metaheuristic algorithms is how to select the optimal metaheuristic algorithm during the iteration process, which is a dynamic decision-making problem. Reinforcement learning is one of the powerful methods for solving dynamic decision-making problems. Reinforcement learning obtains the mapping relationship between the agent's states and actions through continuous interaction between the agent and the environment. This mapping relationship can guide the agent to select appropriate actions for the current state, thereby maximizing the cumulative reward. Fig. 1 illustrates the interaction process between the agent and the environment. In this study, we use reinforcement learning to combine multiple metaheuristic algorithms. Q-learning is one of the most widely used reinforcement learning methods [57], so we adopted Q-learning to combine multiple metaheuristic algorithms.

Q-learning records the Q-values of state–action pairs in a Q-value table, and then selects the optimal action for a given state based on these Q-values. The size of the Q-value table is equal to the number of states multiplied by the number of actions. The rows of the Q-value table represent states, and the columns represent actions. In this study, we used diversity, convergence, and fitness as the three indicators to construct the state of the environment. The diversity of the population represents the distribution of individuals within the population, which is represented by the evolutionary factor Ef [58]. The larger the Ef, the farther the distance among individuals in the population, indicating better diversity; the smaller the Ef, the closer the distance among individuals, indicating worse diversity. First, calculate the average Euclidean distance between the ith individual and other individuals, which is given by Eq. (2):

$$d_i = \frac{1}{N-1}\sum_{j=1,j\neq i}^{N}\sqrt{\sum_{k=1}^{D}\left(x_i^k - x_j^k\right)^2} \tag{2}$$

where N is the population size, and D is the dimension of the problem. Next, normalize the Euclidean distance d_i to obtain the evolutionary factor for each individual, and then average these values to get the evolutionary factor Ef for the population, as shown in Eq. (3):

$$Ef = \frac{1}{N}\sum_{i=1}^{N}\frac{d_i - d_{\min}}{d_{\max} - d_{\min}} \tag{3}$$

where d_{max} and d_{min} represent the longest and shortest distances of d_i ($i \in \{1,\ldots,N\}$), respectively.

Table 1
State definitions.

State	Metrics
s_1	$\Delta Ef \geq 0, \Delta C \geq 0$ and $\Delta F_{mean} \geq 0$
s_2	$\Delta Ef \geq 0, \Delta C < 0$ and $\Delta F_{mean} \geq 0$
s_3	$\Delta Ef \geq 0, \Delta C \geq 0$ and $\Delta F_{mean} < 0$
s_4	$\Delta Ef \geq 0, \Delta C < 0$ and $\Delta F_{mean} < 0$
s_5	$\Delta Ef < 0, \Delta C \geq 0$ and $\Delta F_{mean} \geq 0$
s_6	$\Delta Ef < 0, \Delta C < 0$ and $\Delta F_{mean} \geq 0$
s_7	$\Delta Ef < 0, \Delta C \geq 0$ and $\Delta F_{mean} < 0$
s_8	$\Delta Ef < 0, \Delta C < 0$ and $\Delta F_{mean} < 0$

Convergence measures the distance between each individual and the optimal position [59]. First, calculate the Euclidean distance $conv_i$ between the position x_i of each individual in the population and the global optimal position $Gbest$ at the current iteration, as shown in Eq. (4):

$$conv_i = \sqrt{\sum_{k=1}^{D}\left(x_i^k - Gbest^k\right)^2} \tag{4}$$

Then, normalize the Euclidean distance $conv_i$ to obtain the convergence measure for each individual, and average these values to get the convergence of the population C, as shown in Eq. (5):

$$C = \frac{1}{N}\sum_{i=1}^{N}\frac{conv_i - conv_{\min}}{conv_{\max} - conv_{\min}} \tag{5}$$

where $conv_{max}$ and $conv_{min}$ represent the longest and shortest distances of $conv_i$ ($i \in \{1,\ldots,N\}$), respectively. The smaller the C, the better convergence of the population; while the larger the C, the worse convergence of the population.

In addition to calculating diversity and convergence, we also represent the state of the solutions in the population by computing the normalized average fitness value F_{mean}, as shown in Eq. (6):

$$F_{mean} = \frac{1}{N}\sum_{i=1}^{N}\frac{F_i - F_{\min}}{F_{\max} - F_{\min}} \tag{6}$$

where F_i represents the fitness value of the ith individual, F_{max} represents the maximum individual fitness value in the population, and F_{min} represents the minimum individual fitness value in the population. By combining these three metrics, we define the population state based on the changes in diversity Ef, convergence C, and fitness value F_{mean} after the population performs a action. Specifically, we calculate the differences between the current and previous iteration values of these three metrics, as shown in Eq. (7):

$$\begin{aligned}\Delta Ef &= Ef(t+1) - Ef(t)\\ \Delta C &= C(t) - C(t+1)\\ \Delta F_{mean} &= F_{mean}(t) - F_{mean}(t+1)\end{aligned} \tag{7}$$

where ΔE_f, ΔC and ΔF_{mean} represent the variation in population diversity, convergence, and fitness value, respectively. For example, $\Delta Ef \geq 0, \Delta C \geq 0$ and $\Delta F_{mean} \geq 0$ represent the improvement in population diversity, convergence, and fitness value, respectively; $\Delta Ef < 0, \Delta C < 0$ and $\Delta F_{mean} < 0$ represent the deterioration in population diversity, convergence, and fitness value, respectively.

Therefore, based on the changes in the three metrics, the states can be defined as one of eight types, $State = \{s | s = s_1, s_2, \ldots, s_8\}$. Each state is defined as shown in Table 1. To better illustrate the transitions between different states, we have constructed a state transition diagram, as shown in Fig. 2, where s_i ($i = 1, 2, \ldots, 8$) represents an arbitrary state.

After defining the states, Q-learning can select the corresponding action based on the state. In this study, we defined four actions, each corresponding to one of the four metaheuristic algorithms: PSO, AVOA, MPA, and WOA. These four algorithms were chosen because they each have distinctive search capabilities [60]. Detailed information about these four metaheuristic algorithms is provided in Appendix.

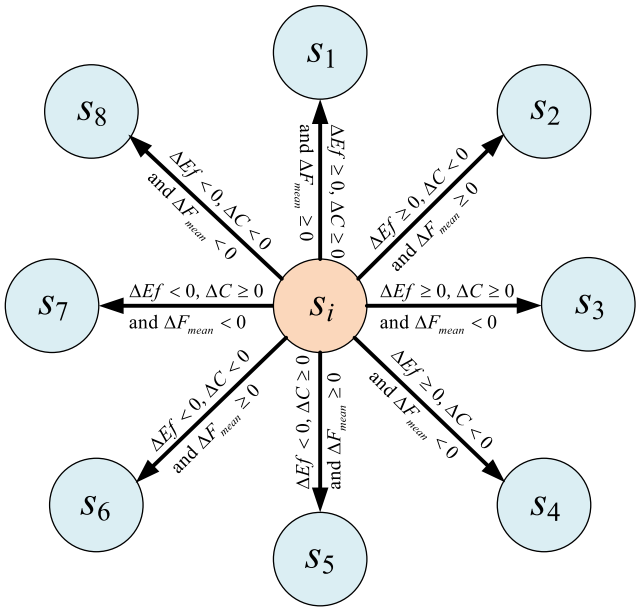


Fig. 2. State transition diagram.

In Q-learning, the Q-table is updated using the Bellman equation, as shown in Eq. (8):

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_a \left\{ Q_t(s_{t+1}, a) \right\} - Q_t(s_t, a_t) \right] \tag{8}$$

where s_t and a_t represent the current state and current action, respectively; s_{t+1} represents the next state; r_{t+1} is the immediate reward obtained from executing a action; γ is the discount factor within the range [0,1]; α is the learning rate within the range [0,1]; and $max_a \left\{ Q_t \left(s_{t+1}, a_t \right) \right\}$ is the maximum Q-value in the Q-table for state s_{t+1}. The value of the learning rate α directly affects the convergence speed of the algorithm. A higher learning rate may lead to faster convergence but can also cause the algorithm to overshoot the optimal solution due to large update steps, resulting in instability. A lower learning rate can make the algorithm more stable, but it will slow down the convergence speed. Therefore, α is usually set to a higher value at the beginning of the search process and gradually decreases during the iterations process [61]. The control method for α used in this study is shown in Eq. (9):

$$\alpha(t) = 1 - (0.9 * \frac{FEs}{MaxFEs}) \tag{9}$$

where FEs is the current number of function evaluations, and $MaxFEs$ is the maximum number of function evaluations. In addition, the discount factor γ plays an important role in balancing current rewards and future rewards. The larger the γ, the more the algorithm focuses on future rewards, while the smaller the γ, the more the algorithm focuses on current rewards [62].

Based on the Q-value table, the ε-Greedy strategy is used to select the action corresponding to the current state. In this study, both the reward function and the control method for the value of ε are associated with the iteration state. We divided the iteration process into three stages: the exploration stage ($Stage_e$), the stage combining both exploration and exploitation ($Stage_{eu}$), and the exploitation stage ($Stage_u$). The method for dividing the three stages is shown in Eq. (10). On this basis, we proposed an adaptive action selection method and a reward function based on state transitions. Section 3.3 provides a detailed introduction to the proposed adaptive action selection method, while Section 3.4 provides a detailed introduction to the reward function based on state transitions. Based on the proposed action selection method and reward function, we proposed a three-stage reinforcement learning for combining multiple metaheuristic algorithms. The overall framework of TSRL-CMM is shown in Fig. 3, and its pseudocode is provided in Algorithm 1. For the PSO algorithm, the individual update process requires updating both position x_i and velocity v_i, whereas for the other algorithms, only the position x_i is updated.

$$\begin{cases} Stage_e & \text{if } FEs < \frac{MaxFEs}{3} \\ Stage_{eu} & \text{elseif } \frac{MaxFEs}{3} \leq FEs < \frac{2*MaxFEs}{3} \\ Stage_u & \text{otherwise} \end{cases} \tag{10}$$

3.3. Proposed adaptive action selection method

Under the ε-Greedy strategy, the agent selects a random action with a probability of ε and selects the action corresponding to the maximum Q-value with a probability of 1-ε. This approach helps to avoid getting trapped in local optima while ensuring that the algorithm converges to the optimal value. The ε-Greedy action selection strategy is shown in Eq. (11):

$$Action_t = \begin{cases} \text{best action} = Max \left[Q \left(\begin{matrix} \text{current state,} \\ \text{all actions} \end{matrix} \right) \right], \\ \qquad\qquad \text{if } rand \geq \varepsilon \\ \text{random action} = randi(\text{all actions}), \\ \qquad\qquad \text{if } rand < \varepsilon \end{cases} \tag{11}$$

Algorithm 1: The pseudo code of TSRL-CMM

```
Data: Population number N, dimension D, maximum number of function evaluations MaxFEs
Result: Globally optimal position Gbest
Initialize the population POP = {X_1, X_2, ..., X_N}, position vector x_i and velocity vector v_i, Each individual X_i contains position x_i and velocity v_i;
Define the set of states State = {s|s = s_1, s_2, ..., s_8}, Define the set of actions Action = {a|a = a_1, a_2, a_3, a_4} and set Q(State, Action) = zero in the Q-table;
for i = 1 to N do
    Calculate the fitness of the individual X_i;
while FEs ≤ MaxFEs do
    Population diversity is calculated by Equation (3);
    Population convergence is calculated by Equation (5);
    Population fitness is calculated by Equation (6);
    Population diversity, convergence, and fitness variation are calculated by Equation (7);
    Determine the current state s_t;
    Use Algorithm 2 to obtain action a_t;
    for i = 1 to N do
        switch metaheuristic algorithms do
            case a_1: PSO do
                Update x_i using the classical PSO equations from Ref. Kennedy and Eberhart [8];
            case a_2: AVOA do
                Update PBestVulture(x_i) using the classical AVOA equations from Ref. Abdollahzadeh et al. [5];
            case a_3: WOA do
                Update x_i using the classical WOA equations from Ref. Mirjalili and Lewis [6];
            case a_4: MPA do
                Update prey_i(x_i) using the classical MPA equations from Ref. Faramarzi et al. [7];
    Calculate the fitness of the individual X_i;
    Use Algorithm 3 to obtain reward;
    Update Q-value using Equation (8);
    Update Global optimal individual Gbest;
```

From Eq. (11), it can be seen that the setting of the ε value is crucial for Q-learning to balance exploration and exploitation. The smaller the ε value, the greater the probability of selecting the action

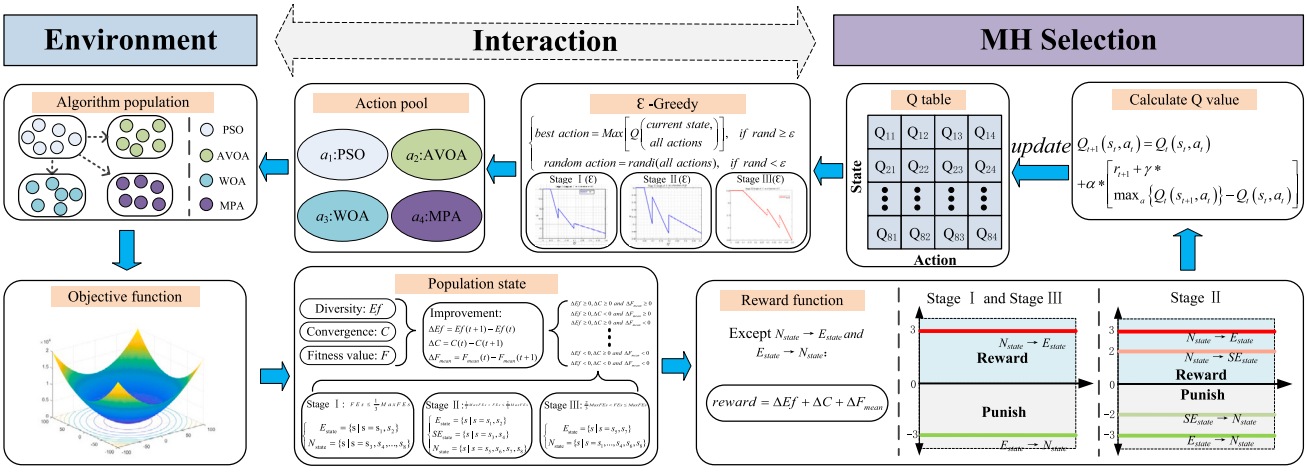


Fig. 3. The overall framework of TSRL-CMM.

corresponding to the maximum Q-value, resulting in better exploitation but worse exploration capability. The larger the ε value, the greater the probability of randomly selecting an action, resulting in better exploration but worse exploitation capability. In this study, we designed an adaptive ε-Greedy action selection strategy (AEAS). This method dynamically adjusts the value of ε based on the iteration state, diversity, and convergence state.

In the $Stage_e$ stage, the population needs to have good diversity. When Ef is larger, the population diversity is better, and the probability of selecting a random action should be reduced; when Ef is smaller, the diversity is worse, and the probability of selecting a random action should be increased. Therefore, ε should be inversely proportional to Ef. When Ef takes different values, the relationship between Ef and ε exhibits different functional forms. Additionally, random exploration must be controlled to reduce the waste of computational resources caused by excessive exploration. Therefore, the value of ε should not be set too high (an upper limit is needed). Finally, we set ε as a piecewise linear function of Ef, as shown in Eq. (12).

Specifically, in $Ef \in [0, 0.05)$, diversity is extremely low, indicating high similarity among individuals, so ε is maintained at its highest value to increase exploration. In $Ef \in [0.05, 0.1)$, diversity starts to improve, allowing a decrease in ε to encourage convergence. In $Ef \in [0.1, 0.2)$, the population maintains relatively good diversity, so the decline in ε slows down to balance exploration and exploitation. When $Ef \geq 0.2$, the population diversity is high, favoring exploitation, and thus ε decreases at the slowest rate to stabilize the search process. To prevent abrupt changes in exploration probability, we introduce an offset adjustment between different segments. This ensures that the final ε value of one segment is slightly higher than the initial value of the next, allowing a smooth transition between exploration and exploitation phases. A similar offset adjustment is also applied in $Stage_{eu}$ and $Stage_u$ to maintain consistency and ensure a stable adaptive control of exploration probability. Fig. 4(a) illustrates the variation curve of ε with Ef in the $Stage_e$ stage.

$$\varepsilon = \begin{cases} -0.25E_f + 0.175, & \text{if } E_f \geq 0.2 \\ -E_f + 0.3, & \text{if } 0.2 > E_f \geq 0.1 \\ -3E_f + 0.45, & \text{if } 0.1 > E_f \geq 0.05 \\ 0.3, & \text{if } 0.05 > E_f \geq 0 \end{cases} \tag{12}$$

In the $Stage_{eu}$ stage, the population requires less exploration capability compared to the $Stage_e$ stage, while the exploitation capability should be increased. Similarly, the probability of selecting a random action should be lower than in the $Stage_e$ stage, and the probability of selecting the optimal action should be higher. Therefore, in this stage, we also set the value of ε as a piecewise linear function of Ef, as shown in Eq. (13). However, compared to the ε values in the $Stage_e$ stage, the ε values in this stage are lower. Fig. 4(b) illustrates the variation curve of ε with Ef in the $Stage_{eu}$ stage.

$$\varepsilon = \begin{cases} -0.25E_f + 0.175, & \text{if } E_f \geq 0.2 \\ -E_f + 0.275, & \text{if } 0.2 > E_f \geq 0.1 \\ -2E_f + 0.3, & \text{if } 0.1 > E_f \geq 0.05 \\ 0.2, & \text{if } 0.05 > E_f \geq 0 \end{cases} \tag{13}$$

In the $Stage_u$ stage, the population needs to ensure good convergence. When C is smaller, the population convergence is better, and the probability of selecting a random action should be increased; when C is larger, the population convergence is poorer, and the probability of selecting a random action should be reduced. Therefore, ε should be inversely proportional to C. When C takes different values, the relationship between C and ε exhibits different functional forms. Thus, we set ε as a piecewise linear function of C, as shown in Eq. (14).

Specifically, in $C \in [0, 0.1)$, the population has good convergence, but a certain level of exploration is still required to refine the solution space, so ε is maintained at its highest value. In $C \in [0.1, 0.3)$, as convergence weakens, exploration is gradually reduced to prioritize exploitation, ensuring more stable optimization. In $C \in [0.3, 0.4)$, the search process becomes more focused on exploitation, leading to a faster decline in ε, further reducing random exploration. Finally, in $C \geq 0.4$, the search enters the final convergence phase, where exploration should be minimized to ensure that the algorithm settles into the optimal solution. Fig. 4(c) illustrates the variation curve of ε with C in the $Stage_u$ stage. Algorithm 2 presents the pseudocode of the proposed AEAS in this study.

$$\varepsilon = \begin{cases} -C + 0.55, & \text{if } C \geq 0.4 \\ -0.75C + 0.4, & \text{if } 0.4 > C \geq 0.3 \\ -0.5C + 0.3, & \text{if } 0.3 > C \geq 0.1 \\ 0.3, & \text{if } 0.1 > C \geq 0 \end{cases} \tag{14}$$

3.4. Reward function based on state transitions

In TSRL-CMM, the agent and the environment exchange information and interact with each other. The agent selects an action (i.e., metaheuristic algorithm) based on the current state and applies it to the environment. After executing the selected action, the environment transitions to the next state and returns a reward to the agent. The reward reflects the quality of the current action. Through continuous interaction with the environment, the agent can gradually improve its strategy. The setting of the reward is crucial, as it determines how

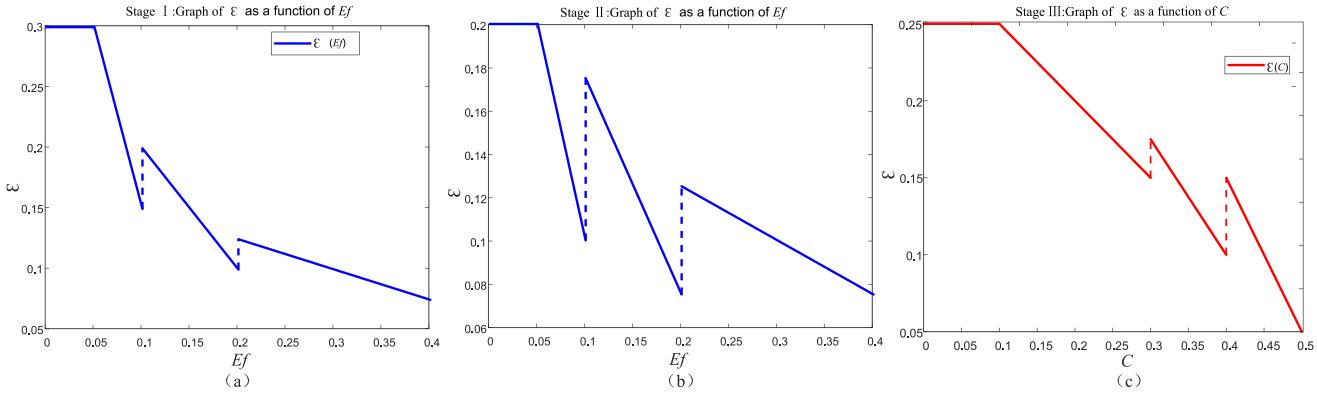


Fig. 4. Adaptive ε curves in the three stages.

Algorithm 2: Adaptive ϵ-Greedy Action Selection Strategy (AEAS)

Data: Population diversity Ef, convergence C, FEs and $MaxFEs$, Q-table
Result: Current action a_t
Initialize probability ϵ;
if $FEs \leq MaxFEs/3$ **then**
  if $Ef \geq 0.2$ **then**
    $\varepsilon = -0.25 * Ef + 0.175$;
  else if $0.2 > Ef \geq 0.1$ **then**
    $\varepsilon = -Ef + 0.3$;
  else if $0.1 > Ef \geq 0.05$ **then**
    $\varepsilon = -3 * Ef + 0.45$;
  else
    $\varepsilon = 0.3$;
else if $MaxFEs/3 < FEs \leq 2 * MaxFEs/3$ **then**
  if $E_f \geq 0.2$ **then**
    $\varepsilon = -0.25 * Ef + 0.175$;
  else if $0.2 > Ef \geq 0.1$ **then**
    $\varepsilon = -Ef + 0.275$;
  else if $0.1 > Ef \geq 0.05$ **then**
    $\varepsilon = -2 * Ef + 0.3$;
  else
    $\varepsilon = 0.2$;
else if $FEs > 2 * MaxFEs/3$ **then**
  if $C \geq 0.3$ **then**
    $\varepsilon = 0.35 * C$;
  else if $0.3 > C \geq 0.2$ **then**
    $\varepsilon = 0.3 * C + 0.05$;
  else
    $\varepsilon = 0.1 * C$;
Generate a random number r_ε between 0 and 1;
if $r_\varepsilon < \varepsilon$ **then**
  a_t = random action;
else
  a_t = action with the maximum Q value for the current state;
return a_t

the agent evaluates the quality of each action, thereby optimizing the agent's strategy.

The setting of the reward depends on the specific optimization objectives and the characteristics of the environment. In this study, the reward should reflect the changes in diversity, convergence, and fitness values. The changes in diversity, convergence, and fitness values are reflected in two aspects: the changes in their values, which is the difference in each indicator's value before and after the action is executed; and the state transitions, which is the change in state after the action is executed. In this study, both aspects are considered when returning the reward. First, we use the changes in the three indicators as one form of the reward, as shown in Eq. (15):

$$CR = \Delta Ef + \Delta C + \Delta F_{mean} \tag{15}$$

The state reflects the performance of the algorithm, so state transitions indicate changes in the algorithm's performance. When returning the reward, state transitions should be considered to better evaluate the impact of actions on the algorithm's performance. Additionally, the state is related to the iteration stage, meaning that the performance of the algorithm reflected by the state varies at different iteration stages. Therefore, we define the state as multiple types based on the three iteration stages. The $Stage_e$ and $Stage_u$ stages have clear objectives: $Stage_e$ ensures good exploration capability, and $Stage_u$ ensures good exploitation capability. Therefore, we divide these two stages into two states: the expected state and the normal state. The $Stage_{eu}$ stage needs to ensure both a certain level of exploration capability and convergence, making it a transitional stage. We divide the state of this stage into three categories: the expected state, the second expected state, and the normal state.

In the $Stage_e$ stage, the population needs better diversity. Therefore, states s_1 and s_2 are considered the expected states (E_{state}) for the current stage; the remaining states are regarded as normal states (N_{state}). as shown in Eq. (16):

$$\begin{cases} E_{state} = \{s | s = s_1, s_2\} \\ N_{state} = \{s | s = s_3, s_4, ..., s_8\} \end{cases} \tag{16}$$

In the $Stage_{eu}$ stage, both the diversity and convergence of the population need to be considered simultaneously. Therefore, states s_1 and s_2 are regarded as the expected state (E_{state}) of the current stage; states s_3 and s_4 are regarded as the second expected state (SE_{state}); and the remaining states are considered as the normal state (N_{state}), as shown in Eq. (17):

$$\begin{cases} E_{state} = \{s | s = s_1, s_2\} \\ SE_{state} = \{s | s = s_3, s_4\} \\ N_{state} = \{s | s = s_5, s_6, s_7, s_8\} \end{cases} \tag{17}$$

In the $Stage_u$ stages, the population needs good convergence; therefore, states s_5 and s_7 are regarded as the expected state (E_{state}) for the current stage, while the remaining states are considered as the normal state (N_{state}). The details are as follows:

$$\begin{cases} E_{state} = \{s | s = s_5, s_7\} \\ N_{state} = \{s | s = s_1, \ldots, s_4, s_6, s_8\} \end{cases} \tag{18}$$

Algorithm 3: Reward Function

Data: current state s_t, FEs, $MaxFEs$, $\Delta Ef, \Delta C$ and ΔF_{mean}
Result: reward
Define expected states (E_{state}) and normal states (N_{state}) for each stage;
if $FEs \leq MaxFEs/3$ **then**
  $E_{state} = \{s_1, s_2\}$;
  N_{state} = {all other states};
else if $MaxFEs/3 < FEs \leq 2 * MaxFEs/3$ **then**
  $E_{state} = \{s_1, s_2\}$;
  $SE_{state} = \{s_3, s_4\}$;
  N_{state} = {all other states};
else if $FEs > 2 * MaxFEs/3$ **then**
  $E_{state} = \{s_5, s_7\}$;
  N_{state} = {all other states};
Calculate reward CR by Equation (15);
Calculate $reward$ based on state transition;
if $FEs \leq MaxFEs/3$ *and* $FEs > 2 * MaxFEs/3$ **then**
  if state transitions from E_{state} to N_{state} **then**
    $reward = -3$;
  else if state transitions from N_{state} to E_{state} **then**
    $reward = 3$;
  else
    $reward = CR$;
else if $MaxFEs/3 < FEs \leq 2 * MaxFEs/3$ **then**
  if state transitions from E_{state} to N_{state} **then**
    $reward = -3$;
  else if state transitions from N_{state} to E_{state} **then**
    $reward = 3$;
  else if state transitions from SE_{state} to N_{state} **then**
    $reward = -2$;
  else if state transitions from N_{state} to SE_{state} **then**
    $reward = 2$;
  else
    $reward = CR$;
return $reward$

In the $Stage_e$ and $Stage_u$ stages, the transition between E_{state} and N_{state} has a significant impact on the performance of the algorithm. When E_{state} transitions to N_{state}, it indicates a decline in algorithm performance, and the environment will return a negative reward to the agent. When N_{state} transitions to E_{state}, it indicates an improvement in algorithm performance, and the environment will return a positive reward to the agent. Therefore, the impact of transitions between these two states is greater than that of transitions between other states. To more effectively guide the transition towards the expected states in each stage, we specifically designed the reward mechanism. The reward CR is determined by the sum of the changes in three normalized indicators (Ef, C, F_{mean}), where each indicator's variation falls within the range of [−1,1]. Consequently, the sum of their variations, CR, is bounded within [−3,3]. To amplify the incentive effect of transitioning to expected states, we assign the maximum reward of 3 when transitioning from a non-expected state to an expected state, and the minimum reward of −3 when deviating from an expected state. This enhancement strengthens the reinforcement learning mechanism in driving optimal state transitions. Hence, when a transition occurs between E_{state} and N_{state}, the reward is set to 3 or −3. For transitions between other states, the reward is set to CR. Eq. (19) shows the calculation method for rewards in the $Stage_e$ and $Stage_u$ stages.

$$reward = \begin{cases} 3, & \text{if } N_{state} \text{ transition to } E_{state} \\ -3, & \text{if } E_{state} \text{ transition to } N_{state} \\ CR, & \text{otherwise} \end{cases} \tag{19}$$

$$reward = \begin{cases} 3, & \text{if } N_{state} \text{ transition to } E_{state} \\ 2, & \text{if } N_{state} \text{ transition to } SE_{state} \\ -2, & \text{if } SE_{state} \text{ transition to } N_{state} \\ -3, & \text{if } E_{state} \text{ transition to } N_{state} \\ CR, & \text{otherwise} \end{cases} \tag{20}$$

In the $Stage_{eu}$ stage, the transitions between E_{state} and N_{state} have the greatest impact on the algorithm's performance, while the transitions between SE_{state} and N_{state} have a lesser impact. When a transition occurs between E_{state} and N_{state}, we set the reward to 3 or −3. When a transition occurs between SE_{state} and N_{state}, we set the reward to 2 or −2. For transitions between other states, the reward is set to CR. Eq. (20) shows the calculation method for rewards in the stage. Algorithm 3 presents the pseudocode for the reward calculation method proposed in this study.

3.5. Computational complexity analysis

In the proposed TSRL-CMM, the computational complexity of the algorithm is mainly composed of three parts: RL-CMM, AEAS, and reward function based on state transitions. To analyze the computational complexity, we assume that the dimension of the problem f is D and the population size is N. RL-CMM consists of Q-learning and four metaheuristic algorithms. Q-learning only needs to be computed once per iteration, so its computational complexity is O(1). The computational complexity of each of the four metaheuristic algorithms is O($N \times D$), and since one of the metaheuristic algorithms is selected for the current population in each iteration, its computational complexity is O($N \times D$). AEAS and the reward function based on state transitions are also computed once per iteration, both having a computational complexity of O(1). Therefore, the overall computational complexity of the proposed algorithm is O($N \times D$).

4. Results and analysis of the artificial benchmark test set

This section can be divided into six parts. The first part verifies the performance of the proposed TSRL-CMM. In this experiment, the CEC 2017 benchmark test suite [63] is used to evaluate the algorithm's performance. The CEC 2017 test suite is widely used for performance testing of evolutionary algorithms and includes 29 test functions, each with four dimensions: 10D, 30D, 50D, and 100D. In each test instance, all variables are restricted to the range [−100, 100]. The 29 test functions are divided into four categories: unimodal functions (F1–F3), multimodal functions (F4–F10), hybrid functions (F11–F20), and composition functions (F21–F30). We compared TSRL-CMM with 10 algorithms, which include: four basic algorithms: AVOA [5], WOA [6], MPA [7], PSO [8]; four combination algorithms: GWOCS [64], CSC-AHHO [65], PSOGSA [66], PSOGWO [67]; and two reinforcement learning-based combination evolutionary algorithms: QLPSO [68] and QLSOPSO [69]. The second part evaluates the competitiveness of TSRL-CMM. TSRL-CMM is compared with three winner algorithms from CEC competitions, including HyDE [70] from CEC 2019, COLSHADE [71] from CEC 2020, and MadDE [72] from CEC 2021, to verify its competitive advantage. The third part analyzes the contribution of the basic algorithms in TSRL-CMM. A quantitative analysis of TSRL-CMM's structure, which consists of four basic algorithms, is conducted to investigate the role of each basic algorithm in the overall framework. The fourth part explores the search characteristics of TSRL-CMM. The exploration/exploitation capability and population diversity of TSRL-CMM during the search process are analyzed to evaluate its search efficiency and convergence properties. The fifth part verifies the performance of the proposed AEAS. In this experiment, AEAS is compared with five action selection strategies: one fixed value, one linearly decreasing, and three adaptive methods. The sixth part provides a detailed sensitivity analysis of the parameters in TSRL-CMM.

Table 2
Parameter settings of algorithms.

Algorithm	Year	Parameter
TSRL-CMM	–	$N = 50$, $\gamma = 0.8$.
AVOA	2021	$N = 30$, $L_1 = 0.8$, $L_2 = 0.2$, $P_2 = 0.4$, $P_3 = 0.6$.
MPA	2020	$N = 50$, $FADs = 0.2$, $P = 0.5$, $CF = 1$(nonlinearly decreased).
WOA	2016	$N = 30$, $b = 1$, $a_1 = [2,0]$, $a_2 = [-2,-1]$.
PSO	1995	$N = 50$, $c_1 = c_2 = 2$, $w_{max} = 0.8$, $w_{min} = 0.4$.
GWOCS	2020	$N = 30$, α = linearly decreasing[2,0], $\alpha_1 = 0.01$, $\beta = 1.5$, $Pa = 0.25$.
CSCAHHO	2022	$N = 30$, $a = 2$, $r \in [0,1]$, $r_1 \in [0,1]$, $r_2 \in [0, 2\pi]$, $r_3 \in [0,2]$, $\alpha_1 = 0.01$.
PSOGSA	2010	$N = 30$, $c_1 = c_2 = 2$, $w_{max} = 0.8$, $w_{min} = 0.4$, $G_0 = 1$.
PSOGWO	2016	$N = 40$, $c_1 = c_2 = 2$, $w_{max} = 0.8$, $w_{min} = 0.4$, α = Linearly decreasing[2,0].
QLPSO	2020	$N = 30$, $w = 0.729844$, $c = 1.49618$, a linearly decreases from 1 to 0.1, $\gamma = 0.8$.
QLSOPSO	2019	$N = 50$, $w = 0.4, 0.6, 0.8, 1.0$, $c_1 = 0.5, 1, 2, 2.5$, $c_2 = 0.5, 1, 2, 2.5$.

To ensure a fair comparison of the 11 algorithms and the six action selection strategies, all parameters were set according to the values reported in the original papers. Specifically, we extracted the parameter values directly from the respective studies and, where necessary, followed the methodologies described in those papers to determine the appropriate settings. The parameter configurations for each algorithm, summarized in Table 2, were directly adopted from the original papers without modification to maintain consistency with previous studies. The difference between the optimal value obtained by each algorithm and the actual optimal value was used as the evaluation metric for each algorithm (error value). Each algorithm was run independently 51 times on each test function, and the average of all runs was used as the result. When the error value is less than 10^{-8}, it is set to zero. Additionally, statistical tests (Wilcoxon rank-sum test) were used to analyze the significance of differences between algorithms, with a significance level set at 0.05. The number of function evaluations was used as the stopping criterion for the algorithms, with a maximum of $D * 10^4$ function evaluations. TSRL-CMM and AEAS were implemented in MATLAB and run on a PC with a 3.50 GHz 13th Gen Intel(R) Core(TM) i5-13600K processor, 32 GB RAM, and a 64-bit operating system.

4.1. Comparison results and analysis of TSRL-CMM with other algorithms on CEC 2017 benchmark experiments

In Appendix, Tables A.1 to A.4 list the mean and standard deviation of each function at $10D, 30D, 50D$ and $100D$ for TSRL-CMM and other algorithms in detail, with the best results highlighted in bold. Table 3 summarizes the comparison results of TSRL-CMM with other algorithms. The symbols "+", "−", and "=" indicate that, according to the Wilcoxon rank-sum test, the mean error value obtained by TSRL-CMM is significantly better ("+"), significantly worse ("−"), or not significantly different ("=") from the corresponding comparison algorithm's mean error value. The detailed results of the Wilcoxon rank-sum test are shown in Tables A.5 to A.8.

As shown in Table 3, TSRL-CMM's performance is significantly better than the other algorithms. The closest competitor to TSRL-CMM is QLPSO, with TSRL-CMM outperforming QLPSO in 80 cases and being outperformed by QLPSO in only 19 cases. Although both are reinforcement learning-based for combining metaheuristic algorithms, TSRL-CMM benefits from the three-stage reinforcement learning approach, providing more precise control over states and action selection, giving it an advantage over other reinforcement learning methods. In each problem dimension, TSRL-CMM showed an average improvement in about 20 cases, while the number of cases with decreased performance was only about 5. Notably, TSRL-CMM demonstrated significant improvements compared to AVOA, MPA, WOA and PSO, suggesting that the combination of these four metaheuristic algorithms leverages their individual strengths and mitigates their weaknesses, leading to enhanced overall performance. TSRL-CMM also outperformed the four combination algorithms, showing that using reinforcement learning to combine multiple metaheuristic algorithms is superior to traditional methods. As we know, with the increase in problem dimensions, the difficulty of solving the problem also rises, and the optimization performance of algorithms faces greater challenges. However, the results in Table 3 indicate that TSRL-CMM has demonstrated strong competitiveness even in high-dimensional problems, and its performance is even better than in lower-dimensional cases.

As shown in Tables A.1 to A.4, for $10D$, TSRL-CMM significantly outperforms other algorithms on functions F1, F3, F4, F6, F9, F12, F15, F18, and F19. However, TSRL-CMM performs worse than AVOA on functions F29 and F30, and its performance is only slightly inferior to QLPSO on functions F11 and F21. For $30D$, TSRL-CMM significantly outperforms other algorithms on functions F1, F3, F4, F12–F15, F18, F19, F21–F23, F25, and F27–F30. However, QLPSO performs better than TSRL-CMM on functions F6, F7, F9, and F26. For $50D$, TSRL-CMM significantly outperforms other algorithms on functions F1, F3, F4, F7, F9–F16, F18, F19, F22, F25, and F27–F30. However, TSRL-CMM performs worse than QLPSO on functions F5 and F6. For $100D$, TSRL-CMM performs exceptionally well across all functions compared to other algorithms.

Figs. 5 to 8 illustrate the convergence curves of the 11 algorithms on four types of functions. For the case study, one test function from each type was selected. It is evident that in all 16 cases, TSRL-CMM demonstrates superior convergence performance compared to the other algorithms, further highlighting its excellent convergence speed and stability.

To observe the distribution of solutions obtained by all algorithms, one test function from each type was selected for a case study. Figs. 9 to 12 display the box plots of the solutions obtained by all algorithms. It is evident that in all 16 cases, TSRL-CMM exhibits lower standard deviation in its results compared to the other algorithms, further demonstrating its superior stability. For the multimodal function F7, TSRL-CMM exhibits higher standard deviations in results for $10D$ and $100D$ compared to $30D$ and $50D$, indicating that the algorithm's stability needs improvement. In the case of the hybrid function F15 and the composition function F30, although there are some outliers, the overall results show that the median, maximum, and minimum values remain consistently low.

To further compare the performance of TSRL-CMM with other algorithms, the Friedman test was used to rank TSRL-CMM and the other algorithms. The detailed rankings can be found in Tables A.9 to A.12. In the Friedman test, a lower average rank value indicates better algorithm performance. Fig. 13 shows the average ranking of the 11 algorithms. For $10D$, $30D$, $50D$, and $100D$, TSRL-CMM ranks first, AVOA ranks second, and QLPSO ranks third. PSOGSA ranks the last, and GWOCS ranks the second to last. Additionally, for $10D$ and $30D$, the rankings of PSO and QLSOPSO are quite similar. WOA, MPA, CSCAHHO, and PSOGWO have similar rankings but perform worse than PSO and QLSOPSO. For $50D$, MPA, PSO, CSCAHHO, PSOGWO, and QLSOPSO have similar rankings. For $100D$, MPA, PSO, CSCAHHO, and QLPSO have similar rankings, while PSOGWO and QLSOPSO have similar rankings but perform worse than MPA, PSO, CSCAHHO, and QLPSO. WOA's ranking on $50D$ and $100D$ is worse compared to its ranking on $10D$ and $30D$.

To compare the performance differences among multiple algorithms across different dimensions, we used the Scheffe's procedure for post-hoc analysis. First, based on the results of the Friedman test mentioned above, we determined the average performance of different algorithms

Table 3
Wilcoxon test results (10D, 30D, 50D and 100D).

	10D			30D			50D			100D			Total		
	+	=	−	+	=	−	+	=	−	+	=	−	+	=	−
AVOA	**17**	9	3	**25**	4	0	**27**	2	0	**25**	3	1	**94**	18	4
MPA	**27**	1	1	**29**	0	0	**29**	0	0	**28**	1	0	**113**	2	1
WOA	**28**	0	1	**29**	0	0	**29**	0	0	**29**	0	0	**115**	0	1
PSO	**24**	4	1	**25**	4	0	**29**	0	0	**29**	0	0	**107**	8	1
GWOCS	**26**	3	0	**29**	0	0	**29**	0	0	**29**	0	0	**113**	3	0
CSCAHHO	**22**	6	1	**29**	0	0	**29**	0	0	**29**	0	0	**109**	6	1
PSOGSA	**29**	0	0	**29**	0	0	**29**	0	0	**29**	0	0	**116**	0	0
PSOGWO	**26**	2	1	**28**	1	0	**29**	0	0	**29**	0	0	**112**	3	1
QLPSO	**18**	6	5	**20**	6	3	**21**	4	4	**21**	1	7	**80**	17	19
QLSOPSO	**26**	3	0	**27**	2	0	**28**	1	0	**29**	0	0	**110**	6	0
Total	**243**	34	13	**270**	17	3	**279**	7	4	**277**	5	8	**1069**	63	28

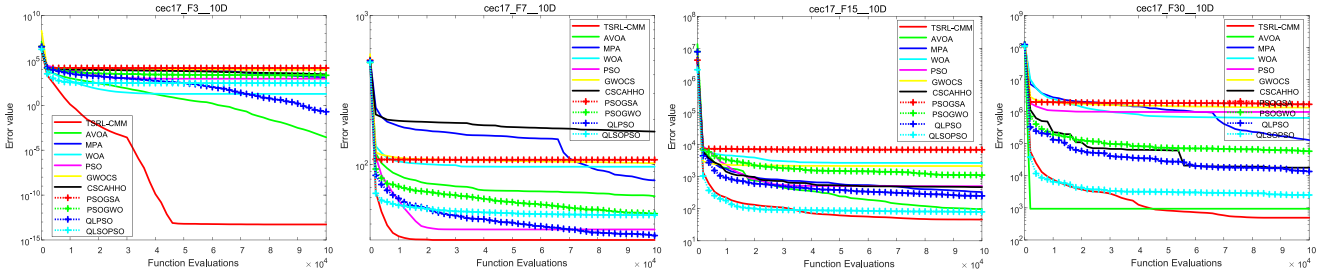


Fig. 5. Convergence curves of F3, F7, F15 and F30 (10D).

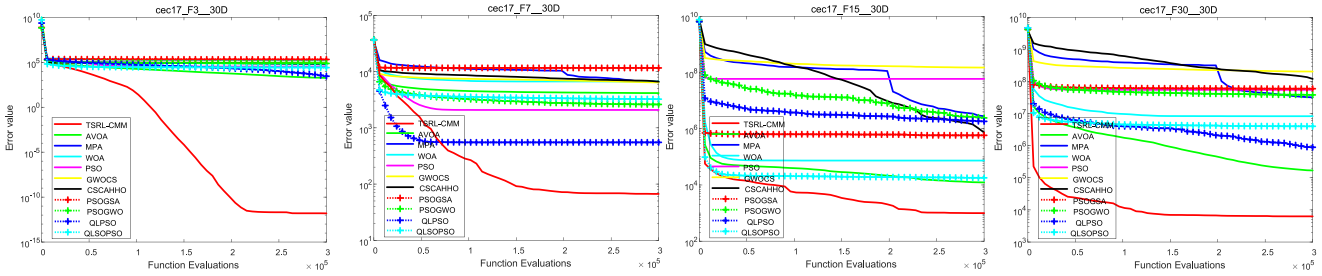


Fig. 6. Convergence curves of F3, F7, F15 and F30 (30D).

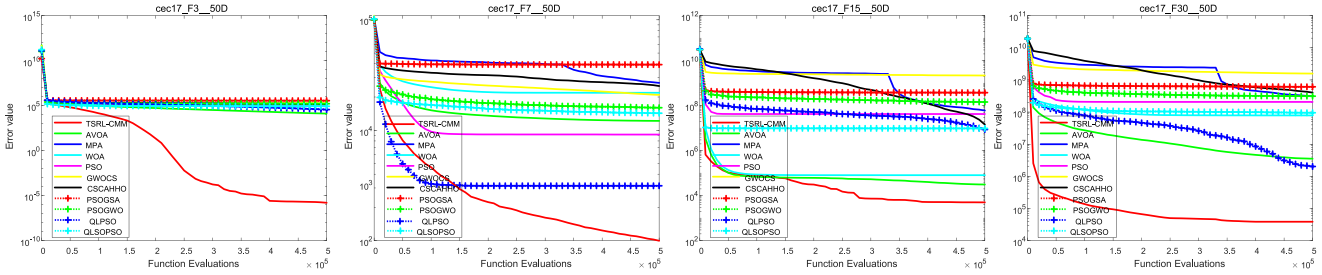


Fig. 7. Convergence curves of F3, F7, F15 and F30 (50D).

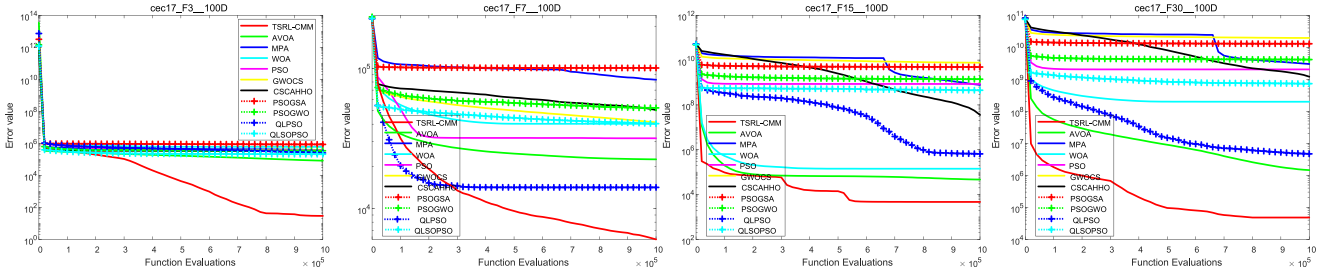


Fig. 8. Convergence curves of F3, F7, F15 and F30 (100D).

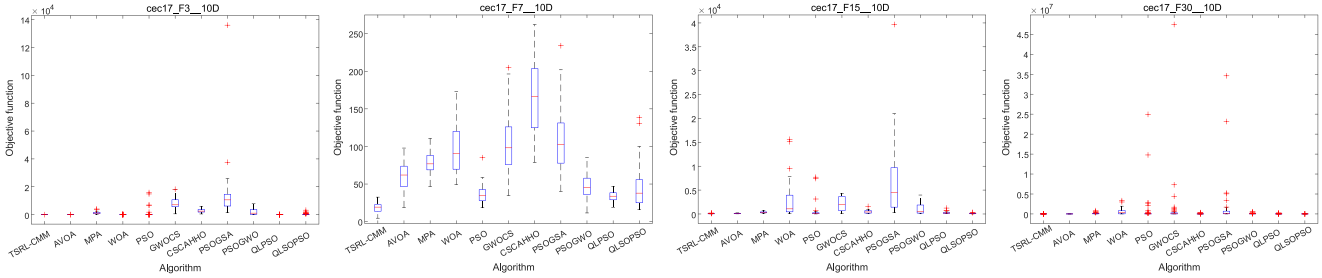


Fig. 9. Box plots of 11 algorithms (10D).

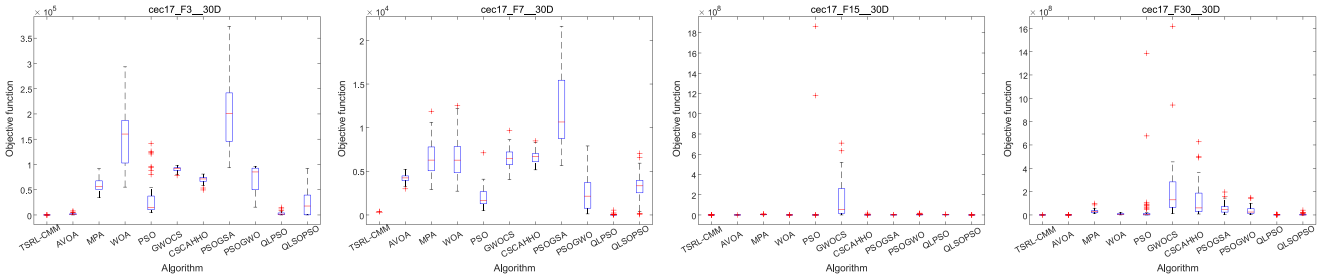


Fig. 10. Box plots of 11 algorithms (30D).

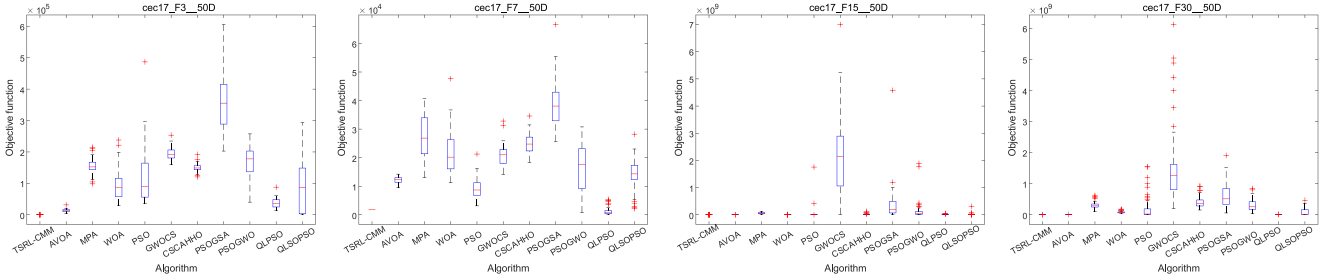


Fig. 11. Box plots of 11 algorithms (50D).

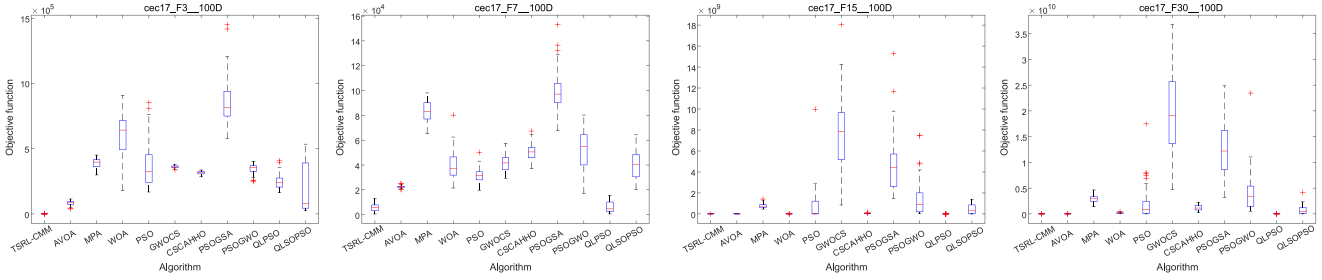


Fig. 12. Box plots of 11 algorithms (100D).

in each dimension. Subsequently, we applied the Scheffe's procedure for multiple comparisons to identify which specific algorithms have significant differences. As shown in Fig. 14, based on the Scheffe's procedure method for comparing the 11 algorithms on the CEC 2017 text suite, the blue ones refer to TSRL-CMM, the gray ones indicates that there is no significant difference (significance level of 0.05) between TSRL-CMM and the corresponding algorithm, and the red ones indicates there is a significant difference between TSRL-CMM and the corresponding algorithm. For dimensions 10D, 30D, 50D, and 100D, TSRL-CMM shows significant differences from the other 10 algorithms, and TSRL-CMM's performance is superior to the other 10 algorithms.

Table 4 summarizes all performance results of TSRL-CMM and other algorithms shown in Tables A.1–A.4 by a useful metric named overall effectiveness (OE) [32]. The OE of each algorithm is computed by Eq. (21), where M is the total number of tests, and L is the total number of losing tests for each algorithm. TSRL-CMM achieves the highest OE of 74.1%, outperforming all compared algorithms.

$$\mathrm{OE} = \left(\frac{M - L}{M}\right) \times 100 \tag{21}$$

4.2. Comparison results and analysis of TSRL-CMM with CEC winner algorithms

In Appendix, Tables A.13 to A.16 list the mean and standard deviation of each function at 10D, 30D, 50D, and 100D for TSRL-CMM and three CEC winner algorithms in detail, with the best results highlighted in bold. Table 5 summarizes the results of the Wilcoxon rank-sum test

Table 4
Effectiveness of the TSRL-CMM and other state-of-the-art competitor algorithms.

Dimensions	TSRL-CMM (w/t/l)	AVOA (w/t/l)	MPA (w/t/l)	WOA (w/t/l)	PSO (w/t/l)	GWOCS (w/t/l)	CSCAHHO (w/t/l)	PSOGSA (w/t/l)	PSOGWO (w/t/l)	QLPSO (w/t/l)	QLSOPSO (w/t/l)
$D = 10$	16/4/9	0/6/23	2/0/27	0/0/29	0/0/29	0/0/29	0/4/25	0/0/29	0/0/29	4/1/24	0/0/29
$D = 30$	20/1/8	0/0/29	0/0/29	1/0/28	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	7/1/21	0/0/29
$D = 50$	22/1/6	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	6/1/22	0/0/29
$D = 100$	21/1/7	2/0/27	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	0/0/29	4/1/24	0/0/29
Total	79/7/30	2/6/108	2/0/114	1/0/115	0/0/116	0/0/116	0/4/112	0/0/116	0/0/116	21/4/91	0/0/116
OE	**74.1%**	7%	2%	1%	0%	0%	3%	0%	0%	21.6%	0%

Table 5
Comparison results between TSRL-CMM and CEC winner algorithms.

	10D			30D			50D			100D			Total		
	+	=	−	+	=	−	+	=	−	+	=	−	+	=	−
COLSHADE	**27**	0	2	**26**	0	3	**25**	0	4	**25**	0	4	**103**	0	13
HyDE	**9**	3	17	**19**	1	9	**21**	3	5	**22**	3	4	**71**	10	35
MadDE	**6**	2	21	**11**	3	15	**14**	4	11	**23**	2	4	**54**	11	51
Total	**42**	5	40	**56**	4	27	**60**	7	20	**70**	5	12	**228**	21	99

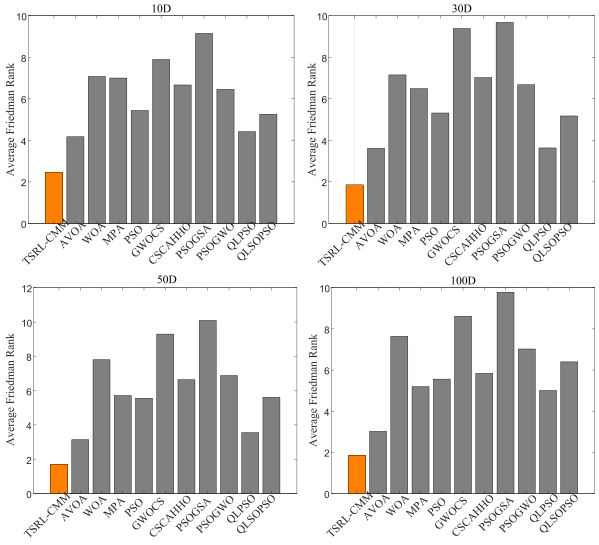


Fig. 13. Average rankings of Friedman test.

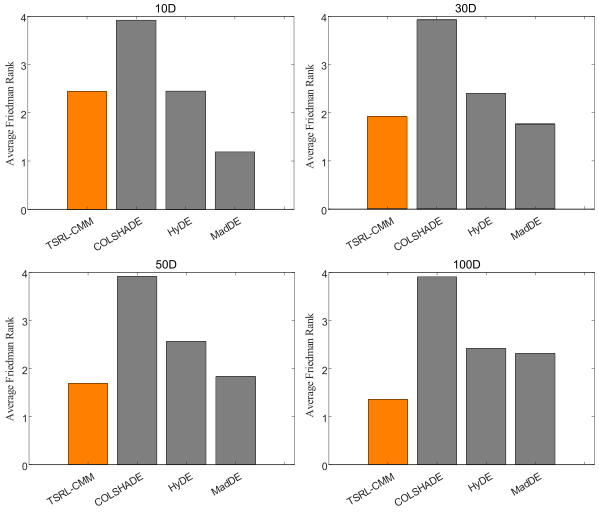


Fig. 15. Average rankings of Friedman test.

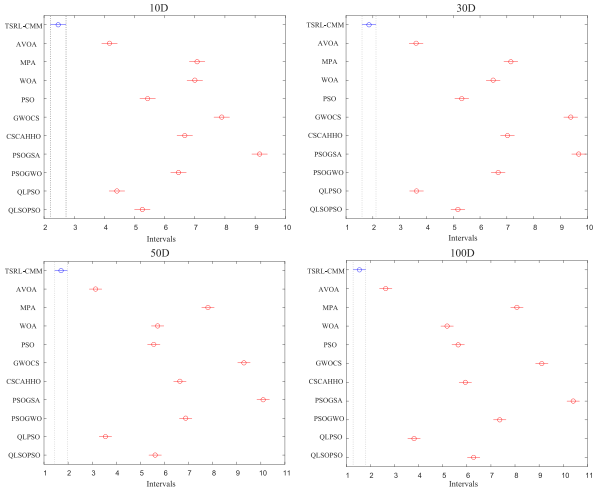


Fig. 14. Scheffe test results.

comparing our algorithm with CEC winner algorithms. Detailed results of the Wilcoxon rank-sum test can be found in Tables A.17 to A.20.

As shown in Table 5, TSRL-CMM performs significantly better than COLSHADE and HyDE, with its performance being closest to that of MadDE. For 10D, TSRL-CMM outperforms COLSHADE in 27 cases while being inferior in only 2 cases. However, TSRL-CMM is outperformed by HyDE and MadDE in 17 and 21 cases, respectively, indicating that its performance in lower dimensions still needs improvement. For 30D, TSRL-CMM outperforms the other three algorithms in approximately 19 cases on average, while it is inferior in about 9 cases. Specifically, compared to MadDE, TSRL-CMM outperforms it in 11 cases and is outperformed in 15 cases, suggesting that the performance gap between the two algorithms is narrowing. For 50D and 100D, TSRL-CMM outperforms the other three algorithms in approximately 22 cases on average, while being inferior in only about 5 cases. In 100D, TSRL-CMM surpasses the other three algorithms in at least 22 cases and is outperformed in only 4 cases, demonstrating that our algorithm exhibits strong competitiveness in high-dimensional problems.

To further compare the performance of TSRL-CMM with the three CEC winner algorithms, we employed the Friedman test to rank the four algorithms. The detailed rankings are presented in Tables A.21 to A.24. Fig. 15 illustrates the average rankings of the four algorithms. For 10D and 30D, MadDE ranks first, TSRL-CMM ranks second, and HyDE ranks third. Notably, in 30D, the ranking gap between TSRL-CMM and MadDE is smaller. For 50D and 100D, TSRL-CMM ranks first, followed by MadDE in second place and HyDE in third place. Overall, TSRL-CMM

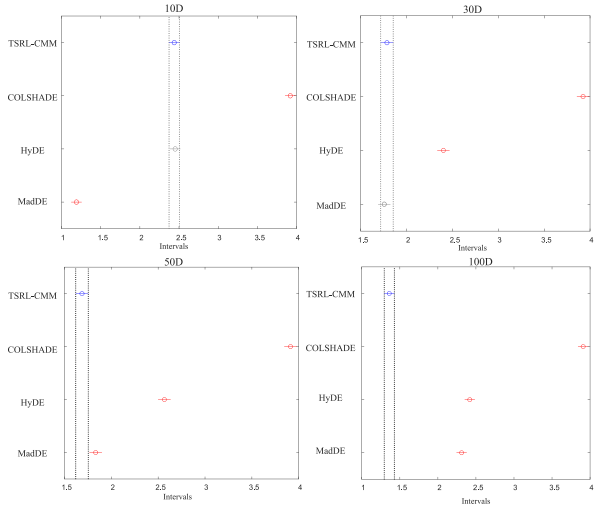


Fig. 16. Scheffe test results.

Table 6
Effectiveness of the TSRL-CMM and CEC winner algorithms.

Dimensions	TSRL-CMM (w/t/l)	COLSHADE (w/t/l)	HyDE (w/t/l)	MadDE (w/t/l)
$D = 10$	1/2/26	0/0/29	3/0/26	23/2/4
$D = 30$	9/0/20	0/0/29	5/0/24	15/0/14
$D = 50$	15/1/13	0/0/29	4/0/25	9/1/19
$D = 100$	22/0/7	0/0/29	4/0/25	3/0/26
Total	47/3/66	0/0/116	16/0/100	50/3/63
OE	43.1%	0%	13.8%	**45.7%**

achieves better rankings as the problem dimension increases, demonstrating its superior adaptability and performance in high-dimensional optimization problems.

To compare the performance differences of multiple algorithms across different dimensions, we applied the Scheffe post-hoc test. As shown in Fig. 16, the test results indicate that at $10D$, TSRL-CMM performs similarly to HyDE, better than COLSHADE, but worse than MadDE. At $30D$, TSRL-CMM exhibits a performance similar to MadDE and outperforms the other two algorithms. In $50D$ and $100D$, TSRL-CMM demonstrates significantly better performance than the other three algorithms, further reinforcing its effectiveness in high-dimensional optimization problems.

Table 6 summarizes all performance results of TSRL-CMM and CEC winner algorithms presented in Tables A.13 to A.16. The OE is calculated using Eq. (21) The OE of TSRL-CMM is 43.1%, which is slightly lower than that of MadDE (45.7%).

4.3. Quantitative analysis of the four basic algorithms

Based on the data and experimental results provided in Tables A.1–A.4 and Figs. 5–12, this section presents a quantitative analysis of the four fundamental algorithms that constitute TSRL-CMM: PSO, AVOA, MPA, and WOA. The analysis focuses on their differences in optimization performance, convergence behavior, and stability. By examining their effectiveness across unimodal, multimodal, hybrid, and composition functions, we aim to highlight the strengths and limitations of each algorithm, providing insights into their suitability for different types of optimization problems.

According to Tables A.1–A.4, AVOA demonstrates superior optimization performance and stability on multimodal functions (F4–F10). Specifically, on F6 and F9, the mean fitness values of AVOA in $50D$ and $100D$ are 18.5% and 21.3% lower than those of PSO, and 15.7% and 19.6% lower than those of WOA, indicating its stronger search capability and faster convergence speed in high-dimensional complex multimodal optimization problems. Additionally, for F5–F8, AVOA exhibits lower standard deviation, reducing optimization result fluctuations by 12.4% and 19.8% compared to PSO and WOA, respectively, demonstrating higher stability. The convergence curves (Figs. 5–8) show that on F7, AVOA has a significantly faster descent rate in the early stages of optimization than WOA and MPA, while still maintaining continuous improvement in the later iterations. In contrast, PSO stagnates after an initial rapid decline. The box plots (Figs. 9–12) further validate this trend, showing that on F7, AVOA's solution distribution is more concentrated, with outliers reduced by 17.6% and 22.3% compared to PSO and WOA, respectively, making its optimization results more stable and reliable.

WOA demonstrates better optimization performance on unimodal functions (F1 and F3) and certain hybrid functions (F18 and F19). On F1 and F3, WOA achieves mean fitness values that are 9.2% and 14.7% lower than those of PSO in $10D$ and $30D$, respectively. On F19, its mean fitness values are 12.1% and 15.3% lower than those of PSO in $30D$ and $50D$, indicating its stronger local search capability for hybrid optimization problems. Additionally, on F18, WOA shows less fluctuation in optimization results in $50D$, with its standard deviation reduced by 10.3% and 18.9% compared to PSO and MPA, respectively, reflecting better overall search stability. The convergence curves indicate that for F3, WOA converges faster in the early stage than PSO and AVOA, allowing it to approach the optimal solution more quickly. The box plots further illustrate that on F3 in $10D$, WOA's solution distribution is more concentrated, with outliers reduced by 14.5% and 19.2% compared to PSO and MPA, making its optimization results more stable.

MPA exhibits strong search capability and stability on composite functions (F20–F24). On F20, MPA achieves a mean fitness value in $50D$ that is 16.8% lower than that of WOA; on F22, its mean fitness value in $100D$ is 29.4% lower than that of WOA; and on F24, its mean fitness value in $100D$ is 40.0% lower than that of PSO, with a lower standard deviation, indicating smaller fluctuations in optimization results and higher stability during the optimization process. The convergence curves show that on F15 ($100D$), MPA continues to improve solution quality in the later optimization stages, whereas PSO and WOA tend to stabilize earlier. The box plots further demonstrate that on F3 ($10D$), F15 ($50D$), and F30 ($100D$), MPA has fewer outliers compared to PSO and WOA, indicating a more concentrated solution distribution and stronger solution stability.

PSO exhibits a certain level of optimization capability on unimodal functions (F1 and F3) and some multimodal and composite functions (F5, F15, F30). On F1 and F3, PSO achieves mean fitness values that are 13.4% and 18.2% lower than those of WOA in $10D$ and $30D$, respectively. Notably, in $50D$ and $100D$, PSO has the lowest standard deviation in most test functions, indicating a faster convergence speed and better population convergence. The convergence curves show that PSO has a rapid initial descent rate; for F3, it converges faster than WOA and MPA during the first 30% of iterations, allowing it to quickly approach the optimal solution. However, on F15 and F30, PSO's convergence trend stabilizes earlier, with its later-stage optimization capability being limited. Compared to AVOA and WOA, its optimization performance decreases by 18.4% and 22.7%, respectively. The box plots show that on F3, PSO's solution distribution is relatively concentrated, with the number of outliers reduced by 17.3% compared to MPA, indicating more stable overall optimization results.

4.4. Exploration/exploitation evaluation and diversity analysis

To evaluate the exploration and exploitation capabilities of TSRL-CMM during the optimization process, we conducted experiments on the CEC 2017 benchmark set across different dimensions ($10D$, $30D$, $50D$, and $100D$). Unimodal test functions (F1, F3) were used to assess

Table 7
Comparison Results between AEAS and Other ε-Greedy Methods.

	10D			30D			50D			100D			Total		
	+	=	−	+	=	−	+	=	−	+	=	−	+	=	−
AEAS-1	**15**	12	2	**8**	21	0	**7**	22	0	**8**	20	1	**38**	75	3
AEAS-2	**18**	9	2	**14**	15	0	**10**	18	1	**10**	19	0	**52**	61	3
AEAS-3	**17**	11	1	**13**	16	0	**13**	16	0	**10**	19	0	**53**	62	1
AEAS-4	**13**	14	2	**16**	11	2	**10**	17	2	**14**	13	2	**53**	55	8
AEAS-5	**17**	9	3	**11**	18	0	**12**	17	0	**8**	20	1	**48**	64	4
Total	**80**	55	10	**62**	81	2	**52**	90	3	**50**	91	4	**244**	317	19

the algorithm's exploitation capability, determining whether it could rapidly converge to the optimal solution in a smooth, single-optimum search space. Meanwhile, multimodal test functions (F4–F10) were employed to evaluate exploration capability, examining whether the algorithm could effectively avoid local optima and continuously search for better solutions in complex search spaces.

According to the results in Tables A.1–A.4, TSRL-CMM demonstrates strong exploitation capability in the unimodal function tests at $10D$ and $30D$. The algorithm converges rapidly to high-quality solutions within a short search process while maintaining a low optimization error. This indicates that in relatively simple search spaces, TSRL-CMM effectively utilizes available information to efficiently approximate the optimal solution. As the dimensionality increases to $50D$ and $100D$, the optimization accuracy remains at a competitive level, demonstrating that TSRL-CMM retains strong exploitation capability even in high-dimensional scenarios. However, the characteristics of high-dimensional search spaces may lead to increased sensitivity of search paths to initial parameters, potentially affecting convergence efficiency. Therefore, future improvements could focus on optimizing search strategies to enhance adaptability in more complex optimization problems.

For multimodal functions, TSRL-CMM exhibits strong exploration capability in the $10D$ and $30D$ function tests. The optimization results indicate that the algorithm effectively avoids local optima and identifies high-quality solutions within a broader search space. As the dimensionality increases to $50D$ and $100D$, the exploration performance varies, potentially due to the increased complexity of the search space, which enhances the sensitivity of parameters to search directions and affects the algorithm's global search capability in high-dimensional scenarios. Nevertheless, TSRL-CMM maintains a relatively broad search scope, avoiding local optima and demonstrating effective global exploration.

To further evaluate the population diversity of TSRL-CMM during the optimization process, we analyzed its performance on the Hybrid (F11–F20) and Composition (F21–F30) functions. These functions consist of multiple sub-functions with varying characteristics, resulting in complex search landscapes with numerous local optima. As such, they are well-suited for assessing an algorithm's ability to maintain population diversity throughout the search process. Higher population diversity enhances global search capability and reduces the risk of premature convergence, whereas insufficient diversity may lead to stagnation in local optima and hinder optimization performance.

From Figs. 5–12 and Tables A.1–A.4, it can be observed that in $10D$ and $30D$, TSRL-CMM effectively maintains population diversity and avoids premature convergence to a certain extent. However, in $50D$ and $100D$, the complexity of the search space may constrain the overall distribution of the population, reducing the migration ability of individuals between different regions and thus affecting diversity maintenance. Overall, TSRL-CMM demonstrates a good ability to escape local optima in Hybrid (F11–F20) and Composition (F21–F30) functions, maintaining population diversity and effectively exploring new solutions.

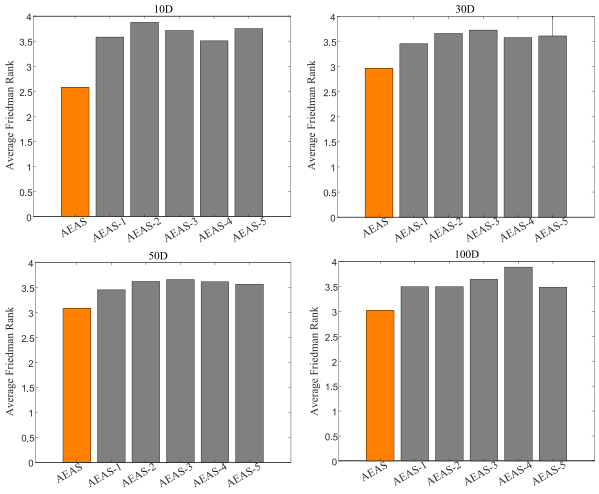


Fig. 17. Average rankings of Friedman test.

4.5. Comparative experiments on adaptive ε-Greedy action selection strategy

To comprehensively evaluate AEAS, we compared it with five different action selection strategies. These strategies were selected to cover a range of adaptive and non-adaptive mechanisms for balancing exploration and exploitation. AEAS-1 (fixed value) [49] serves as a control case, illustrating the limitations of a non-adaptive selection mechanism. AEAS-2 [50] and AEAS-3 [51] employ linear and nonlinear decay functions, respectively, representing widely used heuristic approaches for gradually transitioning from exploration to exploitation. AEAS-4 [52] adopts an exponential decay scheme, providing a more aggressive transition. Finally, AEAS-5 [22] utilizes a staged adaptation mechanism, making it particularly relevant for comparison with AEAS, as both approaches dynamically adjust exploration based on different optimization phases. The mathematical definitions of these five action selection strategies are provided in Appendix Eqs. (A.1)–(A.5).

In Appendix, Tables A.25 to A.28 detail the mean and standard deviations of AEAS compared with five other ε-Greedy methods for each function across $10D$, $30D$, $50D$, and $100D$. Table 7 summarizes the results of the Wilcoxon rank-sum test comparing our method with the other five methods. Detailed results of the Wilcoxon rank-sum test can be found in Tables A.29 to A.32.

As shown in Table 7, AEAS performs significantly better than the other methods. For $10D$, AEAS outperforms the other five methods in an average of 16 cases, while it performs worse in only 2 cases. For $30D$, AEAS outperforms the other five methods in approximately 12 cases on average, and performs worse only in 2 cases compared to AEAS-4. For $50D$ and $100D$, AEAS outperforms the other five methods in approximately 10 cases on average, while it performs worse in about 1 case. Although AEAS's advantage diminishes with increasing problem dimensions, it still significantly outperforms the other five methods. It is worth noting the performance differences between AEAS and AEAS-5. Since both methods consider adaptive action selection designed

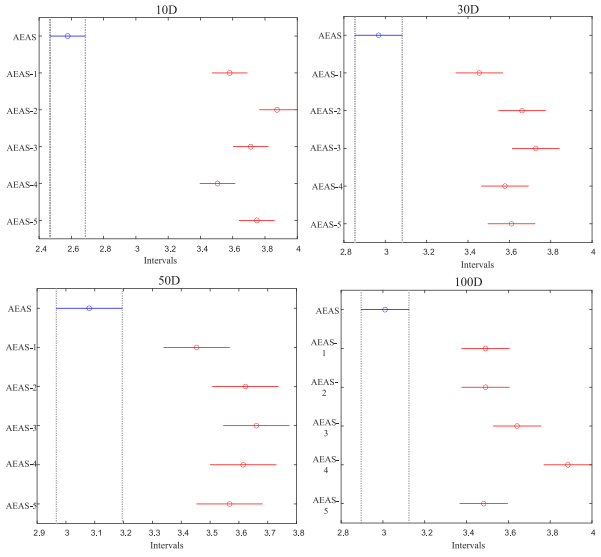


Fig. 18. Scheffe test results.

Table 8
Friedman Test Rankings of Six Parameters.

	$\gamma = 1$	$\gamma = 0.8$	$\gamma = 0.6$	$\gamma = 0.4$	$\gamma = 0.2$	$\gamma = 0$
10D	3.486	**3.457**	3.530	3.513	3.466	3.547
30D	3.513	3.466	**3.418**	3.505	3.565	3.534
50D	3.511	**3.425**	3.445	3.562	3.466	3.592
100D	3.470	**3.455**	3.528	3.457	3.532	3.558
Mean	3.495	**3.451**	3.480	3.509	3.507	3.558

for different stages, but with significant performance differences, the results from Table 7 show that AEAS significantly outperforms AEAS-5. AEAS performs better than AEAS-5 in 48 cases, while it only performs worse in 4 cases. On each problem dimension, AEAS improves the results in an average of 12 cases, while the performance declines in only 1 case.

To further compare the performance of AEAS with other methods, the Friedman test was used to rank AEAS compared to the other methods. The detailed rankings can be found in Tables A.33 to A.36. Fig. 17 shows the average rankings of the six ϵ-Greedy methods. AEAS ranks the best in $10D$, $30D$, $50D$, and $100D$. Except for AEAS, all algorithms have their own advantages in different dimensions. For $10D$, AEAS-4 ranks second, while AEAS-2 ranks last. For $30D$ and $50D$, AEAS-1 ranks second, and AEAS-3 ranks last. For $100D$, AEAS-5 ranks second, and AEAS-4 ranks last.

To compare the performance differences of multiple algorithms across different dimensions, we used the Scheffe's procedure for post hoc testing. The test results, as shown in Fig. 18, indicate that AEAS significantly outperforms the other five methods on $10D$, $30D$, $50D$, and $100D$.

4.6. Parameter sensitivity analysis

The performance of metaheuristic algorithms is often influenced by their parameter settings, as different parameter values can lead to variations in search behavior and convergence efficiency [73]. While parameter tuning plays a crucial role in many metaheuristic-based optimization frameworks [17], our study primarily focuses on the RL-driven [74] combination of metaheuristic algorithms rather than fine-tuning their internal parameters. However, RL itself also relies on key hyperparameters that affect its decision-making process, among which the discount factor γ is particularly important.

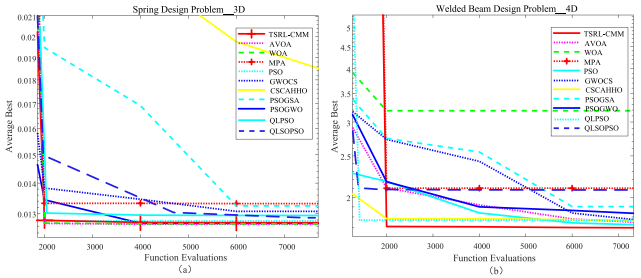


Fig. 19. (a) Convergence curves for the compression/tension spring design case; (b) Convergence curves for the welded beam design case.

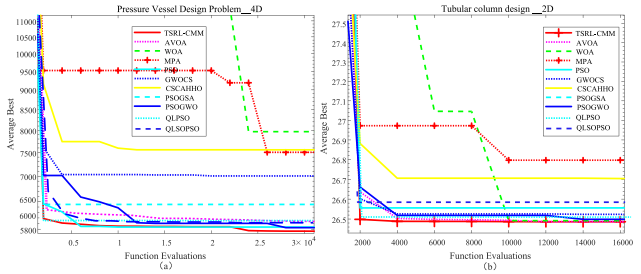


Fig. 20. (a) Convergence curves for the pressure vessel design case; (b) Convergence curves for the tubular column design case.

In TSRL-CMM, γ determines how much future rewards influence the current decision-making process when selecting metaheuristic algorithms. A larger γ places greater emphasis on long-term rewards, whereas a smaller γ prioritizes immediate rewards. The choice of an appropriate γ value directly impacts TSRL-CMM's ability to balance exploration and exploitation throughout the optimization process. Although our work does not focus on automated hyperparameter tuning, analyzing the sensitivity of γ helps us understand how different settings affect algorithmic performance. To evaluate this, we conducted experiments using six different values of γ: $\gamma = \{1, 0.8, 0.6, 0.4, 0.2, 0\}$, and evaluated their performance on the CEC 2017 test suite across $10D$, $30D$, $50D$, and $100D$ on TSRL-CMM's optimization performance.

Table 8 summarizes the Friedman test results for the six different parameter values of γ. Overall, it is observed that TSRL-CMM with $\gamma = 0.8$ performs the best. $\gamma = 0.6$ is the second best. There is little difference between $\gamma = 1$, $\gamma = 0.4$ and $\gamma = 0.2$. $\gamma = 0$ is the least effective. For $10D$, $50D$, and $100D$, $\gamma = 0.8$ ranks first, but for $30D$, $\gamma = 0.8$ performs worse than $\gamma = 0.6$. To analyze the differences between $\gamma = 0.8$ and other parameter values, the Wilcoxon rank-sum test is used for comparison.

As shown in Table 9, there are no significant differences between $\gamma = 1$, $\gamma = 0.6$, and $\gamma = 0.4$ compared to $\gamma = 0.8$. However, $\gamma = 0.2$ and $\gamma = 0$ perform relatively worse, with $\gamma = 0.8$ outperforming them in five cases. The experiments indicate that a small γ does not adequately consider future rewards, while a large γ overly focuses on future rewards, leading to worse performance of TSRL-CMM. Therefore, $\gamma = 0.8$ is chosen in this study as it effectively balances both current and future rewards while giving sufficient consideration to future rewards.

Beyond the impact of the discount factor γ on the reinforcement learning process, the internal parameters of the combined metaheuristic algorithms also influence optimization performance. Each algorithm in TSRL-CMM, such as PSO, WOA, and MPA, has specific control parameters that affect its search behavior. While these parameters are set to commonly used values from the literature to ensure fair comparisons, further adjustments may be needed in different problem environments to achieve optimal performance. Therefore, incorporating insights from existing adaptive parameter tuning methods can contribute to more robust and competitive optimization results.

Table 9
Comparison of parameters using Wilcoxon rank-sum test results.

	10D			30D			50D			100D			Total		
	+	=	−	+	=	−	+	=	−	+	=	−	+	=	−
$\gamma = 1$	**0**	29	0	**0**	29	0	**1**	27	1	**1**	28	0	**2**	113	1
$\gamma = 0.6$	**1**	28	0	**0**	29	0	**1**	28	0	**0**	28	1	**2**	113	1
$\gamma = 0.4$	**2**	27	0	**1**	27	1	**0**	29	0	**0**	29	0	**3**	112	1
$\gamma = 0.2$	**1**	28	0	**0**	29	0	**3**	26	0	**1**	28	0	**5**	111	0
$\gamma = 0$	**1**	28	0	**1**	28	0	**2**	26	1	**3**	25	1	**7**	107	2
Total	**5**	140	0	**2**	142	1	**7**	136	2	**5**	138	2	**19**	556	5

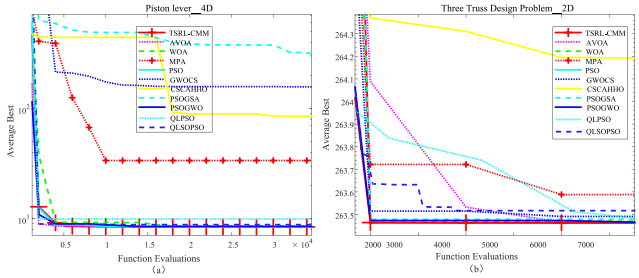


Fig. 21. (a) Convergence curves for the piston lever design case; (b) Convergence curves for the three-bar truss design case.

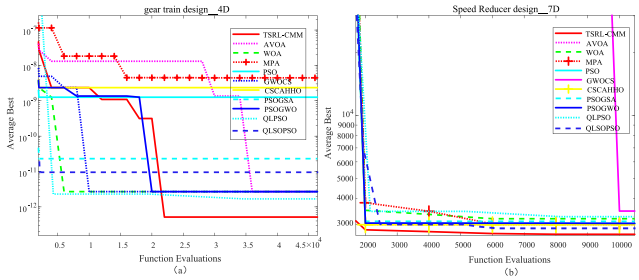


Fig. 22. (a) Convergence curves for the gear train design case; (b) Convergence curves for the speed reducer design case.

5. Real-world engineering optimization problems

To comprehensively evaluate the performance of TSRL-CMM, we conduct two sets of real-world engineering optimization [75] experiments: (1) nine real-world engineering optimization problems, including eight standard engineering design tasks and a photovoltaic module parameter estimation problem, all of which have low-dimensional decision spaces with at most 10 decision variables, and (2) complex power system optimization problems that involve high-dimensional decision spaces with more than 100 decision variables. The first set includes well-established benchmark problems commonly used in engineering design, while the second set consists of large-scale power system optimization tasks that involve significantly more decision variables and constraints. To ensure fair algorithm comparison, we employ a unified constraint-handling approach using the penalty function method, where constraint violations are converted into penalty terms within the objective function. Each problem is independently run 30 times, with the average of all runs taken as the result. The maximum function evaluations are determined based on problem dimensionality, following the recommendations outlined in Ref. Kumar et al. [76].

For detailed descriptions of these engineering optimization problems, we refer readers to the relevant literature. The optimal values of all test functions are known in advance. Furthermore, the difference between the optimal values obtained by each algorithm and the actual optimal values is used as a metric (error value) to evaluate each algorithm. The results of TSRL-CMM are compared with those of the 10 comparison algorithms, with the algorithm parameters remaining unchanged.

5.1. Standard engineering design optimization problems

To evaluate the performance of the proposed TSRL-CMM, this section considers eight standard engineering design problems: compression and tension spring design (C/TSD) [77,78], welded beam design (WBD), pressure vessel design (PVD) [79,80], tubular column design (TCD), piston lever design (PLD) [81], three-bar truss design (TTD), gear train design (GTD), and speed reducer design (SRD), as well as a photovoltaic module parameter estimation problem [82]. According to the experimental setting in Ref. Kumar et al. [76], for optimization problems where the number of decision variables is at most 10, the maximum number of function evaluations is set to 10^5.

In Appendix, Tables A.37–A.47 provide detailed parameter solutions for the 11 algorithms across nine engineering optimization problems. Table 10 summarizes the mean and standard deviations of the 11 algorithms on these nine engineering optimization problems. TSRL-CMM outperforms the comparison algorithms on all nine engineering optimization problems, with the smallest average error values, further demonstrating its effectiveness in solving engineering optimization problems. Although TSRL-CMM's standard deviations are worse than PSO on PVD, TCD, TTD, GTD, SRD, and TDM, it remains competitive, with standard deviations all below 10^{-5}.

As shown in Figs. 19 to 25, TSRL-CMM exhibits better convergence compared to the comparison algorithms for the eight constrained engineering optimization problems, further demonstrating its good convergence speed and stability. Additionally, the I–V and P–V characteristic curves in Figs. 23 to 25 indicate that the I–V and P–V values estimated by TSRL-CMM are consistent with the actual values, highlighting the effectiveness of TSRL-CMM in parameter estimation.

5.2. High-dimensional power system optimization problems

To evaluate the performance of the proposed TSRL-CMM on large-scale optimization tasks, this section considers six high-dimensional power system optimization problems: Optimal Sizing of Single-Phase Distributed Generation with Reactive Power Support for Phase Balancing at Main Transformer/Grid, Optimal Sizing of Distributed Generation for Active Power Loss Minimization, Optimal Sizing of Distributed Generation (DG) and Capacitors for Reactive Power Loss Minimization, Optimal Power Flow (Minimization of Active Power Loss), Optimal Power Flow (Minimization of Fuel Cost), and Optimal Power Flow (Minimization of Active Power Loss and Fuel Cost). To simplify the notation, these six optimization problems are denoted as PSP1 to PSP6, respectively. According to the experimental setting in Ref. Kumar et al. [76], for optimization problems where the number of decision variables $D \in (50, 150]$, the maximum number of function evaluations is set to 8×10^5. For problems where $D > 150$, the maximum number of function evaluations is set to 10^6.

Table 11 summarizes the mean and standard deviation of 11 algorithms across six power system optimization problems. TSRL-CMM outperforms the comparison algorithms in all six high-dimensional

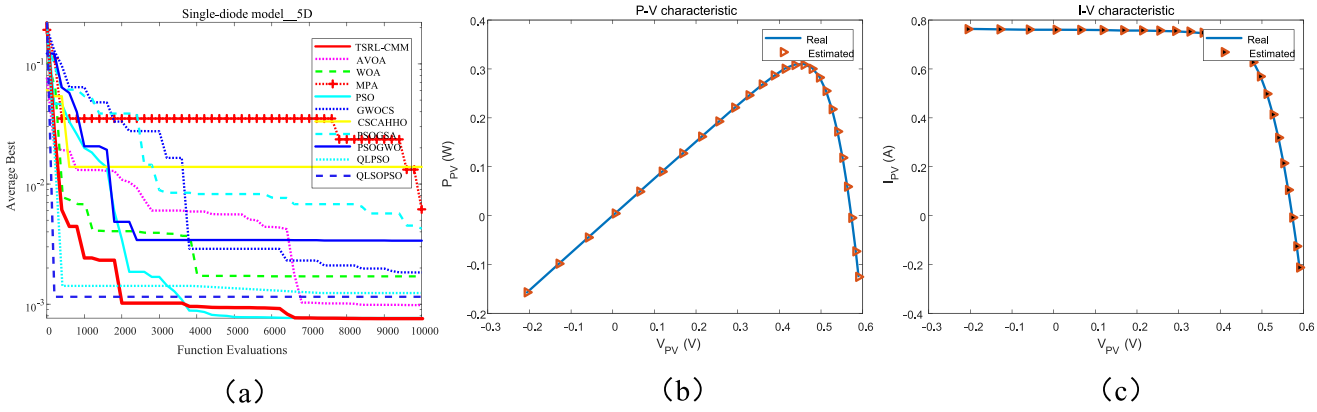


Fig. 23. SDM: (a) Convergence curves; (b) I–V characteristic curve; (c) P–V characteristic curve.

Table 10
Real-world engineering optimization problems results.

Algorithms	Metrics	Standard engineering design problems								Photovoltaic module parameter estimation		
		C/TSD	WBD	PVD	TCD	PLD	TTD	GTD	SRD	SDM	DDM	TDM
TSRL-CMM	Mean	**1.270E−02**	**1.700E+00**	**5.820E+03**	**2.650E+01**	**8.410E+00**	**2.630E+02**	**5.150E−13**	**2.470E+03**	**7.650E−04**	**3.100E−03**	**2.380E−03**
	Std	**2.080E−14**	**0.000E+00**	1.060E+01	3.690E−15	**1.790E−15**	2.570E−14	2.090E−25	1.110E−05	**1.300E−11**	**6.290E−05**	3.410E−03
AVOA	Mean	**1.270E−02**	1.710E+00	8.840E+24	**2.650E+01**	**8.410E+00**	**2.630E+02**	2.700E−12	2.990E+03	9.830E−04	4.000E−01	4.020E−01
	Std	3.578E+01	9.430E+29	7.440E+24	9.630E+07	1.360E+17	9.630E+07	2.815E−01	2.580E+15	3.005E+01	3.052E−03	6.660E−02
WOA	Mean	**1.270E−02**	1.720E+00	6.240E+18	**2.650E+01**	8.570E+00	**2.630E+02**	2.700E−12	3.000E+03	1.710E−03	4.000E−01	4.020E−01
	Std	4.750E−04	2.320E+16	2.130E+19	9.350E+02	1.810E+11	9.350E+02	**0.000E+00**	7.450E+09	6.340E−03	4.220E−03	2.510E−02
MPA	Mean	**1.270E−02**	1.720E+00	3.500E+24	**2.650E+01**	8.520E+00	**2.630E+02**	2.700E−12	3.030E+03	6.160E−03	4.000E−01	4.010E−01
	Std	7.930E+01	1.760E+24	5.930E+24	4.390E+07	1.230E+17	4.390E+07	1.370E+01	7.220E+14	5.140E+01	5.450E−03	1.370E−01
PSO	Mean	1.286E−02	1.702E+00	7.700E+05	2.656E+01	8.423E+00	2.635E+02	1.260E−09	3.042E+03	7.670E−04	4.000E−01	4.000E−01
	Std	1.150E−13	**0.000E+00**	**1.180E−10**	**0.000E+00**	2.050E+05	**0.000E+00**	**0.000E+00**	**0.000E+00**	1.610E−01	8.350E−01	**6.840E−16**
GWOCS	Mean	**1.270E−02**	1.710E+00	4.680E+18	**2.650E+01**	1.540E+02	**2.630E+02**	2.700E−12	3.100E+03	1.840E−03	4.220E−01	4.510E−01
	Std	8.480E−06	3.280E+15	9.420E+18	1.020E+02	8.430E+10	1.020E+02	7.480E−06	4.620E+09	2.480E−03	5.940E−04	4.220E−04
CSCAHHO	Mean	**1.270E−02**	1.720E+00	1.820E+24	**2.650E+01**	9.250E+00	**2.630E+02**	2.700E−12	3.000E+03	1.390E−02	4.270E−01	4.690E−01
	Std	3.601E+01	1.180E+26	3.950E+24	5.100E+07	4.940E+16	5.100E+07	8.216E+00	9.570E+14	2.183E+03	4.790E−04	4.210E−04
PSOGSA	Mean	1.320E−02	1.830E+00	2.200E+23	2.770E+01	1.790E+01	**2.630E+02**	2.310E−11	3.110E+03	4.260E−03	4.100E−01	4.120E−01
	Std	7.020E+01	8.660E+20	1.060E+24	2.230E+04	7.750E+16	2.230E+04	1.450E−01	6.630E+02	2.380E+02	6.330E−03	4.330E−04
PSOGWO	Mean	**1.270E−02**	**1.700E+00**	7.970E+21	**2.650E+01**	**8.410E+00**	**2.630E+02**	2.700E−12	2.990E+03	3.380E−03	4.020E−01	4.020E−01
	Std	9.820E−05	1.090E+21	2.870E+22	1.080E+03	7.260E+14	1.080E+03	1.260E−05	1.510E+12	6.010E−03	1.590E−02	3.790E−03
QLPSO	Mean	**1.270E−02**	1.715E+00	5.871E+03	**2.650E+01**	8.437E+00	2.635E+02	2.310E−11	2.529E+03	8.540E−04	4.103E−01	1.050E−02
	Std	1.020E−02	5.250E+00	2.360E+00	7.140E−15	1.340E+14	4.030E+04	**0.000E+00**	8.230E−07	2.300E−08	8.830E−01	6.060E−01
QLSOPSO	Mean	1.290E−02	1.820E+00	5.890E+03	**2.650E+01**	9.320E+00	**2.630E+02**	9.930E−11	2.530E+03	1.310E−03	4.010E−01	9.440E−03
	Std	2.160E+00	2.450E+01	1.590E−08	4.790E−01	1.410E+12	2.110E+05	2.040E−15	1.710E−06	4.090E−09	5.130E−01	4.430E−01

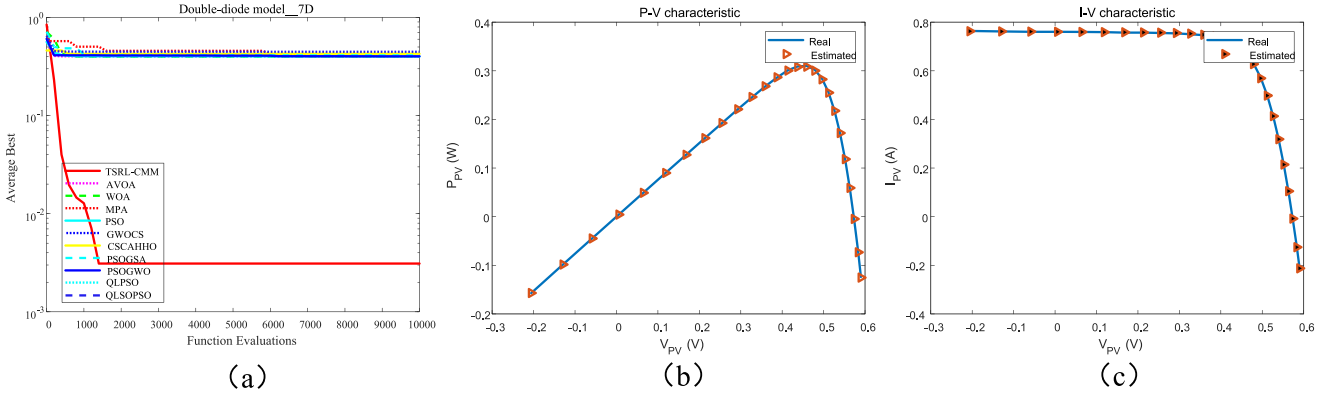


Fig. 24. DDM: (a) Convergence curves; (b) I–V characteristic curve; (c) P–V characteristic curve.

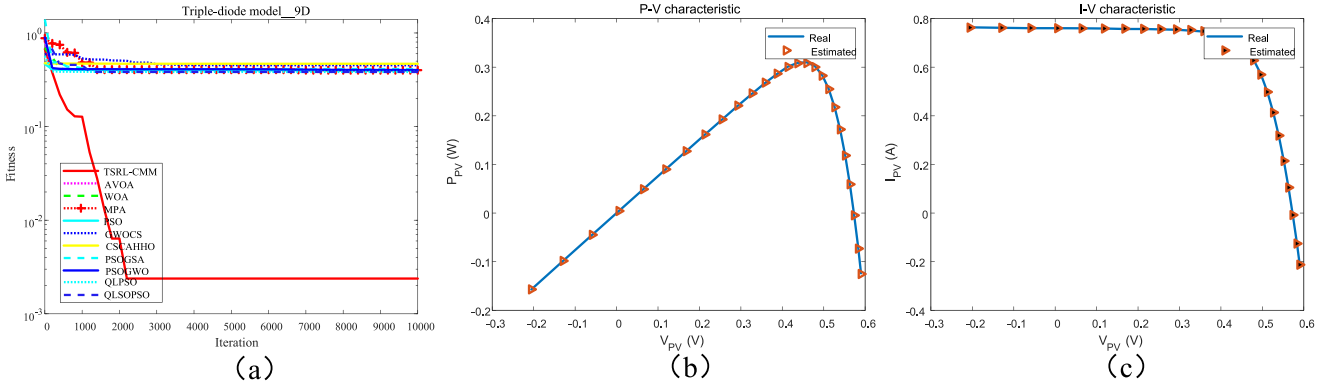


Fig. 25. TDM: (a) Convergence curves; (b) I–V characteristic curve; (c) P–V characteristic curve.

Table 11
Power system optimization problems results.

Algorithms	Metrics	Power system optimization problems					
		PSP1	PSP2	PSP3	PSP4	PSP5	PSP6
TSRL-CMM	Mean	**1.853E+01**	**3.968E+02**	**3.063E+02**	**1.161E+00**	**6.881E+01**	**9.716E+00**
	Std	**4.704E−10**	**2.488E−02**	**2.408E−07**	4.357E+02	8.080E+02	**4.513E−01**
AVOA	Mean	8.905E+01	2.116E+03	1.357E+03	1.569E+01	2.193E+02	4.598E+01
	Std	6.218E+06	3.007E+05	4.938E+06	5.948E+06	7.918E+02	1.895E+03
WOA	Mean	1.374E+02	1.317E+03	1.284E+03	8.643E+01	8.525E+03	7.897E+01
	Std	2.457E+03	6.026E+06	3.778E+06	3.637E+02	1.322E+02	4.544E+02
MPA	Mean	9.242E+03	9.788E+05	1.008E+05	3.457E+03	1.465E+05	3.875E+03
	Std	8.177E+08	9.896E+08	9.237E+08	1.499E+05	2.205E+05	1.790E+06
PSO	Mean	3.402E+02	3.895E+04	3.968E+04	2.280E+02	3.171E+04	5.094E+02
	Std	8.666E−02	1.012E+05	2.060E+04	**6.656E−01**	**2.324E+01**	6.060E+00
GWOCS	Mean	4.378E+02	6.765E+03	7.709E+03	8.365E+01	8.686E+03	9.183E+01
	Std	2.321E+02	1.515E+05	1.750E+05	2.733E+02	2.564E+02	2.853E+02
CSCAHHO	Mean	1.544E+03	1.102E+04	5.780E+03	1.281E+02	8.508E+03	1.611E+02
	Std	5.794E+08	7.833E+08	7.788E+06	8.989E+08	8.436E+05	8.238E+05
PSOGSA	Mean	1.178E+03	1.864E+04	1.055E+04	2.023E+02	8.888E+03	1.475E+02
	Std	2.532E+07	2.846E+07	6.306E+05	2.178E+06	1.580E+06	1.359E+05
PSOGWO	Mean	2.991E+02	5.696E+03	2.577E+03	7.562E+01	6.885E+03	8.296E+01
	Std	9.352E+03	4.495E+06	5.070E+06	3.345E+02	7.502E+03	1.434E+03
QLPSO	Mean	1.178E+03	1.842E+05	9.917E+03	2.728E+03	7.430E+04	3.192E+03
	Std	9.722E+08	9.165E+07	8.640E+07	1.383E+05	1.651E+05	1.950E+06
QLSOPSO	Mean	9.410E+01	6.489E+03	6.987E+03	2.182E+02	1.255E+04	1.798E+02
	Std	8.095E+01	4.906E+02	8.565E+01	7.782E+01	1.264E+03	3.420E+02

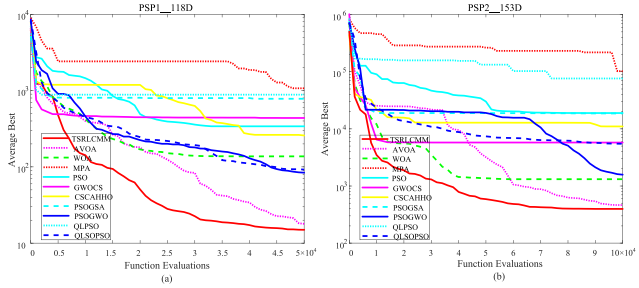


Fig. 26. (a) Convergence curves for PSP1; (b) Convergence curves for PSP2.

optimization problems, further demonstrating its effectiveness in solving complex high-dimensional engineering optimization problems. For PSP1, PSP4, PSP5, and PSP6, which have dimensions between 100D and 150D, TSRL-CMM achieves relatively low mean values, showing strong competitiveness. In PSP2 and PSP3, which exceed 150D, the mean values of TSRL-CMM increase slightly but still remain the best compared to the other algorithms. Regarding standard deviation, TSRL-CMM exhibits higher variability in PSP4 and PSP5, performing slightly worse than AVOA, WOA, PSO, and GWOCS, indicating some instability in certain experimental runs. However, for other optimization problems, TSRL-CMM maintains a relatively low standard deviation, demonstrating overall robustness.

As shown in Figs. 26–28, TSRL-CMM demonstrates a faster convergence speed and superior optimization performance compared to the benchmark algorithms across six power system optimization problems. Specifically, the convergence curves of TSRL-CMM exhibit a more rapid decline, indicating that it can find better solutions in the early stages of function evaluations. Additionally, compared to other algorithms, TSRL-CMM maintains a more stable convergence

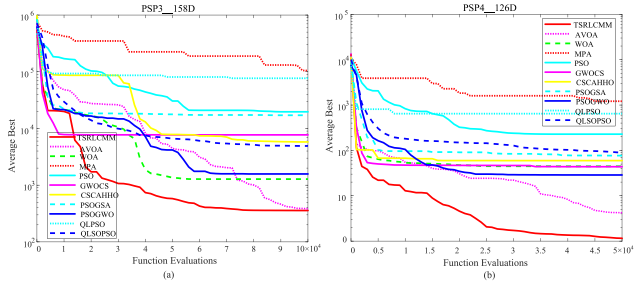


Fig. 27. (a) Convergence curves for PSP3; (b) Convergence curves for PSP4.

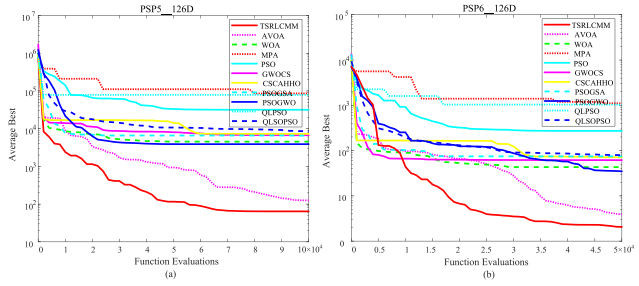


Fig. 28. (a) Convergence curves for PSP5; (b) Convergence curves for PSP6.

trend, showcasing its adaptability and robustness in high-dimensional optimization problems.

6. Conclusion and future work

This paper proposes a three-stage reinforcement learning for combining multiple metaheuristic algorithms (TSRL-CMM). In TSRL-CMM, the adaptive action selection strategy and reward function based on state transitions in the three stages improve the efficiency of combining multiple metaheuristic algorithms using reinforcement learning. The four metaheuristic algorithms, guided by the three-stage reinforcement learning, achieve effective collaboration. The experimental results of TSRL-CMM compared with 10 state-of-the-art algorithms on the CEC 2017 test suite demonstrate that TSRL-CMM exhibits superior convergence. Additionally, TSRL-CMM was compared with 10 algorithms using the Wilcoxon rank-sum test, Friedman test, and Scheffe's procedure for post-hoc analysis. The results show that there are significant differences between TSRL-CMM and the other 10 algorithms, with TSRL-CMM showing significant performance advantages. Additionally, TSRL-CMM was further evaluated using eight constrained engineering optimization problems and a photovoltaic module parameters estimation problem, demonstrating its effectiveness in solving real-world engineering optimization problems. This study indicates that three-stage reinforcement learning has significant potential in combining multiple metaheuristic algorithms.

Future research can be explored in three key directions: (1) This study employs Q-learning to combine multiple metaheuristic algorithms, primarily focusing on reinforcement learning applications in discrete spaces. Future work could extend its applicability to continuous spaces, such as parameter configuration and topology optimization. (2) Integrating adaptive parameter tuning mechanisms into TSRL-CMM could enable it to not only select the most suitable metaheuristic algorithm but also dynamically adjust its internal parameters based on the optimization environment. (3) The definition of states, selection of actions, and design of reward functions in reinforcement learning could be further refined to enhance TSRL-CMM's generalization ability in solving complex real-world problems.

CRediT authorship contribution statement

Xiaotong Liu: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Tianlei Wang:** Visualization, Supervision, Funding acquisition, Conceptualization. **Zhiqiang Zeng:** Writing – original draft, Visualization, Validation, Supervision, Software. **Ye Tian:** Software, Resources, Investigation. **Jun Tong:** Supervision, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Five ε-Greedy update functions

AEAS-1 [49]: Using a fixed value method, $\varepsilon = 0.1$;

AEAS-2 [50]: As the number of iterations increases, ε linearly decreases from 0.99 to 0.05, and then it remains at the lower limit of 0.05.

$$\varepsilon = (0.99 - 0.05) * (1 - t/T) \tag{A.1}$$

AEAS-3 [51]: Using a nonlinear cooling schedule about probability ε with the following expression:

$$\varepsilon = \varepsilon_f + \frac{\varepsilon_0 - \varepsilon_f}{1 + e^{\beta(t - T/2)}} \tag{A.2}$$

where $\beta = 0.5$, $\varepsilon_0 = 0.0001$ and $\varepsilon_f = 0.4$.

AEAS-4 [52]: Using an inverse decay method.

$$\varepsilon = e^{-0.1 * t} \tag{A.3}$$

AEAS-5 [22]: The whole exploration process for the Q-learning algorithm is divided into two steps. The first step:

$$\varepsilon = \chi - e^{(t/T)^2}, t \le \mu T \tag{A.4}$$

where μ is the self-adaptive exploration factor which can be [0,0.3] and $\chi = 2$.

The second step:

$$\varepsilon = \begin{cases} \varepsilon_f + (\varepsilon_m - \varepsilon_f) \times \frac{t - \mu T}{T - \mu T} \times \rho, \\ \qquad if\ \varepsilon_f + (\varepsilon_m - \varepsilon_f) \times \frac{t - \mu T}{T_1 - \mu T} \times \rho \ge \varepsilon_m \\ \varepsilon_m, \text{ elseif} \end{cases} \tag{A.5}$$

where $\epsilon_f = 1$, $\epsilon_m = 0.01$ and $\rho = 0.9$.

Data availability

No data was used for the research described in the article.

Table A.1
Comparison of optimization algorithms on benchmark functions (10D).

Function	Metric	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	Mean	**1.444E−01**	1.164E+03	1.261E+08	4.981E+04	9.311E+08	3.670E+09	1.475E+09	5.162E+08	5.845E+07	4.662E+05	1.264E+07
	Std	**3.870E−03**	1.519E+10	8.588E+10	1.975E+04	5.845E−06	1.036E+05	4.236E+10	1.081E+09	8.258E+08	6.493E+09	1.528E+08
F3	Mean	**4.347E−14**	2.729E−04	1.265E+03	1.814E+01	9.053E+02	8.935E+03	2.739E+03	1.386E+04	2.153E+03	1.887E−01	2.891E+02
	Std	**4.663E−15**	2.240E+06	1.711E+12	1.015E−02	2.712E−11	4.893E−03	1.651E+12	2.473E+08	4.251E+10	1.955E+10	9.986E+07
F4	Mean	**8.582E−14**	2.373E+01	3.723E+01	3.581E+01	6.989E+01	3.693E+02	1.630E+02	8.725E+01	4.710E+01	5.091E+00	3.628E+01
	Std	**1.441E−14**	3.487E+03	6.161E+04	2.241E−02	8.451E−12	1.376E−02	3.141E+04	4.441E+02	1.896E+02	4.611E+03	1.216E+01
F5	Mean	2.848E+01	3.458E+01	4.561E+01	4.630E+01	2.680E+01	4.514E+01	4.981E+01	5.426E+01	2.611E+01	**2.161E+01**	2.633E+01
	Std	**1.126E−13**	7.031E+01	2.007E+02	7.272E−02	9.690E−12	2.382E−04	1.173E+02	1.661E+01	7.605E+00	5.774E+01	1.072E+01
F6	Mean	**3.325E−01**	1.013E+01	1.406E+01	2.799E+01	5.518E+00	3.333E+01	4.026E+01	3.591E+01	6.949E+00	1.048E−04	1.467E+01
	Std	**1.531E−12**	3.554E+01	1.105E+02	9.462E−02	1.952E−05	1.533E−02	5.453E+01	1.202E+01	5.831E+00	3.925E−03	2.756E+00
F7	Mean	**3.199E+01**	6.147E+01	7.844E+01	9.683E+01	3.677E+01	1.024E+02	1.664E+02	1.078E+02	4.694E+01	3.348E+01	4.589E+01
	Std	**8.106E−14**	1.123E+02	9.383E+02	1.214E−01	1.572E−10	3.591E−03	3.813E+02	2.600E+01	4.788E+01	2.403E+02	1.040E+01
F8	Mean	2.468E+01	3.635E+01	4.530E+01	4.868E+01	2.619E+01	4.908E+01	4.882E+01	5.633E+01	2.900E+01	**2.434E+01**	2.697E+01
	Std	1.316E−07	5.725E+01	1.594E+02	1.609E−02	**1.897E−11**	1.429E−04	8.166E+01	2.098E+01	3.116E+00	6.159E+01	9.798E+00
F9	Mean	5.293E−01	6.244E+02	1.543E+02	9.443E+02	5.421E+01	8.245E+02	8.487E+02	1.041E+03	1.249E+02	**2.733E−09**	2.738E+02
	Std	**5.855E−14**	2.180E+03	1.829E+04	7.660E+00	5.922E−08	7.120E−01	1.121E+04	1.382E+03	3.514E+02	4.151E−05	5.297E+01
F10	Mean	**7.096E+02**	7.434E+02	1.097E+03	9.302E+02	7.667E+02	1.270E+03	1.068E+03	1.321E+03	1.103E+03	8.443E+02	7.850E+02
	Std	**3.451E−04**	1.572E+03	7.680E+02	8.130E+00	5.122E+00	1.690E+00	6.341E+02	6.596E+02	2.916E+01	6.534E+02	4.558E+02
F11	Mean	**1.385E+01**	2.360E+01	7.388E+01	3.734E+01	1.125E+02	9.476E+02	3.497E+02	1.506E+02	6.699E+01	7.917E+00	2.506E+01
	Std	3.639E−08	4.402E+06	3.236E+08	6.397E−02	**4.301E−11**	2.566E−03	2.668E+08	1.250E+05	4.172E+05	8.863E+06	2.815E+03
F12	Mean	**5.804E+03**	8.639E+05	5.927E+06	3.996E+06	1.394E+05	8.645E+06	2.003E+07	5.818E+06	3.896E+06	8.056E+05	1.002E+05
	Std	**7.818E−01**	1.855E+09	1.688E+10	8.813E+03	8.495E−01	1.350E+03	1.124E+10	6.604E+08	1.775E+08	7.394E+09	3.973E+07
F13	Mean	**1.687E+02**	5.436E+03	5.670E+04	1.371E+04	1.122E+04	8.666E+03	1.586E+04	3.003E+04	1.308E+04	7.688E+03	1.457E+03
	Std	4.353E−03	2.984E+09	1.734E+10	1.453E+04	**1.305E−09**	1.519E+02	1.304E+10	2.248E+08	1.373E+08	5.870E+09	1.772E+07
F14	Mean	**5.345E+01**	2.250E+02	1.939E+02	1.444E+02	5.027E+02	6.843E+02	6.467E+02	4.027E+03	5.681E+02	1.075E+02	1.530E+02
	Std	**6.533E+01**	1.258E+09	8.732E+09	3.672E+04	6.911E−03	5.907E+01	8.035E+09	1.045E+08	7.825E+07	3.380E+09	8.184E+06
F15	Mean	**4.618E+01**	9.792E+01	3.321E+02	2.688E+03	5.059E+02	2.146E+03	4.734E+02	6.893E+03	1.123E+03	2.519E+02	7.993E+01
	Std	1.233E+01	2.391E+08	5.230E+09	4.221E+03	**3.064E−06**	1.558E+01	4.966E+09	3.093E+08	1.809E+08	1.593E+09	1.881E+07
F16	Mean	8.583E+01	1.009E+02	**6.449E+01**	1.437E+02	1.065E+02	2.768E+02	2.084E+02	2.526E+02	1.332E+02	7.669E+01	1.463E+02
	Std	**5.327E−03**	5.460E+02	5.968E+03	1.813E−01	1.954E−02	2.855E−02	4.681E+03	2.652E+02	9.959E+01	5.391E+03	1.526E+02
F17	Mean	**4.630E+01**	4.987E+01	7.398E+01	1.204E+02	9.887E+01	7.933E+01	7.450E+01	1.598E+02	9.063E+01	7.513E+01	6.138E+01
	Std	**3.085E−09**	4.670E+02	1.506E+07	7.356E−01	5.585E+00	8.735E−02	7.654E+06	4.430E+03	2.884E+05	4.263E+06	8.617E+01
F18	Mean	**1.475E+02**	1.296E+04	2.779E+04	1.252E+04	3.968E+03	9.652E+03	1.258E+04	1.302E+04	1.266E+04	1.846E+04	1.876E+03
	Std	3.154E−02	4.479E+08	7.783E+09	1.725E+03	**1.064E−05**	2.579E+01	6.365E+09	1.904E+08	1.725E+08	5.287E+09	2.716E+07
F19	Mean	**1.841E+01**	3.004E+02	4.175E+02	5.815E+02	1.049E+03	2.276E+04	5.654E+04	7.087E+03	7.415E+03	5.421E+01	2.802E+02
	Std	1.240E−01	1.444E+09	1.025E+10	1.615E+03	**3.575E−04**	5.339E+01	8.934E+09	7.544E+07	4.307E+07	3.274E+09	1.297E+07
F20	Mean	8.025E+01	8.385E+01	9.575E+01	1.396E+02	1.021E+02	2.004E+02	1.832E+02	2.209E+02	1.104E+02	**6.817E+01**	1.042E+02
	Std	**1.892E−02**	4.579E+02	4.423E+02	7.094E−01	4.914E+00	1.262E−01	3.454E+02	2.718E+02	3.926E+01	3.426E+02	1.423E+02
F21	Mean	**1.000E+02**	1.001E+02	1.403E+02	1.004E+02	1.000E+02	1.171E+02	1.038E+02	1.494E+02	1.043E+02	**1.000E+02**	1.154E+02
	Std	**1.584E−13**	1.694E+01	2.255E+01	3.655E−03	1.254E−11	6.327E−06	8.114E+00	5.622E−01	9.671E−01	1.854E−12	1.706E+00
F22	Mean	1.000E+02	1.001E+02	**7.251E+01**	9.906E+01	9.555E+01	1.188E+02	1.036E+02	1.494E+02	9.705E+01	8.212E+01	1.157E+02
	Std	**1.621E−13**	1.678E+01	3.285E+01	4.572E−03	6.760E−10	1.025E−05	5.939E+00	8.175E−01	5.832E+00	5.140E+00	1.215E+00
F23	Mean	**2.000E+02**	**2.000E+02**	3.933E+02	4.098E+02	4.755E+02	5.053E+02	**2.000E+02**	5.170E+02	3.931E+02	4.111E+02	5.169E+02
	Std	1.801E−14	**0.000E+00**	1.447E+03	2.904E−01	1.826E−02	6.693E−02	7.645E+02	1.875E+02	5.151E+00	4.910E+01	3.749E+01
F24	Mean	**1.941E+02**	2.000E+02	2.063E+02	3.316E+02	3.874E+02	2.000E+02	2.000E+02	3.942E+02	3.008E+02	3.541E+02	2.124E+02
	Std	1.028E−14	**0.000E+00**	1.461E+03	4.286E−02	2.066E−10	0.000E+00	9.015E+02	6.651E+01	4.037E+01	3.609E+02	1.958E+01
F25	Mean	**2.000E+02**	**2.000E+02**	4.470E+02	4.540E+02	4.325E+02	4.643E+02	**2.000E+02**	4.943E+02	4.614E+02	4.013E+02	4.436E+02
	Std	**0.000E+00**	**0.000E+00**	1.947E+04	1.427E−01	6.526E−11	3.290E−03	1.322E+04	1.045E+02	2.717E+02	2.761E+03	1.879E+01
F26	Mean	**1.922E+02**	2.000E+02	3.721E+02	5.516E+02	6.019E+02	2.000E+02	2.000E+02	7.518E+02	4.228E+02	4.872E+02	3.363E+02
	Std	**0.000E+00**	**0.000E+00**	4.805E+03	7.167E−02	4.923E−04	3.628E−14	3.974E+03	5.740E+02	4.506E+01	1.270E+03	8.572E+01
F27	Mean	2.494E+02	**2.000E+02**	4.559E+02	5.207E+02	5.958E+02	3.232E+02	**2.000E+02**	6.279E+02	5.879E+02	4.491E+02	7.384E+02
	Std	5.657E+00	**0.000E+00**	1.937E+03	1.886E+00	5.158E−01	1.387E−01	2.113E+03	5.545E+02	4.569E+01	2.010E+03	9.089E+01
F28	Mean	**2.000E+02**	**2.000E+02**	3.849E+02	3.919E+02	4.591E+02	3.003E+02	**2.000E+02**	4.815E+02	4.128E+02	3.837E+02	3.588E+02
	Std	**0.000E+00**	**0.000E+00**	2.566E+03	1.003E−02	3.995E−11	4.665E−03	2.050E+03	2.687E+02	1.664E+02	5.108E+03	2.348E+01
F29	Mean	2.164E+02	**2.000E+02**	3.082E+02	3.323E+02	3.058E+02	2.868E+02	2.118E+02	3.871E+02	2.928E+02	2.893E+02	3.040E+02
	Std	5.916E−01	**0.000E+00**	6.368E+06	8.286E−01	3.487E+00	2.737E−01	1.101E+07	1.898E+02	3.314E+03	1.227E+06	6.500E+02
F30	Mean	7.435E+02	**2.000E+02**	1.299E+05	6.387E+05	9.770E+05	1.340E+06	1.792E+04	1.696E+06	5.851E+04	1.368E+04	2.524E+03
	Std	4.920E+03	**0.000E+00**	3.390E+09	1.353E+04	1.005E+05	1.146E+02	3.837E+09	6.913E+08	5.083E+07	3.735E+09	1.693E+07

Table A.2
Comparison of optimization algorithms on benchmark functions (30D).

Function	Metric	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	Mean	**4.011E+00**	1.016E+04	2.256E+06	8.485E+09	1.475E+09	2.533E+10	2.924E+08	2.919E+10	7.198E+09	2.101E+04	2.123E+09
	Std	**1.100E−03**	2.657E+10	7.632E+03	1.916E+11	6.502E+01	2.264E+03	5.873E+10	2.253E+08	4.415E+09	1.745E+08	5.791E+08
F3	Mean	**3.109E−12**	2.299E+03	1.587E+05	5.764E+04	2.759E+04	8.918E+04	6.791E+04	2.033E+05	7.464E+04	3.385E+03	2.197E+04
	Std	**4.923E−14**	3.281E+08	1.194E+02	1.046E+15	1.332E−07	4.779E−02	1.642E+15	3.640E+14	1.721E+13	4.084E+13	5.681E+10
F4	Mean	4.797E+01	9.458E+01	1.332E+02	5.380E+02	4.074E+02	5.346E+03	3.027E+02	4.831E+03	8.459E+02	**4.162E+01**	3.436E+02
	Std	4.056E−01	1.080E+04	9.751E−03	2.759E+05	**9.690E−11**	7.526E−03	8.636E+04	3.696E+03	1.140E+03	7.147E+03	7.340E+01
F5	Mean	1.628E+02	2.046E+02	2.954E+02	3.140E+02	1.631E+02	2.960E+02	2.597E+02	3.745E+02	2.274E+02	**1.600E+02**	2.157E+02
	Std	2.948E−03	1.689E+02	5.626E−03	5.287E+02	**1.492E−06**	8.032E−03	1.847E+02	4.620E+01	2.339E+01	1.513E+02	3.605E+01
F6	Mean	3.322E+01	4.287E+01	6.947E+01	5.723E+01	3.217E+01	6.523E+01	6.794E+01	8.345E+01	3.954E+01	**3.484E−01**	5.418E+01
	Std	5.825E−05	4.157E+01	6.526E−03	9.025E+01	**6.124E−10**	1.187E−03	3.221E+01	2.006E+01	2.720E+00	6.550E−01	2.603E+00
F7	Mean	1.639E+02	4.193E+02	5.201E+02	5.255E+02	2.399E+02	5.612E+02	4.813E+02	7.699E+02	3.601E+02	**1.290E+02**	3.361E+02
	Std	**1.470E−13**	2.031E+02	6.376E−03	2.809E+03	6.489E−10	8.406E−03	7.406E+02	4.665E+01	6.877E+01	5.295E+02	2.021E+01
F8	Mean	1.378E+02	1.426E+02	2.238E+02	2.853E+02	1.403E+02	2.323E+02	1.857E+02	3.333E+02	1.914E+02	**1.299E+02**	1.521E+02
	Std	2.414E−08	1.590E+02	6.576E−02	4.013E+02	**7.174E−10**	1.004E−04	1.592E+02	3.706E+01	2.601E+01	1.264E+02	4.272E+01
F9	Mean	3.978E+02	4.274E+03	7.626E+03	6.656E+03	2.397E+03	6.069E+03	7.422E+03	1.180E+04	3.271E+03	**3.196E+01**	3.353E+03
	Std	**9.777E−12**	1.129E+04	2.759E+01	5.758E+04	1.444E−07	2.524E+00	2.425E+04	7.749E+03	2.630E+03	1.172E+02	2.601E+02
F10	Mean	**3.507E+03**	4.091E+03	5.022E+03	6.809E+03	5.184E+03	6.153E+03	4.450E+03	6.901E+03	6.402E+03	4.656E+03	4.492E+03
	Std	8.296E+00	3.625E+03	3.211E+01	1.177E+03	**7.790E+00**	1.215E+01	1.068E+03	1.244E+03	1.301E+02	1.224E+03	8.115E+02
F11	Mean	**1.174E+02**	1.281E+02	4.078E+02	1.208E+03	1.076E+03	8.296E+03	1.280E+03	5.818E+03	3.762E+03	1.363E+02	4.638E+02
	Std	1.013E−01	1.918E+08	3.533E−02	7.396E+10	**7.619E−10**	4.968E−03	6.329E+10	3.688E+08	6.664E+08	1.878E+09	5.350E+06
F12	Mean	**1.511E+05**	2.880E+06	4.115E+07	4.323E+08	6.621E+08	3.257E+09	1.722E+08	1.651E+09	5.885E+08	7.284E+06	2.171E+08
	Std	8.603E+03	9.259E+09	7.407E+03	5.743E+10	**1.312E+00**	1.661E+04	2.212E+10	5.811E+09	5.300E+08	6.181E+09	1.729E+08
F13	Mean	**5.306E+03**	7.290E+04	1.253E+05	7.695E+07	7.106E+08	4.632E+08	3.373E+07	1.199E+08	5.703E+07	5.210E+06	5.313E+06
	Std	1.782E−03	1.311E+10	2.041E+03	9.258E+10	**2.959E−04**	2.831E+03	4.364E+10	3.963E+09	1.209E+09	1.962E+10	2.313E+08
F14	Mean	**2.859E+02**	3.996E+04	7.333E+05	9.053E+04	6.308E+05	1.923E+06	1.078E+06	1.297E+06	7.604E+05	6.720E+04	1.843E+04
	Std	4.856E−02	3.708E+08	2.516E+02	5.701E+09	**6.391E−08**	8.290E+00	5.691E+09	6.033E+08	2.871E+08	3.066E+09	1.231E+07
F15	Mean	**1.394E+03**	1.410E+04	6.807E+04	2.942E+06	2.622E+04	1.071E+08	5.086E+06	3.847E+06	2.281E+06	1.600E+06	1.843E+04
	Std	2.686E−01	2.054E+09	1.336E+03	3.633E+10	**7.694E−08**	7.455E+01	2.360E+10	3.382E+09	1.168E+09	1.480E+10	1.387E+08
F16	Mean	**1.022E+03**	1.253E+03	1.929E+03	1.984E+03	1.644E+03	2.430E+03	1.998E+03	2.519E+03	1.652E+03	1.396E+03	1.542E+03
	Std	**3.510E−01**	7.592E+03	1.729E+01	1.424E+05	8.938E+00	1.071E+01	6.833E+04	4.055E+03	2.351E+02	1.393E+04	2.481E+02
F17	Mean	**5.657E+02**	6.020E+02	7.891E+02	7.387E+02	8.039E+02	9.973E+02	7.636E+02	1.098E+03	6.921E+02	5.596E+02	6.891E+02
	Std	**2.675E−03**	8.883E+04	4.956E−01	4.014E+08	1.717E+00	7.522E−02	8.439E+07	4.374E+05	1.255E+04	4.969E+07	2.345E+02
F18	Mean	**5.348E+03**	6.310E+05	3.140E+06	2.034E+06	7.979E+06	2.717E+07	7.358E+06	7.813E+06	5.641E+06	9.137E+05	2.880E+05
	Std	2.479E−03	1.434E+09	7.012E+02	1.933E+10	**8.666E−05**	3.068E+02	1.823E+10	1.592E+09	8.938E+07	1.034E+10	8.072E+07
F19	Mean	**6.509E+02**	1.459E+04	3.185E+06	1.526E+07	1.650E+07	6.349E+07	1.383E+07	2.614E+07	8.616E+06	1.585E+06	1.258E+05
	Std	1.167E−01	2.008E+09	3.811E+03	3.975E+10	**1.183E−04**	8.800E+01	2.459E+10	2.473E+09	5.287E+08	1.529E+10	1.808E+08
F20	Mean	**5.334E+02**	5.356E+02	7.492E+02	6.413E+02	6.105E+02	8.498E+02	6.648E+02	9.471E+02	6.886E+02	5.711E+02	5.807E+02
	Std	2.698E−01	1.273E+03	5.719E+00	6.646E+02	**1.858E−03**	3.572E+00	4.984E+02	5.295E+02	4.817E+01	5.370E+02	3.410E+02
F21	Mean	3.575E+02	3.925E+02	4.745E+02	4.601E+02	3.668E+02	4.733E+02	4.637E+02	5.817E+02	4.111E+02	**3.546E+02**	4.229E+02
	Std	2.910E−02	2.519E+02	1.049E−02	1.747E+03	**1.394E−09**	4.530E−03	9.703E+02	6.721E+01	9.168E+00	1.225E+02	2.518E+01
F22	Mean	1.734E+03	1.583E+03	4.604E+03	**1.098E+03**	4.289E+03	6.318E+03	3.261E+02	5.847E+03	4.204E+03	4.740E+03	3.103E+03
	Std	**1.297E+00**	3.409E+03	3.720E+01	5.047E+03	5.051E+00	3.908E+00	1.142E+03	1.038E+03	1.550E+02	1.044E+03	2.482E+02
F23	Mean	**5.638E+02**	6.162E+02	7.389E+02	6.475E+02	9.101E+02	1.025E+03	7.915E+02	1.009E+03	6.561E+02	6.340E+02	9.443E+02
	Std	3.403E−01	1.639E+03	4.547E−01	3.430E+03	**2.800E−02**	2.485E−01	1.615E+03	2.175E+02	1.096E+01	1.072E+02	2.852E+01
F24	Mean	**6.203E+02**	7.040E+02	7.867E+02	6.869E+02	9.647E+02	1.145E+03	8.362E+02	1.057E+03	7.173E+02	7.635E+02	9.265E+02
	Std	3.719E−02	6.586E+02	5.391E−02	5.173E+03	**5.572E−10**	6.037E−03	1.281E+03	2.818E+02	8.390E+00	6.018E+01	1.915E+01
F25	Mean	**3.793E+02**	3.997E+02	4.499E+02	7.874E+02	4.356E+02	1.146E+03	5.062E+02	1.516E+03	7.495E+02	**3.793E+02**	5.837E+02
	Std	9.596E−02	1.922E+03	1.742E−03	1.115E+05	**5.911E−09**	2.045E−04	3.652E+04	4.457E+02	6.413E+02	4.310E+03	4.434E+01
F26	Mean	**2.300E+03**	3.068E+03	5.114E+03	2.857E+03	3.695E+03	6.365E+03	4.036E+03	6.563E+03	3.907E+03	2.632E+03	4.645E+03
	Std	9.578E−01	3.496E+03	1.558E−01	3.757E+04	**3.851E−10**	5.820E−02	2.031E+04	8.992E+02	7.727E+01	1.193E+03	1.696E+02
F27	Mean	**4.973E+02**	5.580E+02	6.475E+02	5.757E+02	8.061E+02	1.282E+03	7.558E+02	8.776E+02	7.060E+02	4.997E+02	1.034E+03
	Std	7.482E+01	2.188E+03	6.347E−01	5.881E+03	**1.102E−01**	9.183E−01	3.579E+03	1.043E+03	4.104E+00	5.702E+03	1.147E+01
F28	Mean	**3.804E+02**	4.187E+02	4.917E+02	8.273E+02	7.850E+02	2.507E+03	5.823E+02	2.347E+03	1.121E+03	5.162E+02	6.560E+02
	Std	4.518E+01	2.228E+03	2.685E−03	1.752E+04	**2.914E−10**	1.710E−03	1.569E+04	1.238E+03	2.319E+02	3.962E+04	4.960E+01
F29	Mean	**8.082E+02**	1.163E+03	1.950E+03	1.474E+03	1.675E+03	3.306E+03	1.891E+03	2.675E+03	1.745E+03	1.095E+03	1.853E+03
	Std	5.948E+00	7.311E+04	8.419E+00	1.152E+09	**2.552E+00**	4.463E+00	7.052E+08	1.441E+06	4.071E+05	2.153E+08	2.712E+02
F30	Mean	**4.715E+03**	1.957E+05	9.724E+06	3.225E+07	6.087E+07	1.319E+08	3.224E+07	6.902E+07	5.228E+07	7.063E+05	3.070E+06
	Std	2.530E+04	3.162E+09	1.687E+03	2.455E+10	**1.473E−03**	1.002E+03	1.365E+10	1.917E+09	2.214E+08	8.056E+09	6.310E+07

Table A.3
Comparison of optimization algorithms on benchmark functions (50D).

Function	Metric	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	Mean	**2.045E+03**	2.097E+04	8.784E+06	2.867E+10	1.373E+09	5.928E+10	8.186E+08	8.391E+10	2.608E+10	1.622E+08	1.290E+10
	Std	**5.643E−03**	4.096E+10	5.415E+03	3.037E+11	1.237E−02	2.105E+03	7.144E+10	1.106E+09	1.595E+09	3.033E+09	5.683E+08
F3	Mean	**4.343E−07**	1.396E+04	7.219E+04	1.531E+05	1.154E+05	1.895E+05	1.010E+09	3.482E+05	1.421E+05	3.643E+04	6.482E+04
	Std	**3.785E−09**	6.151E+13	1.719E+01	5.218E+16	8.673E−04	2.202E−01	4.362E+16	3.084E+14	1.876E+14	1.779E+15	2.928E+12
F4	Mean	**7.577E+01**	1.425E+02	2.788E+02	2.700E+03	6.707E+02	1.724E+04	6.080E+02	1.678E+04	3.976E+03	1.648E+02	1.294E+03
	Std	6.187E−01	1.729E+04	7.134E−03	4.671E+05	**2.606E−08**	1.826E−02	1.080E+05	1.333E+04	1.658E+03	9.370E+03	1.431E+02
F5	Mean	3.120E+02	3.479E+02	4.230E+02	6.136E+02	3.573E+02	5.157E+02	4.108E+02	6.983E+02	4.852E+02	**2.159E+02**	3.990E+02
	Std	**1.379E−13**	2.386E+02	1.932E−03	6.690E+02	1.839E−08	9.101E−05	2.001E+02	6.075E+01	2.085E+01	1.395E+02	4.950E+01
F6	Mean	4.385E+01	4.940E+01	7.871E+01	7.714E+01	4.830E+01	7.765E+01	8.281E+01	9.922E+01	5.662E+01	**2.354E+00**	6.579E+01
	Std	1.013E−06	4.125E+01	2.696E−03	7.779E+01	**2.025E−09**	2.416E−03	2.512E+01	1.967E+01	2.383E+00	9.362E−01	5.327E+00
F7	Mean	3.266E+02	8.612E+02	1.024E+03	1.170E+03	6.206E+02	1.079E+03	9.647E+02	1.632E+03	8.242E+02	**1.804E+02**	7.469E+02
	Std	**1.651E−13**	2.686E+02	3.489E−03	4.738E+03	4.858E−06	2.761E−03	1.036E+03	3.503E+01	3.442E+02	3.064E+02	3.148E+01
F8	Mean	3.154E+02	3.491E+02	3.963E+02	6.069E+02	3.442E+02	4.793E+02	4.206E+02	6.857E+02	4.677E+02	**2.340E+02**	4.127E+02
	Std	6.025E−06	2.594E+02	2.021E−03	7.236E+02	**1.415E−08**	1.278E−03	2.128E+02	5.332E+01	1.913E+01	1.503E+02	4.737E+01
F9	Mean	2.971E+03	1.192E+04	1.956E+04	2.823E+04	8.596E+03	2.074E+04	2.959E+04	4.045E+04	1.736E+04	**1.604E+03**	1.432E+04
	Std	**4.319E−09**	3.063E+04	3.112E+01	1.130E+05	1.022E−06	2.507E−01	4.276E+04	1.637E+04	1.616E+03	1.407E+04	2.103E+03
F10	Mean	**6.325E+03**	6.730E+03	9.395E+03	1.333E+04	9.097E+03	1.058E+04	8.581E+03	1.278E+04	1.188E+04	1.008E+04	8.745E+03
	Std	1.168E+01	6.184E+03	5.448E+01	1.504E+03	**9.598E+00**	1.586E+01	1.376E+03	1.991E+03	3.047E+02	1.615E+03	1.286E+03
F11	Mean	**2.265E+02**	2.525E+02	4.571E+02	5.038E+03	2.947E+03	1.888E+04	1.581E+03	2.091E+04	6.575E+03	3.630E+02	1.865E+03
	Std	1.027E−01	6.348E+05	5.611E−03	3.996E+11	**1.872E−08**	9.711E−03	5.332E+11	2.210E+06	4.516E+09	2.246E+08	2.775E+07
F12	Mean	**8.049E+05**	9.562E+06	1.999E+08	5.051E+09	4.882E+09	2.998E+10	7.297E+08	1.879E+10	5.248E+09	3.146E+07	1.982E+09
	Std	3.071E+03	4.377E+10	1.039E+04	2.622E+11	**1.671E+00**	7.537E+03	8.085E+10	3.063E+10	4.060E+09	1.321E+10	5.638E+08
F13	Mean	**8.567E+03**	9.986E+04	2.450E+05	6.186E+08	3.675E+09	7.790E+09	2.104E+08	4.685E+09	1.117E+09	3.133E+06	3.679E+08
	Std	2.076E−02	3.577E+10	3.709E+03	2.024E+11	**3.483E−03**	4.058E+03	7.847E+10	2.022E+10	5.831E+08	1.149E+10	4.642E+08
F14	Mean	**1.392E+04**	1.960E+05	7.342E+05	1.258E+06	7.953E+06	2.386E+07	3.741E+06	8.335E+06	4.144E+06	4.950E+05	6.053E+05
	Std	2.680E+00	4.515E+08	1.369E+02	4.636E+09	**5.985E−07**	4.357E+02	5.282E+09	5.780E+08	6.123E+07	3.034E+09	1.365E+07
F15	Mean	**6.019E+03**	3.146E+04	7.753E+04	6.383E+07	1.176E+08	1.484E+09	4.867E+07	2.885E+08	1.242E+08	6.764E+06	7.065E+06
	Std	**9.476E−03**	7.307E+09	1.239E+03	7.879E+10	8.450E−05	2.636E+04	3.702E+10	1.071E+10	9.260E+08	2.096E+10	4.057E+08
F16	Mean	**1.830E+03**	2.248E+03	3.198E+03	4.045E+03	2.826E+03	4.167E+03	3.245E+03	4.833E+03	3.175E+03	3.005E+03	2.938E+03
	Std	**1.727E+00**	6.387E+03	1.796E+01	8.747E+04	8.961E+00	1.189E+01	2.790E+04	2.804E+03	2.836E+02	1.109E+04	3.963E+02
F17	Mean	**1.418E+03**	1.702E+03	2.266E+03	3.030E+03	2.099E+03	3.006E+03	2.101E+03	3.085E+03	1.934E+03	2.133E+03	1.921E+03
	Std	1.108E−01	5.678E+04	1.769E+01	5.051E+09	**1.260E−04**	7.986E+00	1.153E+09	1.731E+07	2.083E+06	6.777E+08	6.565E+02
F18	Mean	**5.172E+04**	1.173E+06	5.360E+06	7.881E+06	1.901E+07	6.528E+07	8.543E+06	2.878E+07	1.802E+07	5.631E+06	3.464E+06
	Std	9.750E−01	6.583E+08	4.753E+02	1.245E+10	**3.184E−06**	2.030E+02	1.339E+10	2.613E+09	1.714E+08	8.980E+09	2.490E+07
F19	Mean	**2.781E+03**	3.312E+04	2.125E+06	1.201E+08	1.791E+08	3.929E+08	2.132E+07	9.000E+07	2.206E+07	2.523E+05	2.103E+06
	Std	**1.804E−05**	4.199E+09	3.181E+02	3.552E+10	1.633E−02	3.614E+02	1.661E+10	3.151E+09	3.705E+08	3.013E+09	8.531E+07
F20	Mean	**1.221E+03**	1.325E+03	1.562E+03	1.837E+03	1.413E+03	1.606E+03	1.396E+03	2.069E+03	1.492E+03	1.570E+03	1.281E+03
	Std	**1.507E+00**	1.460E+03	1.329E+01	8.015E+02	8.569E+00	3.557E+00	6.672E+02	8.130E+02	1.107E+02	7.105E+02	3.288E+02
F21	Mean	5.400E+02	6.026E+02	7.881E+02	7.779E+02	6.033E+02	8.001E+02	7.576E+02	9.948E+02	7.006E+02	**4.864E+02**	7.209E+02
	Std	2.578E−03	5.289E+02	6.693E−03	3.667E+03	**4.034E−09**	1.374E−04	1.258E+03	1.301E+02	2.266E+01	1.559E+02	4.932E+01
F22	Mean	**6.669E+03**	7.664E+03	9.567E+03	1.205E+04	9.739E+03	1.148E+04	9.884E+03	1.318E+04	1.192E+04	1.110E+04	9.576E+03
	Std	1.154E+01	5.987E+03	3.397E+01	2.551E+03	**3.234E+00**	1.080E+01	1.457E+03	1.853E+03	3.145E+02	1.610E+03	1.387E+03
F23	Mean	**9.039E+02**	9.936E+02	1.320E+03	1.063E+03	1.640E+03	1.843E+03	1.376E+03	1.795E+03	1.116E+03	1.158E+03	1.752E+03
	Std	1.839E−01	1.943E+03	1.055E+00	4.986E+03	**3.682E−02**	1.236E+00	1.890E+03	3.072E+02	1.479E+01	1.301E+02	3.852E+01
F24	Mean	**9.367E+02**	1.124E+03	1.297E+03	1.066E+03	1.779E+03	1.875E+03	1.431E+03	1.916E+03	1.152E+03	1.304E+03	1.565E+03
	Std	1.262E−01	1.037E+03	4.678E−02	8.485E+03	**1.067E−10**	4.455E−03	1.632E+03	5.229E+02	1.492E+01	7.826E+01	2.685E+01
F25	Mean	**4.698E+02**	5.629E+02	6.465E+02	2.759E+03	7.138E+02	7.474E+03	8.575E+02	8.719E+03	3.116E+03	5.013E+02	1.825E+03
	Std	1.082E+00	5.268E+03	6.754E−03	1.911E+05	**9.092E−09**	2.073E−03	5.056E+04	3.108E+03	1.968E+03	1.238E+04	9.982E+01
F26	Mean	**4.071E+03**	5.042E+03	1.058E+04	6.999E+03	8.513E+03	1.192E+04	7.458E+03	1.390E+04	7.981E+03	5.151E+03	8.679E+03
	Std	3.522E−01	4.218E+03	4.544E−02	1.239E+05	**9.469E−08**	1.112E−01	2.946E+04	2.017E+03	2.470E+02	1.397E+03	3.532E+02
F27	Mean	**5.000E+02**	9.413E+02	1.653E+03	9.654E+02	1.899E+03	3.797E+03	1.671E+03	1.988E+03	1.669E+03	**5.000E+02**	2.417E+03
	Std	9.916E+01	4.870E+03	1.938E+00	1.160E+04	**1.467E−01**	3.432E+00	7.584E+03	4.790E+03	1.187E+01	9.011E+03	2.268E+01
F28	Mean	**4.891E+02**	5.099E+02	6.229E+02	1.951E+03	1.531E+03	6.041E+03	9.849E+02	6.406E+03	3.281E+03	6.082E+02	1.813E+03
	Std	5.164E+02	5.384E+03	4.186E−03	2.770E+04	**3.111E−10**	1.096E−03	2.350E+04	2.384E+03	4.049E+02	4.878E+04	5.584E+01
F29	Mean	**1.218E+03**	1.942E+03	4.154E+03	3.823E+03	3.995E+03	1.323E+04	4.755E+03	8.036E+03	4.370E+03	2.807E+03	4.007E+03
	Std	1.816E+01	2.210E+06	6.026E+00	7.708E+09	**1.771E+00**	2.035E+00	2.941E+09	2.908E+07	9.331E+05	8.359E+08	1.308E+03
F30	Mean	**3.591E+04**	3.617E+06	9.623E+07	3.229E+08	4.609E+07	1.158E+09	1.988E+08	5.476E+08	3.599E+08	5.050E+06	1.141E+08
	Std	4.022E+05	7.542E+09	4.795E+03	6.447E+10	**1.799E−03**	1.766E+03	3.324E+10	7.102E+09	2.573E+08	1.624E+10	2.474E+08

Table A.4
Comparison of optimization algorithms on benchmark functions (100D).

Function	Metric	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	Mean	6.888E+03	4.869E+05	3.631E+07	1.304E+11	7.702E+09	1.664E+11	3.481E+09	2.366E+11	1.090E+11	1.413E+09	5.322E+10
	Std	**1.469E−03**	9.124E+10	2.423E+03	5.927E+11	6.637E+01	2.015E+03	1.069E+11	1.254E+10	1.899E+10	9.377E+09	2.254E+09
F3	Mean	**3.890E+00**	8.103E+04	5.668E+05	3.880E+05	3.586E+05	3.628E+05	3.492E+05	8.013E+05	3.447E+05	2.406E+05	2.004E+05
	Std	9.311E−04	4.675E+13	1.527E+03	1.580E+18	**3.962E−05**	6.314E−02	2.255E+18	8.575E+17	1.437E+17	2.412E+17	9.900E+13
F4	Mean	**2.364E+02**	2.927E+02	6.025E+02	1.610E+04	1.556E+03	4.700E+04	1.399E+03	5.995E+04	2.008E+04	4.459E+02	1.110E+04
	Std	1.032E+00	4.801E+04	5.552E−03	1.203E+06	**1.224E−04**	1.214E−02	2.270E+05	5.032E+04	3.613E+03	9.592E+03	1.548E+02
F5	Mean	7.653E+02	8.323E+02	9.489E+02	1.489E+03	9.233E+02	1.184E+03	1.088E+03	1.644E+03	1.227E+03	**3.477E+02**	1.083E+03
	Std	8.632E−04	4.436E+02	5.523E−04	1.232E+03	**3.212E−06**	1.104E−03	3.126E+02	1.089E+02	1.108E+01	1.093E+02	8.472E+01
F6	Mean	5.297E+01	5.331E+01	8.052E+01	1.007E+02	6.772E+01	8.724E+01	8.466E+01	1.129E+02	8.346E+01	**5.774E+00**	7.677E+01
	Std	1.186E−07	3.657E+01	7.893E−04	5.997E+01	**4.399E−10**	7.273E−04	1.721E+01	2.025E+01	7.101E−01	1.571E+00	5.779E+00
F7	Mean	**1.018E+03**	2.259E+03	2.584E+03	3.394E+03	1.951E+03	2.802E+03	2.296E+03	4.310E+03	2.423E+03	3.816E+02	2.328E+03
	Std	**3.402E−13**	5.729E+02	1.180E−03	9.375E+03	9.813E−09	1.173E−03	1.541E+03	8.790E+01	2.866E+02	2.999E+02	6.305E+01
F8	Mean	8.534E+02	9.140E+02	1.122E+03	1.480E+03	9.603E+02	1.321E+03	1.227E+03	1.766E+03	1.276E+03	**3.595E+02**	1.200E+03
	Std	5.605E−03	4.908E+02	1.964E−03	1.539E+03	**1.172E−08**	8.019E−05	3.553E+02	9.646E+01	4.187E+01	8.697E+01	7.247E+01
F9	Mean	1.372E+04	2.244E+04	3.340E+04	8.212E+04	3.244E+04	4.276E+04	6.233E+04	9.726E+04	5.437E+04	**5.902E+03**	4.225E+04
	Std	**2.248E−08**	4.754E+04	2.393E+00	1.723E+05	1.718E−06	9.227E−02	5.580E+04	2.794E+04	6.258E+03	2.303E+04	3.768E+03
F10	Mean	**1.438E+04**	1.478E+04	2.016E+04	3.013E+04	2.005E+04	2.371E+04	1.967E+04	2.949E+04	2.710E+04	2.481E+04	2.152E+04
	Std	1.066E+01	1.062E+04	5.632E+01	2.066E+03	**7.577E−06**	1.730E+01	1.927E+03	2.737E+03	8.865E+02	2.408E+03	1.601E+03
F11	Mean	1.348E+03	**1.273E+03**	7.389E+03	9.053E+04	4.186E+04	1.904E+05	1.568E+05	2.838E+05	1.495E+05	5.677E+03	4.007E+04
	Std	6.022E−01	8.472E+12	2.421E−02	8.080E+14	**1.821E−07**	6.322E−02	5.196E+14	7.841E+12	4.583E+11	2.506E+12	1.232E+10
F12	Mean	**3.400E+06**	3.707E+07	6.583E+08	2.831E+10	8.548E+09	9.681E+10	1.697E+09	1.160E+11	3.451E+10	6.319E+08	1.079E+10
	Std	2.030E+03	8.155E+10	5.208E+03	5.146E+11	**1.246E−02**	1.237E+04	1.112E+11	6.967E+10	4.744E+09	3.949E+10	1.432E+09
F13	Mean	**1.140E+04**	6.063E+04	7.688E+04	3.493E+09	1.118E+09	1.668E+10	1.543E+08	2.064E+10	4.289E+09	2.819E+06	1.111E+09
	Std	7.214E+03	2.037E+10	1.153E+03	1.363E+11	**4.249E−04**	3.761E+02	3.353E+10	2.217E+10	1.700E+09	5.913E+09	2.649E+08
F14	Mean	**5.096E+03**	5.053E+05	1.713E+06	1.565E+07	1.449E+07	2.277E+07	5.015E+06	3.410E+07	1.538E+07	4.888E+06	5.444E+06
	Std	1.093E−01	4.557E+08	2.290E+01	6.687E+09	**7.995E−05**	3.426E+01	6.040E+09	4.638E+08	1.377E+08	2.519E+09	3.092E+07
F15	Mean	**3.732E+03**	4.065E+04	7.301E+04	8.533E+08	9.547E+08	5.388E+09	5.462E+07	5.164E+09	1.331E+09	2.286E+06	2.719E+08
	Std	5.426E−01	1.241E+10	2.764E+02	8.421E+10	**1.594E−04**	1.907E+02	2.652E+10	1.729E+10	3.346E+08	8.203E+09	2.846E+08
F16	Mean	**3.941E+03**	4.640E+03	8.188E+03	1.057E+04	6.047E+03	1.292E+04	7.896E+03	1.573E+04	9.062E+03	8.292E+03	1.009E+04
	Std	2.469E+00	1.044E+04	1.332E+01	1.567E+05	**1.639E+00**	1.031E+01	3.889E+04	5.431E+03	6.075E+02	1.624E+04	4.971E+02
F17	Mean	**3.399E+03**	4.148E+03	5.266E+03	8.752E+03	1.304E+04	2.568E+05	5.971E+03	1.887E+05	1.628E+04	5.439E+03	9.275E+03
	Std	3.224E+00	5.563E+07	1.948E+01	1.238E+11	**6.665E−06**	5.495E+00	1.158E+10	1.037E+08	1.523E+07	1.410E+09	6.231E+03
F18	Mean	**7.446E+03**	7.947E+05	2.097E+06	2.538E+07	1.201E+07	2.804E+07	5.062E+06	4.244E+07	1.820E+07	1.250E+07	6.375E+06
	Std	2.627E−01	4.509E+08	7.124E+00	7.214E+09	**3.038E−03**	3.068E+01	7.627E+09	2.403E+08	6.659E+07	3.480E+09	4.239E+07
F19	Mean	**3.091E+03**	6.419E+04	1.291E+07	1.229E+09	7.301E+08	4.871E+09	8.163E+07	5.241E+09	9.909E+08	1.022E+06	2.454E+08
	Std	**5.875E−04**	1.285E+10	5.828E+02	8.506E+10	1.622E−03	1.565E+02	2.694E+10	1.419E+10	4.300E+08	3.963E+09	2.341E+08
F20	Mean	**3.140E+03**	3.572E+03	4.364E+03	5.247E+03	3.679E+03	4.333E+03	3.821E+03	5.312E+03	4.451E+03	4.861E+03	3.566E+03
	Std	1.681E+00	2.833E+03	3.626E+01	1.087E+03	**1.311E+00**	1.302E+01	9.103E+02	1.049E+03	1.194E+02	1.035E+03	7.340E+02
F21	Mean	**1.217E+03**	1.333E+03	1.866E+03	1.714E+03	1.519E+03	2.120E+03	1.758E+03	2.485E+03	1.644E+03	8.279E+02	1.998E+03
	Std	1.445E−01	2.354E+03	1.827E−02	5.892E+03	**1.960E−08**	2.312E−03	1.838E+03	4.407E+02	2.951E+01	1.401E+02	8.583E+01
F22	Mean	**1.564E+04**	1.752E+04	2.195E+04	3.106E+04	2.188E+04	2.584E+04	2.347E+04	3.024E+04	2.874E+04	2.497E+04	2.312E+04
	Std	1.504E+01	1.050E+04	8.430E+01	2.227E+03	**3.445E+00**	1.873E+01	1.976E+03	2.782E+03	1.841E+02	2.528E+03	1.840E+03
F23	Mean	1.708E+03	**1.523E+03**	2.429E+03	2.023E+03	3.368E+03	4.800E+03	2.574E+03	3.634E+03	2.418E+03	2.368E+03	3.945E+03
	Std	3.885E−01	3.718E+03	7.339E−01	9.537E+03	**3.387E−02**	1.175E+00	2.718E+03	6.128E+02	1.508E+01	1.351E+02	7.599E+01
F24	Mean	**1.908E+03**	2.296E+03	3.417E+03	2.450E+03	5.693E+03	6.972E+03	3.571E+03	5.889E+03	3.633E+03	4.952E+03	4.711E+03
	Std	4.180E−01	3.830E+03	6.721E−02	2.608E+04	**7.773E−10**	3.535E−02	3.993E+03	1.880E+03	1.696E+01	1.553E+02	4.274E+01
F25	Mean	**8.061E+02**	8.432E+02	1.128E+03	1.446E+04	1.572E+03	1.365E+04	1.645E+03	2.337E+04	8.410E+03	8.336E+02	4.763E+03
	Std	1.778E+00	9.858E+03	7.705E−03	3.517E+05	**3.465E−07**	4.092E−04	8.780E+04	1.983E+03	8.918E+03	4.060E+03	1.776E+02
F26	Mean	**1.555E+04**	1.810E+04	2.856E+04	1.964E+04	3.311E+04	3.871E+04	2.377E+04	4.413E+04	2.486E+04	2.534E+04	3.439E+04
	Std	6.194E−01	1.448E+04	2.402E−01	3.817E+05	**1.681E−08**	9.504E−02	5.807E+04	7.470E+03	1.937E+02	1.765E+03	4.114E+02
F27	Mean	**5.000E+02**	1.085E+03	2.507E+03	1.602E+03	4.724E+03	6.929E+03	2.164E+03	4.910E+03	3.116E+03	**5.000E+02**	4.501E+03
	Std	2.068E+02	6.888E+03	2.043E+00	1.509E+04	**1.555E−01**	1.744E+00	6.061E+03	5.567E+03	2.753E+01	1.214E+04	4.665E+01
F28	Mean	**5.398E+02**	6.118E+02	9.242E+02	1.335E+04	2.492E+03	1.671E+04	1.890E+03	2.355E+04	1.026E+04	2.097E+03	7.540E+03
	Std	1.696E+00	1.234E+04	6.051E−03	6.361E+04	**1.607E−09**	1.128E−03	3.078E+04	8.872E+03	1.718E+03	7.962E+04	1.353E+02
F29	Mean	**3.384E+03**	5.010E+03	1.068E+04	1.324E+04	6.912E+03	3.381E+04	9.109E+03	3.444E+04	1.281E+04	6.528E+03	1.246E+04
	Std	2.546E+01	2.810E+06	2.079E+01	2.889E+10	**8.111E−05**	9.782E+00	4.460E+09	1.670E+07	1.068E+07	1.937E+09	1.351E+03
F30	Mean	**6.129E+04**	1.521E+06	1.960E+08	3.078E+09	1.662E+09	1.619E+10	3.642E+08	1.278E+10	3.400E+09	6.182E+06	9.159E+08
	Std	1.495E+05	1.981E+10	4.476E+03	1.555E+11	**2.180E−04**	3.192E+03	4.373E+10	1.771E+10	6.552E+08	2.903E+10	1.530E+08

Table A.5
Wilcoxon test results (10D).

Function	TSRL-CMM vs.									
	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	6.304E−18	3.304E−18	3.304E−18	3.919E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.301E−18
F3	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	8.022E−06	5.290E−20
F4	6.445E−14	2.120E−18	7.562E−13	1.118E−15	2.120E−18	2.120E−18	2.120E−18	7.790E−18	1.234E−12	2.294E−15
F5	1.141E−02	4.943E−11	7.396E−08	3.487E−01	1.315E−10	5.170E−11	1.405E−13	2.011E−01	1.095E−04	3.806E−01
F6	2.058E−16	3.303E−18	3.303E−18	3.791E−16	3.303E−18	3.303E−18	3.303E−18	8.380E−17	3.366E−12	5.279E−16
F7	2.895E−14	4.434E−18	4.987E−18	2.863E−02	7.485E−17	3.304E−18	8.947E−18	1.288E−07	6.972E−02	1.378E−02
F8	7.799E−04	1.319E−11	3.148E−10	6.635E−01	2.670E−12	6.876E−12	1.123E−12	9.680E−01	2.362E−01	8.070E−01
F9	5.508E−19	2.770E−19	2.770E−19	3.607E−17	2.770E−19	2.770E−19	2.770E−19	3.282E−18	1.856E−03	9.211E−17
F10	2.724E−01	6.029E−09	7.224E−05	4.104E−01	2.985E−13	3.492E−08	2.105E−12	5.924E−08	6.717E−03	1.053E−01
F11	2.128E−05	4.608E−18	3.835E−09	2.199E−14	3.236E−18	1.660E−17	1.173E−17	2.075E−13	2.371E−03	7.229E−04
F12	2.301E−16	3.304E−18	1.508E−17	1.825E−06	3.304E−18	3.504E−18	4.181E−18	3.504E−18	3.504E−18	2.998E−08
F13	3.942E−18	3.304E−18	3.304E−18	2.234E−14	3.504E−18	3.304E−18	3.304E−18	3.504E−18	3.504E−18	3.361E−08
F14	1.447E−09	3.187E−17	1.432E−12	6.570E−12	5.031E−17	1.095E−13	3.304E−18	1.340E−13	3.297E−13	3.015E−11
F15	8.627E−09	7.085E−18	2.392E−17	5.338E−06	8.441E−18	4.241E−17	3.304E−18	2.013E−17	9.927E−17	3.894E−04
F16	1.205E−01	2.497E−01	1.646E−04	4.142E−01	1.823E−12	2.317E−12	1.000E−11	4.940E−03	8.200E−01	1.682E−05
F17	2.609E−01	1.383E−11	1.365E−12	4.349E−07	1.378E−10	7.415E−11	6.012E−15	2.217E−07	2.891E−10	2.328E−02
F18	9.483E−18	4.434E−18	4.434E−18	5.010E−04	3.304E−18	4.703E−18	6.683E−18	6.683E−18	4.987E−18	1.547E−03
F19	7.341E−16	4.181E−18	7.085E−18	1.288E−07	5.607E−18	4.490E−17	4.181E−18	8.441E−18	2.805E−12	1.617E−08
F20	2.107E−01	1.198E−03	5.688E−06	2.287E−02	2.006E−12	2.191E−11	6.576E−13	6.069E−03	4.991E−01	6.854E−03
F21	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20	NAN	7.095E−19
F22	1.391E−20	8.549E−07	4.029E−19	2.279E−16	1.391E−20	1.391E−20	1.391E−20	7.626E−14	9.632E−04	7.116E−19
F23	NAN	1.391E−20	1.391E−20	1.391E−20	3.139E−16	NAN	1.391E−20	1.391E−20	1.391E−20	1.023E−17
F24	8.221E−02	8.965E−06	2.613E−09	2.950E−16	8.221E−02	8.221E−02	8.504E−19	3.685E−07	1.089E−09	6.645E−10
F25	NAN	1.391E−20	1.476E−18	1.391E−20	8.041E−14	NAN	1.391E−20	1.391E−20	1.391E−20	1.391E−20
F26	1.593E−01	1.103E−11	6.384E−13	6.722E−19	1.593E−01	1.593E−01	6.742E−19	2.473E−08	3.237E−16	1.005E−19
F27	1.316E−04	8.414E−14	4.219E−15	7.189E−18	9.199E−01	1.316E−04	2.290E−18	5.675E−18	5.249E−14	2.748E−18
F28	NAN	1.391E−20	6.294E−17	1.388E−20	8.022E−06	NAN	1.391E−20	1.391E−20	1.391E−20	4.007E−19
F29	6.650E−05	9.950E−17	3.249E−15	2.624E−15	1.928E−08	3.100E−01	3.228E−18	9.098E−14	4.071E−15	2.350E−15
F30	3.139E−16	4.619E−18	9.875E−18	1.803E−08	1.106E−10	3.389E−09	6.565E−18	1.156E−16	7.607E−02	7.357E−08

Table A.6
Wilcoxon test results (30D).

Function	TSRL-CMM vs.									
	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	3.942E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.504E−18	3.304E−18
F3	2.017E−18	2.017E−18	2.017E−18	2.017E−18	2.017E−18	2.017E−18	2.017E−18	2.017E−18	2.017E−18	2.017E−18
F4	2.058E−16	3.304E−18	4.181E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.262E−10	7.085E−18
F5	2.014E−09	1.900E−17	8.868E−17	4.616E−01	5.326E−17	1.491E−15	4.987E−18	7.659E−09	1.579E−01	1.748E−08
F6	3.361E−08	4.861E−15	2.392E−17	7.725E−02	5.326E−17	2.683E−17	4.434E−18	7.576E−03	3.304E−18	5.404E−13
F7	6.315E−17	3.304E−18	3.304E−18	4.454E−05	3.304E−18	3.304E−18	3.304E−18	8.498E−14	1.841E−05	4.032E−09
F8	3.374E−03	3.303E−18	6.338E−15	8.543E−02	8.945E−18	4.373E−14	3.303E−18	3.034E−09	5.067E−02	1.285E−03
F9	3.304E−18	3.304E−18	3.304E−18	1.661E−15	3.304E−18	3.304E−18	3.304E−18	1.274E−13	1.931E−16	4.139E−11
F10	1.198E−03	3.304E−18	3.912E−12	5.194E−12	1.343E−17	9.714E−09	3.717E−18	6.304E−18	4.856E−05	3.089E−06
F11	6.880E−01	3.304E−18	1.129E−17	4.369E−15	3.304E−18	3.304E−18	3.304E−18	3.717E−18	8.148E−01	3.454E−11
F12	1.196E−17	3.304E−18	3.304E−18	6.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	9.406E−14	4.987E−18
F13	4.476E−16	3.304E−18	7.073E−17	9.383E−17	3.304E−18	3.304E−18	3.304E−18	3.304E−18	7.511E−18	3.748E−14
F14	3.304E−18	3.304E−18	3.304E−18	2.013E−14	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.742E−11
F15	4.753E−17	3.304E−18	6.304E−18	2.392E−17	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.176E−15
F16	3.677E−03	5.031E−17	3.465E−13	2.292E−08	4.181E−18	1.050E−16	4.703E−18	3.236E−08	1.279E−05	1.787E−05
F17	8.935E−01	3.257E−04	1.983E−04	3.257E−04	3.281E−07	1.316E−03	3.558E−12	1.599E−01	4.822E−01	1.521E−01
F18	3.304E−18	3.304E−18	3.304E−18	7.085E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.013E−17
F19	6.315E−17	3.304E−18	3.504E−18	1.723E−13	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.717E−18	1.050E−16
F20	8.409E−01	1.037E−02	2.015E−05	1.112E−01	2.229E−10	2.656E−03	1.321E−11	1.858E−03	2.362E−01	2.182E−01
F21	2.297E−15	4.861E−15	3.942E−18	1.482E−13	3.304E−18	1.076E−15	3.304E−18	2.058E−16	4.330E−11	6.578E−16
F22	6.318E−05	2.635E−18	4.103E−16	2.889E−17	2.635E−18	1.065E−10	2.635E−18	6.768E−18	3.601E−11	1.196E−13
F23	2.878E−05	1.872E−10	4.374E−14	6.578E−16	3.504E−18	2.842E−17	3.504E−18	2.154E−06	7.555E−06	1.337E−15
F24	6.876E−06	5.952E−07	5.444E−12	2.259E−17	1.129E−17	1.814E−14	5.607E−18	1.518E−04	6.275E−09	2.985E−13
F25	1.076E−15	3.304E−18	3.942E−18	1.005E−17	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.167E−05	1.129E−17
F26	2.791E−05	3.394E−07	3.158E−15	1.180E−08	6.274E−18	7.073E−11	8.905E−18	1.146E−11	9.487E−05	3.075E−12
F27	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	8.777E−01	3.304E−18
F28	2.963E−01	3.205E−18	3.172E−12	3.303E−14	3.205E−18	3.205E−18	3.205E−18	3.205E−18	1.096E−17	2.165E−12
F29	3.614E−11	9.927E−17	4.987E−18	1.136E−15	3.304E−18	3.717E−18	3.304E−18	1.391E−16	3.743E−10	2.301E−16
F30	3.504E−18	3.304E−18	3.304E−18	4.987E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	5.945E−18

Table A.7
Wilcoxon test results (50D).

Function	TSRL-CMM vs.									
	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	4.846E−14	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.783E−04	3.304E−18
F3	3.281E−18	3.281E−18	3.281E−18	3.281E−18	3.281E−18	3.281E−18	3.281E−18	3.281E−18	3.281E−18	3.281E−18
F4	1.638E−09	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.662E−13	2.259E−17
F5	1.632E−05	3.304E−18	1.639E−13	1.955E−05	3.304E−18	2.259E−17	3.304E−18	7.295E−14	1.316E−03	9.603E−08
F6	6.419E−04	5.031E−17	6.683E−18	1.569E−02	7.511E−18	3.717E−18	3.304E−18	1.560E−04	3.304E−18	4.662E−13
F7	3.942E−18	3.304E−18	3.304E−18	1.200E−15	3.304E−18	3.304E−18	3.304E−18	7.921E−17	2.279E−09	1.409E−13
F8	3.914E−07	3.304E−18	1.737E−12	1.831E−04	4.987E−18	7.073E−17	3.304E−18	5.126E−15	1.142E−01	8.595E−08
F9	3.304E−18	3.304E−18	3.304E−18	3.723E−15	3.304E−18	3.304E−18	3.304E−18	8.259E−15	4.310E−04	1.196E−17
F10	7.786E−01	3.304E−18	3.395E−16	1.551E−14	4.703E−18	8.381E−17	3.304E−18	3.304E−18	5.701E−15	7.659E−09
F11	3.663E−01	3.304E−18	9.383E−17	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	6.878E−08	5.184E−11
F12	6.227E−16	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.075E−14	7.963E−18
F13	3.375E−17	3.304E−18	4.181E−18	3.717E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.895E−14	8.947E−18
F14	3.783E−17	3.304E−18	4.987E−18	6.949E−16	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.504E−18	1.200E−15
F15	6.227E−16	3.304E−18	1.129E−17	3.211E−14	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.534E−17	1.661E−15
F16	2.482E−05	3.717E−18	2.297E−15	8.398E−10	7.085E−18	1.954E−15	3.717E−18	7.048E−15	1.211E−13	1.437E−07
F17	1.189E−04	3.504E−18	7.431E−15	4.139E−11	2.842E−17	1.449E−11	3.304E−18	1.437E−07	7.431E−15	2.382E−07
F18	4.476E−16	3.304E−18	5.288E−18	2.749E−14	3.304E−18	3.304E−18	3.304E−18	4.434E−18	3.304E−18	2.985E−13
F19	3.009E−17	3.304E−18	3.304E−18	1.646E−16	3.304E−18	3.304E−18	3.304E−18	3.304E−18	7.998E−13	8.441E−18
F20	3.560E−02	1.397E−14	6.060E−06	4.940E−03	1.681E−08	5.257E−03	2.719E−16	7.224E−05	2.016E−06	4.535E−01
F21	4.407E−06	3.304E−18	7.485E−17	4.268E−06	5.607E−18	5.607E−18	3.304E−18	2.208E−12	5.469E−01	3.015E−11
F22	6.800E−10	2.374E−09	6.683E−18	3.792E−16	3.304E−18	1.076E−15	3.304E−18	6.315E−17	5.607E−18	9.178E−15
F23	1.719E−02	2.673E−08	7.341E−16	9.483E−18	3.504E−18	1.065E−17	3.942E−18	1.518E−04	4.349E−07	1.019E−15
F24	8.561E−06	9.686E−04	2.006E−12	4.434E−18	3.717E−18	2.874E−16	3.304E−18	5.535E−04	9.261E−11	3.927E−15
F25	3.395E−16	3.304E−18	4.703E−18	4.181E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.738E−09	7.511E−18
F26	3.612E−02	7.270E−06	1.964E−17	8.197E−09	3.623E−18	1.489E−09	3.219E−18	2.627E−07	6.083E−02	1.300E−11
F27	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	8.988E−01	3.304E−18
F28	5.739E−10	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	5.701E−15	2.176E−16
F29	1.048E−11	3.304E−18	3.304E−18	5.288E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.196E−17	1.946E−16
F30	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.181E−18	3.304E−18

Table A.8
Wilcoxon test results (100D).

Function	TSRL-CMM vs.									
	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	1.005E−17	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.780E−09	3.304E−18
F3	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18
F4	5.399E−04	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	6.530E−09	8.441E−18
F5	1.263E−06	3.304E−18	4.508E−12	1.383E−11	4.987E−18	2.842E−17	3.304E−18	1.129E−17	2.430E−12	3.361E−08
F6	9.147E−01	3.304E−18	3.304E−18	7.754E−16	3.304E−18	3.304E−18	3.304E−18	6.012E−15	3.304E−18	4.020E−13
F7	3.942E−18	3.304E−18	3.304E−18	2.392E−17	3.304E−18	3.504E−18	3.304E−18	1.794E−17	4.235E−16	7.834E−15
F8	1.946E−16	3.304E−18	3.304E−18	5.652E−14	3.304E−18	3.304E−18	3.304E−18	1.005E−17	2.513E−11	1.723E−13
F9	3.304E−18	3.304E−18	3.304E−18	1.268E−17	3.304E−18	3.304E−18	3.304E−18	6.683E−18	2.003E−13	3.304E−18
F10	1.540E−02	3.304E−18	1.508E−17	5.945E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.065E−17	3.560E−14
F11	8.829E−01	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.987E−18	2.842E−17
F12	1.230E−08	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	5.326E−17	6.304E−18
F13	4.987E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	5.607E−18	3.304E−18
F14	6.304E−18	3.304E−18	3.304E−18	5.607E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.434E−18	4.434E−18
F15	3.717E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.703E−18	3.304E−18
F16	8.697E−12	3.304E−18	5.288E−18	6.683E−17	3.304E−18	3.717E−18	3.304E−18	3.717E−18	3.504E−18	1.315E−16
F17	4.349E−07	3.304E−18	1.019E−15	7.511E−18	3.304E−18	1.065E−17	3.304E−18	4.753E−17	3.187E−17	1.472E−14
F18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18
F19	3.717E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.482E−13	3.717E−18
F20	1.208E−10	3.304E−18	1.841E−16	4.956E−11	1.741E−16	5.144E−13	3.504E−18	1.946E−16	6.227E−16	6.029E−09
F21	6.993E−03	1.129E−17	7.921E−17	1.384E−08	3.942E−18	1.900E−17	3.304E−18	4.032E−09	4.139E−11	5.370E−14
F22	6.626E−08	3.304E−18	1.598E−17	2.683E−17	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.003E−13
F23	7.427E−03	1.397E−14	3.187E−17	3.304E−18	3.304E−18	7.963E−18	3.304E−18	1.277E−09	1.639E−13	3.942E−18
F24	1.321E−11	8.381E−17	3.504E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.013E−14	3.304E−18	3.304E−18
F25	2.878E−05	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	2.411E−02	4.729E−16
F26	9.831E−02	1.304E−01	3.572E−17	2.549E−12	3.303E−18	3.557E−12	3.303E−18	3.797E−04	3.431E−05	4.728E−16
F27	6.315E−17	5.326E−17	1.129E−17	3.717E−18	3.304E−18	2.013E−17	3.304E−18	1.129E−17	3.991E−02	1.005E−17
F28	2.176E−15	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	4.297E−01	3.504E−18
F29	4.608E−15	3.304E−18	3.304E−18	6.304E−18	3.304E−18	3.304E−18	3.304E−18	3.504E−18	2.259E−17	4.434E−18
F30	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	3.304E−18	1.065E−17	3.304E−18

Table A.9
Friedman test results (10D).

Function	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	1.01961	2.31373	7.62745	4.11765	6.19608	10.49020	9.74510	8.37255	6.66667	4.92157	4.52941
F3	1.34314	3.56863	7.52941	5.33333	4.03922	10.27451	8.43137	10.50980	7.70588	1.89216	5.37255
F4	1.43137	4.15686	6.33333	5.66667	5.19608	10.50980	9.52941	8.05882	7.00000	2.64706	5.47059
F5	4.25490	5.72549	8.11765	7.62745	4.01961	8.05882	8.56863	9.09804	3.86275	2.78431	3.88235
F6	1.94118	5.09804	6.05882	8.47059	4.15686	9.15686	10.23529	9.25490	4.47059	1.11765	6.03922
F7	2.47059	5.94118	7.60784	8.35294	3.05882	8.19608	10.43137	8.78431	4.41176	2.78431	3.96078
F8	4.03922	5.78431	7.84314	8.03922	3.62745	8.25490	8.19608	8.74510	4.41176	3.19608	3.86275
F9	1.74510	7.70588	5.25490	8.96078	3.90196	8.74510	8.92157	9.15686	4.66667	1.39216	5.54902
F10	4.00000	3.82353	7.39216	5.68627	4.19608	8.80392	6.82353	8.86275	7.09804	4.86275	4.45098
F11	2.60784	3.94118	7.80392	5.03922	7.07843	10.47059	9.21569	8.21569	6.33333	1.62745	3.66667
F12	1.41176	5.01961	9.05882	6.98039	2.90196	8.13725	8.31373	7.66667	7.82353	5.86275	2.82353
F13	1.23529	5.37255	9.66667	6.82353	5.35294	5.94118	7.52941	8.31373	6.84314	6.37255	2.54902
F14	1.78431	5.13725	7.49020	5.09804	4.90196	8.50980	6.72549	10.23529	6.35294	4.84314	4.92157
F15	1.78431	3.31373	6.52941	8.64706	4.41176	8.64706	6.90196	10.31373	7.37255	5.39216	2.68627
F16	4.07843	4.82353	3.50980	6.45098	5.00000	9.03922	8.33333	8.92157	5.76471	3.96078	6.11765
F17	3.00000	3.09804	6.54902	8.19608	6.54902	6.35294	6.25490	9.07843	6.27451	6.43137	4.21569
F18	1.60784	6.94118	9.35294	6.86275	3.35294	6.47059	7.21569	7.03922	6.92157	7.58824	2.64706
F19	1.60784	5.94118	6.82353	7.17647	4.05882	9.13725	8.13725	9.03922	7.68627	3.31373	3.07843
F20	3.62745	4.47059	5.17647	6.74510	5.09804	9.05882	8.52941	9.21569	5.47059	3.27451	5.33333
F21	1.52941	4.23529	9.80392	5.31373	3.25490	8.62745	6.92157	11.00000	6.72549	1.52941	7.05882
F22	2.56863	5.15686	3.41176	6.07843	4.00000	9.35294	7.68627	11.00000	6.94118	2.17647	7.62745
F23	2.11765	2.11765	6.19608	6.47059	8.88235	8.60784	2.11765	9.05882	5.60784	6.29412	8.52941
F24	3.50980	3.72549	5.66667	7.24510	8.92157	3.72549	3.72549	9.05882	6.53922	7.96078	5.92157
F25	2.15686	2.15686	7.60784	7.93137	6.43137	8.40196	2.15686	9.45098	8.05882	4.62745	7.01961
F26	3.04902	3.16667	7.23529	7.33333	8.76471	3.16667	3.16667	9.47059	6.12745	7.51961	7.00000
F27	3.09804	2.31373	5.80392	7.54902	8.47059	4.00000	2.31373	8.86275	8.19608	5.78431	9.60784
F28	2.41176	2.41176	7.25490	7.36275	8.03922	5.10784	2.41176	9.54902	8.29412	6.98039	6.17647
F29	2.56863	1.94118	7.90196	8.13725	7.17647	6.05882	2.50000	9.66667	6.38235	6.50980	7.15686
F30	3.58824	1.56863	8.60784	9.33333	6.60784	7.59804	2.32353	9.35294	7.31373	4.45098	5.25490

Table A.10
Friedman test results (30D).

Function	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	1.00000	2.43137	8.60784	4.25490	6.03922	10.29412	6.00000	10.68627	7.92157	2.56863	6.19608
F3	1.00000	2.88235	6.43137	9.74510	5.00000	8.74510	6.92157	10.66667	7.62745	2.88235	4.09804
F4	1.15686	3.19608	7.84314	4.39216	6.09804	10.60784	6.54902	10.37255	7.84314	1.96078	5.98039
F5	2.49020	4.56863	9.13725	8.13725	2.78431	8.43137	6.96078	10.60784	5.39216	2.68627	4.80392
F6	3.03922	4.64706	6.94118	8.88235	3.27451	8.19608	8.56863	10.43137	4.60784	1.00000	6.41176
F7	2.39216	6.11765	8.23529	8.13725	3.05882	9.05882	7.21569	10.72549	5.21569	1.33333	4.50980
F8	2.86275	3.45098	9.88235	7.54902	3.45098	8.17647	6.21569	10.72549	6.13725	3.56863	3.98039
F9	2.23529	5.86275	8.11765	8.33333	3.86275	7.45098	9.01961	10.54902	4.84314	1.03922	4.68627
F10	2.15686	3.17647	9.54902	5.50980	5.86275	8.23529	4.19608	9.82353	8.76471	4.56863	4.15686
F11	2.23529	2.09804	7.60784	5.31373	5.29412	10.54902	7.64706	9.88235	8.31373	2.29412	4.76471
F12	1.07843	2.84314	7.84314	4.78431	6.94118	10.60784	6.33333	9.84314	7.56863	2.76471	5.39216
F13	1.09804	3.09804	9.43137	3.64706	6.00000	8.54902	7.96078	8.96078	7.94118	5.68627	3.62745
F14	1.19608	4.09804	5.54902	7.88235	4.56863	9.68627	9.05882	8.68627	7.80392	4.88235	2.58824
F15	1.05882	3.09804	8.86275	4.74510	3.09804	10.54902	8.90196	6.86275	7.94118	7.74510	3.13725
F16	2.19608	3.29412	7.68627	7.00000	5.15686	9.29412	7.47059	9.33333	5.54902	4.07843	4.94118
F17	4.41176	4.27451	6.31373	6.68627	6.84314	7.80392	6.17647	9.05882	5.43137	3.94118	5.05882
F18	1.00000	4.31373	6.86275	7.11765	5.49020	9.43137	8.09804	8.17647	7.52941	5.27451	2.70588
F19	1.09804	2.72549	8.94118	6.25490	4.52941	9.45098	8.27451	9.13725	6.88235	5.33333	3.37255
F20	4.17647	4.17647	5.60784	7.21569	4.88235	8.88235	6.23529	9.27451	5.98039	4.56863	5.00000
F21	1.29412	4.72549	7.88235	8.13725	3.54902	7.84314	7.54902	10.74510	5.31373	3.31373	5.64706
F22	1.61765	3.49020	4.74510	7.17647	7.05882	9.43137	3.13725	9.05882	7.66667	7.26471	5.35294
F23	1.90196	3.43137	4.23529	5.88235	8.37255	9.68627	7.01961	9.43137	4.03922	3.68627	8.31373
F24	2.19608	3.72549	3.41176	5.50980	8.50980	9.92157	6.50980	9.74510	4.00000	4.86275	7.60784
F25	1.21569	3.33333	8.52941	5.15686	4.50980	9.96078	6.39216	10.72549	8.05882	1.78431	6.33333
F26	2.13725	4.41176	4.03922	7.62745	5.56863	9.94118	6.31373	9.94118	5.58824	3.52941	6.90196
F27	1.45098	3.45098	4.17647	5.82353	7.88235	10.33333	7.66667	8.62745	6.64706	1.54902	8.39216
F28	1.74510	1.64706	7.90196	3.84314	5.66667	10.56863	5.84314	10.21569	8.33333	4.60784	5.62745
F29	1.41176	3.37255	4.82353	7.52941	5.70588	10.39216	7.33333	9.41176	6.41176	2.82353	6.78431
F30	1.00000	2.86275	8.13725	5.92157	5.17647	10.07843	8.11765	9.05882	8.25490	3.68627	3.70588

Table A.11
Friedman test results (50D).

Function	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	1.37255	2.72549	8.50980	4.09804	5.70588	9.94118	5.58824	10.90196	8.27451	2.01961	6.86275
F3	1.00000	2.37255	7.54902	4.90196	6.29412	9.01961	8.72549	10.70588	7.19608	3.76471	4.47059
F4	1.17647	2.56863	8.33333	4.23529	5.96078	10.49020	6.21569	10.49020	7.90196	2.74510	5.88235
F5	2.52941	3.66667	10.01961	5.94118	4.11765	8.37255	5.84314	10.72549	7.37255	1.90196	5.50980
F6	3.11765	3.58824	7.66667	7.98039	3.60784	8.01961	9.00000	10.84314	5.03922	1.00000	6.13725
F7	1.98039	5.66667	9.47059	7.68627	3.52941	8.29412	6.88235	11.00000	5.56863	1.13725	4.78431
F8	2.39216	3.80392	10.01961	5.54902	3.80392	7.90196	6.19608	10.86275	7.29412	2.29412	5.88235
F9	1.84314	4.35294	9.15686	7.01961	3.39216	7.21569	9.39216	10.72549	6.05882	1.43137	5.41176
F10	1.84314	2.03922	10.50980	5.37255	4.98039	7.17647	4.41176	9.94118	8.68627	6.31373	4.72549
F11	1.80392	1.92157	8.23529	4.05882	6.21569	10.39216	6.35294	10.45098	8.00000	3.11765	5.45098
F12	1.11765	2.45098	8.27451	4.37255	6.96078	10.70588	5.66667	10.05882	7.78431	2.70588	5.90196
F13	1.13725	2.76471	7.72549	3.62745	8.01961	10.17647	6.13725	9.90196	7.62745	3.21569	5.66667
F14	1.11765	3.15686	6.33333	5.31373	6.62745	9.84314	8.05882	9.23529	7.72549	4.58824	4.00000
F15	1.13725	2.90196	8.31373	4.41176	4.21569	10.62745	7.56863	9.56863	8.01961	5.50980	3.72549
F16	1.70588	3.01961	8.78431	6.19608	4.60784	9.03922	5.92157	10.11765	5.94118	5.58824	5.07843
F17	2.09804	3.41176	9.86275	6.86275	5.09804	8.62745	5.52941	9.60784	4.49020	5.82353	4.58824
F18	1.19608	3.03922	6.72549	5.80392	5.49020	10.33333	7.01961	9.07843	7.45098	6.05882	3.80392
F19	1.07843	3.00000	9.92157	5.35294	5.29412	9.60784	7.33333	9.27451	7.74510	3.00000	4.39216
F20	3.19608	4.31373	8.84314	6.49020	4.92157	6.86275	4.92157	10.00000	5.60784	6.37255	4.47059
F21	2.13725	3.56863	7.88235	7.78431	3.66667	8.31373	7.25490	10.84314	5.86275	2.33333	6.35294
F22	1.35294	2.80392	8.88235	4.66667	5.11765	7.52941	5.35294	9.94118	8.41176	6.96078	4.98039
F23	2.17647	2.78431	3.54902	6.27451	8.47059	9.68627	6.86275	9.49020	3.94118	4.29412	8.47059
F24	2.23529	3.45098	2.84314	5.50980	9.23529	9.60784	6.80392	9.94118	3.78431	5.29412	7.29412
F25	1.15686	3.00000	8.21569	4.60784	5.09804	10.33333	6.47059	10.58824	8.19608	2.03922	6.29412
F26	2.67647	3.47059	4.68627	8.15686	6.21569	9.41176	5.41176	10.64706	5.39216	3.18627	6.74510
F27	1.54902	3.47059	3.78431	6.88235	7.88235	10.74510	7.05882	8.09804	6.76471	1.45098	8.31373
F28	1.19608	2.64706	7.60784	4.37255	6.31373	10.25490	6.01961	10.49020	8.25490	2.49020	6.35294
F29	1.17647	2.29412	6.23529	6.60784	5.62745	10.52941	7.39216	9.92157	6.74510	3.80392	5.66667
F30	1.00000	2.80392	8.41176	5.60784	4.64706	10.58824	7.15686	9.33333	8.37255	2.49020	5.58824

Table A.12
Friedman test results (100D).

Function	TSRL-CMM	AVOA	MPA	WOA	PSO	GWOCS	CSCAHHO	PSOGSA	PSOGWO	QLPSO	QLSOPSO
F1	1.43137	3.45098	8.62745	4.80392	6.47059	9.68627	6.37255	10.98039	8.70588	2.41176	3.05882
F3	1.01961	3.41176	7.68627	8.88235	6.23529	7.01961	5.88235	10.35294	6.54902	4.15686	4.80392
F4	1.35294	1.76471	7.56863	3.25490	4.68627	9.64706	5.01961	10.17647	7.43137	7.03922	8.05882
F5	1.64706	2.52941	8.35294	3.92157	3.58824	6.50980	5.43137	9.25490	6.43137	8.50980	9.82353
F6	1.68627	1.52941	7.84314	5.03922	3.41176	6.09804	5.45098	8.92157	5.64706	9.90196	10.47059
F7	1.27451	4.05882	8.45098	5.80392	2.90196	7.03922	4.19608	9.60784	5.19608	7.98039	9.49020
F8	1.33333	2.80392	7.88235	4.49020	3.19608	6.70588	5.58824	9.33333	6.17647	8.45098	10.03922
F9	2.15686	3.39216	9.68627	5.25490	5.01961	6.49020	8.15686	10.39216	7.35294	3.43137	4.66667
F10	2.41176	2.70588	9.90196	5.43137	5.39216	6.98039	5.31373	9.49020	8.29412	4.27451	5.80392
F11	1.82353	1.84314	6.90196	4.07843	5.33333	9.00000	7.96078	10.07843	8.09804	4.72549	6.15686
F12	2.84314	3.62745	8.47059	4.92157	7.00000	10.07843	6.05882	10.72549	8.37255	1.47059	2.43137
F13	2.17647	3.86275	8.41176	4.21569	6.90196	10.19608	6.27451	10.66667	8.17647	2.05882	3.05882
F14	1.62745	3.80392	8.41176	5.29412	5.68627	9.33333	6.39216	9.96078	8.35294	2.74510	4.39216
F15	1.52941	3.68627	8.15686	4.37255	6.76471	10.25490	6.33333	10.23529	8.05882	2.66667	3.94118
F16	1.86275	2.72549	7.88235	5.86275	3.84314	9.01961	5.54902	9.74510	6.45098	5.74510	7.31373
F17	1.80392	2.84314	7.54902	4.52941	6.21569	9.58824	5.27451	9.84314	5.98039	5.21569	7.15686
F18	1.80392	3.84314	9.11765	5.19608	6.09804	9.19608	6.50980	9.84314	8.13725	2.21569	4.03922
F19	1.49020	3.49020	8.56863	5.09804	6.23529	10.21569	6.37255	10.35294	7.92157	2.41176	3.84314
F20	1.92157	3.82353	8.94118	6.13725	4.15686	6.29412	4.56863	9.05882	6.45098	6.80392	7.84314
F21	2.07843	2.76471	5.60784	6.78431	4.15686	8.19608	5.94118	9.52941	5.33333	6.82353	8.78431
F22	2.33333	3.19608	9.96078	5.07843	5.25490	7.23529	5.82353	9.33333	8.62745	3.80392	5.35294
F23	2.31373	1.84314	3.84314	5.21569	7.82353	9.47059	5.82353	8.50980	5.05882	7.19608	8.90196
F24	1.84314	2.98039	3.58824	5.50980	8.37255	9.41176	5.90196	8.78431	5.45098	6.49020	7.66667
F25	1.66667	2.25490	8.66667	3.56863	4.74510	8.39216	5.43137	10.21569	7.25490	6.09804	7.70588
F26	3.49020	3.92157	3.96078	7.09804	7.78431	9.05882	6.00000	10.07843	5.62745	4.03922	4.94118
F27	1.29412	2.49020	3.72549	5.50980	8.01961	9.58824	5.05882	8.25490	6.19608	7.01961	8.84314
F28	1.45098	2.39216	7.84314	3.52941	5.15686	8.70588	5.15686	10.11765	7.29412	6.58824	7.76471
F29	1.80392	2.86275	7.58824	6.39216	4.03922	9.70588	5.43137	9.76471	6.90196	4.98039	6.52941
F30	2.31373	3.66667	8.41176	5.33333	6.74510	10.72549	6.13725	10.13725	8.13725	1.76471	2.62745

Table A.13
Comparison of optimization algorithms on benchmark functions (10D).

Fun	Metric	TSRLCMM	COLSHADE	HyDE	MadDE
F1	Mean	1.385E+03	1.688E+09	1.317E+03	**4.247E−08**
	Std	8.660E+02	2.037E+10	2.480E+08	3.628E+09
F3	Mean	**0.000E+00**	1.016E+04	1.208E+01	**0.000E+00**
	Std	4.504E−15	3.478E+11	4.214E+05	2.349E+06
F4	Mean	**0.000E+00**	1.024E+02	1.841E−01	**0.000E+00**
	Std	2.364E−14	8.575E+03	3.829E+00	6.491E+02
F5	Mean	2.919E+01	5.207E+01	3.320E+01	**3.969E+00**
	Std	9.457E−14	6.832E+01	2.693E+01	1.688E+01
F6	Mean	2.417E+00	2.811E+01	2.060E−01	**1.146E−07**
	Std	7.174E−11	4.351E+01	2.741E−01	4.136E+00
F7	Mean	3.564E+01	1.374E+02	4.669E+01	**1.508E+01**
	Std	7.881E−14	2.347E+02	3.114E+01	1.981E+01
F8	Mean	2.478E+01	5.410E+01	3.601E+01	**4.924E+00**
	Std	1.756E−13	5.087E+01	2.848E+01	1.076E+01
F9	Mean	1.114E+00	4.667E+02	6.659E−02	**0.000E+00**
	Std	3.828E−14	6.925E+03	2.854E−01	4.628E+02
F10	Mean	7.272E+02	1.326E+03	3.991E+02	**8.140E+01**
	Std	1.731E−01	6.126E+02	5.701E+02	5.086E+02
F11	Mean	2.829E+01	3.545E+02	1.049E+01	**1.223E+00**
	Std	2.029E−13	8.216E+08	7.677E+01	1.463E+05
F12	Mean	9.358E+03	2.360E+07	3.844E+02	**2.178E+01**
	Std	4.528E−03	6.583E+09	3.887E+07	8.647E+08
F13	Mean	2.111E+02	3.232E+04	1.550E+01	**4.198E+00**
	Std	5.505E−02	3.560E+09	2.380E+05	3.047E+08
F14	Mean	6.056E+01	1.560E+02	2.192E+01	**8.193E−01**
	Std	3.514E−06	2.226E+09	6.684E+04	3.972E+07
F15	Mean	5.100E+01	1.566E+03	5.176E+00	**4.449E−01**
	Std	2.326E−04	5.141E+09	3.081E+05	1.410E+07
F16	Mean	1.473E+02	1.841E+02	6.152E+00	**6.071E−01**
	Std	6.549E−04	9.263E+03	8.263E+02	1.751E+02
F17	Mean	6.237E+01	1.025E+02	3.802E+01	**3.239E−01**
	Std	3.373E−13	5.391E+05	7.216E+02	6.891E+01
F18	Mean	3.239E+02	6.681E+04	1.437E+01	**5.900E−01**
	Std	3.356E−04	9.792E+09	5.181E+05	1.303E+09
F19	Mean	4.036E+01	2.155E+03	1.978E+00	**5.011E−02**
	Std	1.196E−05	7.511E+09	9.289E+04	6.288E+07
F20	Mean	7.173E+01	1.051E+02	2.372E+01	**0.000E+00**
	Std	3.324E−13	3.072E+02	3.875E+02	2.913E+02
F21	Mean	1.382E+02	2.034E+02	1.434E+02	**9.412E+01**
	Std	4.601E−04	1.836E+02	1.226E+01	7.460E+01
F22	Mean	9.937E+01	2.855E+02	**7.722E+01**	9.184E+01
	Std	3.963E−13	1.022E+03	8.495E+00	1.226E+02
F23	Mean	3.254E+02	3.622E+02	3.303E+02	**2.875E+02**
	Std	1.353E−02	4.122E+02	2.685E+01	5.597E+01
F24	Mean	3.280E+02	3.628E+02	3.464E+02	**9.440E+01**
	Std	1.557E−04	3.658E+02	2.507E+01	1.439E+02
F25	Mean	4.143E+02	5.202E+02	**3.985E+02**	3.986E+02
	Std	1.252E−06	3.764E+03	4.451E−01	2.262E+02
F26	Mean	3.459E+02	7.981E+02	3.000E+02	**1.392E+02**
	Std	1.610E−03	1.357E+03	1.276E−02	6.030E+02
F27	Mean	**3.735E+02**	4.169E+02	3.890E+02	3.883E+02
	Std	5.998E−01	1.356E+03	9.069E−01	4.100E+01
F28	Mean	3.982E+02	5.790E+02	3.001E+02	**2.647E+02**
	Std	9.133E+00	1.131E+03	7.164E−02	2.815E+02
F29	Mean	3.255E+02	3.521E+02	2.648E+02	**2.487E+02**
	Std	4.973E−01	3.532E+06	5.328E+05	1.685E+03
F30	Mean	9.233E+02	1.272E+06	**5.396E+02**	9.371E+02
	Std	6.724E+04	2.064E+09	1.510E+08	5.511E+07

Table A.14
Comparison of optimization algorithms on benchmark functions (30D).

Fun	Metric	TSRLCMM	COLSHADE	HyDE	MadDE
F1	Mean	**1.652E+02**	3.526E+10	8.848E+07	3.241E+03
	Std	2.915E−06	4.439E+10	7.237E+08	5.293E+09
F3	Mean	**0.000E+00**	1.036E+05	1.124E+05	3.900E+04
	Std	6.237E−14	5.598E+14	1.977E+14	1.896E+12
F4	Mean	**5.642E+01**	4.912E+03	1.006E+02	1.114E+02
	Std	5.159E−01	4.819E+04	2.129E+01	2.117E+03
F5	Mean	1.608E+02	3.570E+02	2.287E+02	**1.041E+02**
	Std	2.417E−03	1.428E+02	5.978E+01	7.971E+01
F6	Mean	2.651E+01	6.951E+01	1.099E+01	**5.043E−01**
	Std	7.046E−13	2.839E+01	5.255E+00	8.974E+00
F7	Mean	2.018E+02	1.113E+03	2.841E+02	**1.362E+02**
	Std	1.514E−13	6.513E+02	8.207E+01	8.505E+01
F8	Mean	1.313E+02	3.287E+02	2.319E+02	**9.400E+01**
	Std	1.502E−04	1.179E+02	6.106E+01	7.835E+01
F9	Mean	1.319E+03	9.308E+03	6.161E+02	**6.050E+01**
	Std	5.305E−11	2.010E+04	1.387E+03	7.640E+02
F10	Mean	**3.489E+03**	6.998E+03	6.736E+03	3.673E+03
	Std	4.813E+00	1.068E+03	1.127E+03	2.015E+03
F11	Mean	1.432E+02	5.183E+03	1.594E+02	**1.077E+02**
	Std	3.441E−01	1.918E+10	1.256E+04	1.210E+07
F12	Mean	**2.376E+05**	3.118E+09	1.129E+07	6.498E+05
	Std	1.651E+03	1.662E+10	1.843E+09	1.635E+09
F13	Mean	7.844E+03	9.097E+08	**1.139E+03**	2.557E+04
	Std	9.962E−03	3.013E+10	4.227E+07	2.919E+09
F14	Mean	7.380E+02	3.949E+05	**8.598E+01**	6.623E+02
	Std	5.556E+01	2.301E+09	4.055E+05	2.215E+08
F15	Mean	5.199E+03	8.239E+07	**1.237E+02**	1.440E+03
	Std	1.332E−01	1.451E+10	2.306E+06	2.554E+08
F16	Mean	9.293E+02	2.306E+03	1.483E+03	**5.365E+02**
	Std	5.532E−01	2.137E+04	1.044E+03	1.060E+03
F17	Mean	6.379E+02	9.531E+02	5.009E+02	**1.014E+02**
	Std	1.222E−02	1.722E+07	8.089E+03	9.207E+02
F18	Mean	1.208E+04	4.921E+06	**7.378E+02**	8.994E+04
	Std	8.405E+03	9.160E+09	6.361E+07	4.294E+08
F19	Mean	7.742E+02	1.334E+08	**4.803E+01**	1.648E+03
	Std	1.254E−06	1.626E+10	1.993E+06	4.957E+08
F20	Mean	5.882E+02	7.051E+02	1.582E+02	**1.243E+02**
	Std	1.489E−01	4.998E+02	8.144E+01	7.121E+02
F21	Mean	3.468E+02	5.247E+02	4.191E+02	**2.174E+02**
	Std	2.514E−02	2.267E+02	5.768E+01	1.755E+02
F22	Mean	2.209E+03	5.551E+03	5.249E+03	**1.000E+02**
	Std	5.317E+00	1.085E+03	8.483E+02	5.994E+02
F23	Mean	5.498E+02	8.272E+02	5.664E+02	**4.423E+02**
	Std	8.986E−02	8.370E+02	5.566E+01	9.958E+01
F24	Mean	6.143E+02	8.932E+02	6.264E+02	**5.083E+02**
	Std	5.636E−02	8.292E+02	5.249E+01	1.515E+02
F25	Mean	**3.785E+02**	2.361E+03	4.012E+02	3.869E+02
	Std	1.177E−01	1.754E+04	2.255E+01	2.855E+02
F26	Mean	2.260E+03	5.948E+03	3.285E+03	**2.807E+02**
	Std	8.100E−02	8.456E+03	5.972E+02	1.462E+03
F27	Mean	**5.000E+02**	8.790E+02	5.102E+02	5.211E+02
	Std	4.221E+01	2.000E+03	6.739E+01	8.058E+01
F28	Mean	**3.861E+02**	2.528E+03	4.602E+02	4.032E+02
	Std	2.068E+01	8.282E+03	8.250E+01	6.784E+02
F29	Mean	7.570E+02	2.316E+03	8.941E+02	**6.098E+02**
	Std	4.250E+00	2.233E+08	4.927E+06	1.573E+03
F30	Mean	**5.775E+03**	1.240E+08	1.807E+04	3.403E+04
	Std	7.217E+03	8.324E+09	2.935E+08	5.876E+08

Table A.15
Comparison of optimization algorithms on benchmark functions (50D).

Fun	Metric	TSRLCMM	COLSHADE	HyDE	MadDE
F1	Mean	**4.767E+03**	1.017E+11	5.570E+08	1.096E+04
	Std	2.269E−06	6.022E+10	8.180E+08	5.520E+09
F3	Mean	**3.570E−06**	2.266E+05	2.606E+05	1.225E+05
	Std	6.966E−11	9.586E+15	5.337E+15	5.708E+13
F4	Mean	**1.016E+02**	1.974E+04	3.209E+02	1.208E+02
	Std	5.098E−01	6.845E+04	2.208E+02	1.593E+03
F5	Mean	**3.106E+02**	7.151E+02	4.432E+02	**3.106E+02**
	Std	7.234E−04	1.703E+02	8.689E+01	1.236E+02
F6	Mean	4.428E+01	8.978E+01	1.460E+01	**2.060E+00**
	Std	5.321E−13	2.240E+01	5.447E+00	7.767E+00
F7	Mean	5.012E+02	2.235E+03	5.297E+02	**3.592E+02**
	Std	2.252E−13	1.049E+03	1.197E+02	1.474E+02
F8	Mean	**3.035E+02**	7.082E+02	4.423E+02	3.130E+02
	Std	2.942E−04	1.792E+02	8.535E+01	1.102E+02
F9	Mean	5.272E+03	3.409E+04	**1.235E+03**	2.776E+03
	Std	8.288E−01	3.612E+04	1.872E+03	3.634E+03
F10	Mean	**6.585E+03**	1.314E+04	1.323E+04	8.289E+03
	Std	7.535E+00	1.395E+03	1.418E+03	3.413E+03
F11	Mean	**2.437E+02**	1.740E+04	2.692E+02	3.344E+02
	Std	7.745E−02	2.912E+11	1.122E+05	1.446E+08
F12	Mean	**1.791E+06**	2.443E+10	2.828E+08	3.283E+06
	Std	4.754E+05	6.003E+10	3.767E+09	7.631E+09
F13	Mean	**1.100E+04**	7.525E+09	7.406E+04	2.257E+04
	Std	5.326E−02	5.478E+10	3.022E+09	6.196E+09
F14	Mean	1.603E+04	3.990E+06	**1.880E+02**	9.818E+04
	Std	2.407E+00	2.484E+09	4.973E+05	1.700E+08
F15	Mean	5.901E+03	1.328E+09	**8.008E+02**	1.541E+04
	Std	5.784E−02	2.785E+10	1.190E+07	1.263E+09
F16	Mean	1.843E+03	4.645E+03	3.322E+03	**1.098E+03**
	Std	2.704E+00	1.333E+04	1.538E+03	2.012E+03
F17	Mean	**1.471E+03**	3.732E+03	2.111E+03	8.320E+02
	Std	1.829E−01	3.133E+08	2.815E+04	1.685E+03
F18	Mean	**5.275E+04**	2.533E+07	2.196E+05	6.896E+05
	Std	8.070E+00	7.335E+09	2.639E+09	4.669E+08
F19	Mean	5.615E+03	6.124E+08	**2.338E+02**	1.751E+04
	Std	1.823E−02	1.169E+10	5.725E+06	6.091E+08
F20	Mean	1.247E+03	1.801E+03	1.797E+03	**5.606E+02**
	Std	6.881E−01	6.680E+02	6.385E+02	1.319E+03
F21	Mean	5.467E+02	9.190E+02	6.463E+02	**4.721E+02**
	Std	3.675E−03	3.458E+02	8.328E+01	1.206E+02
F22	Mean	6.722E+03	1.349E+04	1.363E+04	**1.400E+02**
	Std	1.066E+01	1.400E+03	1.396E+03	9.161E+02
F23	Mean	8.483E+02	1.493E+03	8.711E+02	**7.092E+02**
	Std	4.461E−02	1.025E+03	8.125E+01	1.809E+02
F24	Mean	9.320E+02	1.582E+03	9.155E+02	**7.697E+02**
	Std	1.172E−01	1.121E+03	7.283E+01	2.111E+02
F25	Mean	**4.690E+02**	1.341E+04	6.313E+02	6.079E+02
	Std	8.168E−01	3.301E+04	1.183E+02	6.820E+02
F26	Mean	5.198E+03	1.238E+04	5.512E+03	**3.065E+02**
	Std	8.010E−01	1.360E+04	8.409E+02	2.163E+03
F27	Mean	**5.000E+02**	2.216E+03	5.931E+02	7.165E+02
	Std	1.235E+02	3.501E+03	1.842E+02	5.845E+02
F28	Mean	**4.946E+02**	8.109E+03	7.182E+02	5.544E+02
	Std	6.211E+02	1.137E+04	1.381E+02	8.073E+02
F29	Mean	**1.218E+03**	6.220E+03	2.597E+03	1.132E+03
	Std	8.417E+00	7.300E+08	3.217E+06	2.113E+04
F30	Mean	**3.641E+04**	1.472E+09	3.907E+06	4.076E+06
	Std	8.175E+05	1.998E+10	2.204E+09	1.158E+09

Table A.16
Comparison of optimization algorithms on benchmark functions (100D).

Fun	Metric	TSRLCMM	COLSHADE	HyDE	MadDE
F1	Mean	**5.183E+03**	3.025E+11	4.681E+08	8.954E+09
	Std	7.827E−07	9.850E+10	3.473E+08	1.806E+10
F3	Mean	**1.196E+01**	5.699E+05	7.378E+05	3.676E+05
	Std	1.247E−07	6.569E+17	4.673E+17	3.229E+15
F4	Mean	**2.559E+02**	8.019E+04	4.353E+02	1.384E+03
	Std	6.615E−01	1.405E+05	9.948E+01	4.028E+03
F5	Mean	**7.628E+02**	1.761E+03	9.543E+02	1.091E+03
	Std	2.148E−03	2.709E+02	1.268E+02	1.559E+02
F6	Mean	5.082E+01	1.148E+02	**1.041E+01**	3.812E+01
	Std	1.481E−12	1.597E+01	2.861E+00	7.626E+00
F7	Mean	1.584E+03	5.495E+03	**1.097E+03**	1.278E+03
	Std	1.014E−12	2.319E+03	1.620E+02	2.032E+02
F8	Mean	**8.786E+02**	1.838E+03	9.561E+02	1.083E+03
	Std	4.099E−04	3.021E+02	1.273E+02	1.550E+02
F9	Mean	1.777E+04	1.074E+05	**1.254E+03**	4.762E+04
	Std	5.772E−03	5.071E+04	1.213E+03	1.769E+04
F10	Mean	**1.431E+04**	2.995E+04	3.048E+04	2.570E+04
	Std	1.587E+01	1.984E+03	1.995E+03	3.951E+03
F11	Mean	**1.278E+03**	1.941E+05	2.276E+04	7.307E+04
	Std	2.529E−01	9.255E+13	5.991E+11	5.740E+11
F12	Mean	**1.244E+07**	1.220E+11	7.565E+08	5.505E+08
	Std	8.955E+03	9.190E+10	1.895E+09	1.014E+10
F13	Mean	**1.208E+04**	2.153E+10	1.148E+06	1.012E+06
	Std	4.093E−04	2.836E+10	4.441E+07	2.810E+09
F14	Mean	**1.081E+04**	4.484E+07	7.397E+04	3.304E+06
	Std	9.996E−03	3.172E+09	3.594E+08	1.093E+08
F15	Mean	**4.380E+03**	7.211E+09	6.256E+04	1.721E+05
	Std	3.336E−01	2.181E+10	9.975E+08	2.238E+09
F16	Mean	**3.878E+03**	1.346E+04	8.852E+03	7.033E+03
	Std	3.889E+00	2.300E+04	2.407E+03	2.501E+03
F17	Mean	**3.498E+03**	3.507E+04	5.723E+03	4.087E+03
	Std	4.692E+00	2.798E+09	3.607E+04	5.459E+05
F18	Mean	**5.530E+04**	7.705E+07	1.019E+07	3.444E+06
	Std	5.785E+03	4.420E+09	2.383E+09	2.534E+08
F19	Mean	**3.212E+03**	7.399E+09	4.047E+05	2.938E+04
	Std	8.324E+01	2.180E+10	7.809E+08	2.015E+09
F20	Mean	**3.185E+03**	5.265E+03	5.208E+03	3.448E+03
	Std	2.438E+00	9.294E+02	9.242E+02	2.161E+03
F21	Mean	1.158E+03	2.189E+03	1.187E+03	**1.114E+03**
	Std	4.990E−02	6.650E+02	1.240E+02	1.926E+02
F22	Mean	**1.585E+04**	3.097E+04	3.094E+04	2.510E+04
	Std	1.365E+01	1.984E+03	1.984E+03	4.620E+03
F23	Mean	1.553E+03	3.080E+03	1.511E+03	**1.410E+03**
	Std	7.487E−01	1.754E+03	1.100E+02	2.721E+02
F24	Mean	1.834E+03	5.315E+03	1.833E+03	**1.771E+03**
	Std	2.096E−01	2.998E+03	1.240E+02	4.553E+02
F25	Mean	**8.176E+02**	4.274E+04	1.196E+03	1.581E+03
	Std	3.249E−01	5.866E+04	1.809E+02	9.863E+02
F26	Mean	1.580E+04	4.055E+04	**1.371E+04**	1.806E+04
	Std	4.902E−01	3.162E+04	1.463E+03	2.289E+03
F27	Mean	**5.000E+02**	5.442E+03	1.114E+03	1.258E+03
	Std	3.852E+02	4.504E+03	5.808E+02	7.249E+02
F28	Mean	**5.620E+02**	3.319E+04	1.318E+04	2.078E+03
	Std	1.273E+00	2.093E+04	5.073E+02	2.278E+03
F29	Mean	**3.496E+03**	2.998E+04	7.372E+03	6.120E+03
	Std	1.867E+01	9.060E+08	9.137E+05	6.931E+04
F30	Mean	**6.577E+04**	1.227E+10	9.747E+06	3.545E+06
	Std	5.928E+05	3.472E+10	5.680E+08	2.890E+09

Table A.17
Wilcoxon test results (10D).

Function	TSRL-CMM vs.		
	COLSHADE	HyDE	MadDE
F1	3.304E−18	2.768E−10	4.987E−18
F3	9.325E−19	9.325E−19	4.848E−01
F4	1.187E−18	1.187E−18	7.158E−15
F5	1.395E−14	4.176E−03	3.298E−18
F6	8.441E−18	8.671E−01	6.683E−18
F7	3.504E−18	5.792E−09	5.607E−18
F8	3.779E−17	1.963E−08	8.432E−18
F9	2.077E−18	3.669E−05	1.489E−09
F10	9.675E−15	3.160E−09	4.703E−18
F11	3.302E−18	5.269E−10	3.302E−18
F12	3.304E−18	1.200E−15	8.441E−18
F13	3.304E−18	3.304E−18	3.304E−18
F14	1.243E−16	7.485E−17	3.304E−18
F15	3.304E−18	1.741E−16	3.304E−18
F16	2.836E−03	2.205E−08	2.392E−17
F17	5.135E−09	7.993E−04	3.304E−18
F18	3.304E−18	3.304E−18	3.304E−18
F19	3.942E−18	3.304E−18	3.304E−18
F20	9.111E−06	7.295E−14	3.233E−18
F21	1.581E−07	2.835E−02	5.026E−01
F22	3.304E−18	2.274E−03	7.431E−15
F23	4.987E−18	1.326E−04	3.304E−18
F24	2.513E−04	3.559E−02	7.210E−12
F25	3.296E−18	7.990E−04	2.365E−16
F26	2.492E−18	1.237E−01	1.746E−05
F27	3.304E−18	3.304E−18	3.304E−18
F28	7.350E−17	6.066E−02	2.869E−03
F29	9.975E−03	7.410E−08	2.805E−12
F30	3.304E−18	4.085E−05	1.097E−05

Table A.18
Wilcoxon test results (30D).

Function	TSRL-CMM vs.		
	COLSHADE	HyDE	MadDE
F1	3.304E−18	3.304E−18	3.304E−18
F3	3.297E−18	3.297E−18	3.297E−18
F4	3.304E−18	3.304E−18	5.288E−18
F5	3.304E−18	2.805E−12	4.604E−14
F6	3.304E−18	5.652E−14	3.304E−18
F7	3.304E−18	4.374E−14	1.449E−11
F8	3.304E−18	4.703E−18	5.739E−10
F9	3.304E−18	2.482E−05	2.301E−16
F10	3.304E−18	3.304E−18	3.116E−02
F11	3.304E−18	4.266E−03	9.814E−05
F12	3.304E−18	3.304E−18	7.922E−12
F13	3.304E−18	3.927E−15	1.180E−12
F14	3.304E−18	3.304E−18	3.488E−01
F15	3.304E−18	3.304E−18	6.347E−01
F16	3.717E−18	1.124E−12	4.933E−09
F17	4.257E−10	5.826E−03	2.534E−17
F18	3.304E−18	1.716E−10	8.650E−16
F19	3.304E−18	3.304E−18	9.714E−09
F20	1.279E−02	8.190E−16	3.783E−17
F21	5.288E−18	3.138E−13	1.661E−15
F22	6.564E−16	6.791E−10	2.674E−02
F23	3.504E−18	4.177E−03	2.558E−15
F24	3.304E−18	1.599E−01	1.111E−16
F25	3.304E−18	4.181E−18	6.315E−17
F26	3.287E−18	1.096E−05	6.246E−06
F27	3.304E−18	1.076E−15	3.304E−18
F28	3.289E−18	6.016E−09	1.639E−01
F29	3.304E−18	1.783E−02	2.204E−04
F30	3.304E−18	1.050E−16	4.181E−18

Table A.19
Wilcoxon test results (50D).

Function	TSRL-CMM vs.		
	COLSHADE	HyDE	MadDE
F1	3.304E−18	3.304E−18	9.261E−11
F3	3.304E−18	3.304E−18	3.304E−18
F4	3.304E−18	3.304E−18	5.911E−02
F5	3.304E−18	1.267E−15	3.592E−01
F6	3.304E−18	3.304E−18	3.304E−18
F7	3.304E−18	2.328E−02	7.415E−11
F8	3.304E−18	3.038E−16	2.059E−01
F9	3.304E−18	2.534E−17	1.137E−08
F10	3.304E−18	3.304E−18	9.136E−16
F11	3.304E−18	4.838E−03	4.846E−14
F12	3.304E−18	3.304E−18	9.963E−08
F13	3.304E−18	3.504E−18	2.099E−09
F14	3.304E−18	3.304E−18	2.353E−14
F15	3.304E−18	5.002E−07	4.371E−09
F16	3.304E−18	3.942E−18	1.754E−15
F17	3.304E−18	8.498E−14	5.406E−15
F18	3.304E−18	2.749E−14	2.534E−17
F19	3.304E−18	2.132E−17	3.211E−14
F20	3.558E−12	5.444E−12	1.095E−13
F21	3.304E−18	4.374E−14	1.572E−10
F22	3.304E−18	3.304E−18	1.076E−15
F23	3.304E−18	4.104E−01	1.638E−09
F24	5.288E−18	7.684E−01	2.374E−09
F25	3.304E−18	3.304E−18	3.304E−18
F26	3.297E−18	3.627E−01	1.241E−07
F27	3.304E−18	3.304E−18	3.304E−18
F28	3.304E−18	3.304E−18	3.304E−18
F29	3.304E−18	3.304E−18	2.059E−01
F30	3.304E−18	3.304E−18	3.304E−18

Table A.20
Wilcoxon test results (100D).

Function	TSRL-CMM vs.		
	COLSHADE	HyDE	MadDE
F1	3.304E−18	3.304E−18	3.304E−18
F3	3.304E−18	3.304E−18	3.304E−18
F4	3.304E−18	1.005E−17	3.304E−18
F5	3.304E−18	4.490E−17	3.304E−18
F6	3.304E−18	3.304E−18	8.868E−17
F7	3.304E−18	3.641E−13	4.933E−09
F8	3.304E−18	6.876E−06	7.085E−18
F9	3.304E−18	3.304E−18	3.304E−18
F10	3.304E−18	3.304E−18	3.304E−18
F11	3.304E−18	3.304E−18	3.304E−18
F12	3.304E−18	3.304E−18	3.304E−18
F13	3.304E−18	3.304E−18	3.304E−18
F14	3.304E−18	1.491E−15	3.304E−18
F15	3.304E−18	3.504E−18	3.304E−18
F16	3.304E−18	3.304E−18	3.304E−18
F17	3.304E−18	3.304E−18	2.298E−07
F18	3.304E−18	3.304E−18	3.504E−18
F19	3.304E−18	3.304E−18	4.006E−17
F20	3.942E−18	3.717E−18	4.838E−03
F21	3.304E−18	8.789E−02	1.540E−01
F22	3.304E−18	3.304E−18	2.301E−16
F23	3.304E−18	4.495E−01	1.651E−06
F24	3.304E−18	8.988E−01	6.194E−03
F25	3.304E−18	3.304E−18	3.304E−18
F26	3.304E−18	6.162E−07	1.039E−01
F27	3.304E−18	3.304E−18	3.304E−18
F28	3.304E−18	3.304E−18	3.304E−18
F29	3.304E−18	3.304E−18	3.304E−18
F30	3.304E−18	3.304E−18	3.304E−18

Table A.21
Friedman test results (10D).

Fun	TSRL-CMM	COLSHADE	HyDE	MadDE
F1	2.13725	4.00000	2.86275	1.00000
F3	1.46078	4.00000	3.00000	1.53922
F4	1.11765	4.00000	3.00000	1.88235
F5	2.35294	3.94118	2.70588	1.00000
F6	2.50980	3.98039	2.50980	1.00000
F7	2.15686	4.00000	2.84314	1.00000
F8	2.23529	3.96078	2.80392	1.00000
F9	2.04902	4.00000	2.74510	1.20588
F10	2.86275	3.94118	2.19608	1.00000
F11	2.84314	4.00000	2.15686	1.00000
F12	2.94118	4.00000	2.03922	1.01961
F13	3.00000	4.00000	2.00000	1.00000
F14	2.98039	4.00000	2.01961	1.00000
F15	2.98039	4.00000	2.01961	1.00000
F16	3.15686	3.66667	2.17647	1.00000
F17	2.80392	3.86275	2.33333	1.00000
F18	3.00000	4.00000	2.00000	1.00000
F19	3.00000	4.00000	2.00000	1.00000
F20	3.19608	3.74510	2.05882	1.00000
F21	2.03922	3.62745	2.88235	1.45098
F22	2.29412	4.00000	2.37255	1.33333
F23	2.29412	4.00000	2.70588	1.00000
F24	2.58824	3.37255	2.94118	1.09804
F25	2.68627	4.00000	2.25490	1.05882
F26	2.13725	4.00000	2.58824	1.27451
F27	1.00000	4.00000	2.92157	2.07843
F28	2.27451	3.98039	2.39216	1.35294
F29	3.07843	3.62745	2.09804	1.19608
F30	1.52941	4.00000	2.45098	2.01961

Table A.22
Friedman test results (30D).

Fun	TSRL-CMM	COLSHADE	HyDE	MadDE
F1	1.00000	4.00000	3.00000	2.00000
F3	1.00000	3.31373	3.68627	2.00000
F4	1.00000	4.00000	2.01961	2.98039
F5	2.03922	4.00000	2.90196	1.05882
F6	2.94118	4.00000	2.05882	1.00000
F7	1.94118	4.00000	2.94118	1.11765
F8	1.86275	4.00000	3.00000	1.13725
F9	2.72549	4.00000	2.25490	1.01961
F10	1.35294	3.78431	3.21569	1.64706
F11	2.05882	4.00000	2.56863	1.37255
F12	1.09804	4.00000	3.00000	1.90196
F13	2.03922	4.00000	1.03922	2.92157
F14	2.54902	4.00000	1.01961	2.43137
F15	2.45098	4.00000	1.00000	2.54902
F16	1.94118	4.00000	2.88235	1.17647
F17	2.86275	3.76471	2.35294	1.01961
F18	1.92157	4.00000	1.09804	2.98039
F19	2.17647	4.00000	1.00000	2.82353
F20	3.23529	3.68627	1.56863	1.50980
F21	2.05882	4.00000	2.92157	1.01961
F22	1.80392	3.23529	3.58824	1.37255
F23	2.21569	4.00000	2.70588	1.07843
F24	2.41176	4.00000	2.56863	1.01961
F25	1.01961	4.00000	3.00000	1.98039
F26	2.01961	4.00000	2.74510	1.23529
F27	1.03922	4.00000	2.05882	2.90196
F28	1.74510	4.00000	2.84314	1.41176
F29	2.13725	4.00000	2.52941	1.33333
F30	1.01961	4.00000	2.05882	2.92157

Table A.23
Friedman test results (50D).

Fun	TSRL-CMM	COLSHADE	HyDE	MadDE
F1	1.13725	4.00000	3.00000	1.86275
F3	1.00000	3.19608	3.80392	2.00000
F4	1.33333	4.00000	3.00000	1.66667
F5	1.47059	4.00000	2.96078	1.56863
F6	3.00000	4.00000	2.00000	1.00000
F7	2.21569	4.00000	2.68627	1.09804
F8	1.45098	4.00000	2.98039	1.56863
F9	2.82353	4.00000	1.03922	2.13725
F10	1.00000	3.43137	3.56863	2.00000
F11	1.41176	4.00000	1.68627	2.90196
F12	1.19608	4.00000	3.00000	1.80392
F13	1.19608	4.00000	3.00000	1.80392
F14	2.05882	4.00000	1.00000	2.94118
F15	1.94118	4.00000	1.23529	2.82353
F16	1.94118	4.00000	3.00000	1.05882
F17	2.03922	4.00000	2.92157	1.03922
F18	1.07843	4.00000	1.94118	2.98039
F19	2.03922	4.00000	1.01961	2.94118
F20	2.11765	3.39216	3.39216	1.09804
F21	1.96078	4.00000	2.90196	1.13725
F22	1.96078	3.43137	3.56863	1.03922
F23	2.29412	4.00000	2.54902	1.15686
F24	2.35294	4.00000	2.50980	1.13725
F25	1.00000	4.00000	2.92157	2.07843
F26	2.35294	4.00000	2.45098	1.19608
F27	1.00000	4.00000	2.00000	3.00000
F28	1.00000	4.00000	2.90196	2.09804
F29	1.56863	4.00000	3.00000	1.43137
F30	1.00000	4.00000	2.39216	2.60784

Table A.24
Friedman test results (100D).

Fun	TSRL-CMM	COLSHADE	HyDE	MadDE
F1	1.00000	4.00000	2.00000	3.00000
F3	1.00000	3.01961	3.98039	2.00000
F4	1.01961	4.00000	1.98039	3.00000
F5	1.03922	4.00000	1.96078	3.00000
F6	3.00000	4.00000	1.00000	2.00000
F7	2.72549	4.00000	1.09804	2.17647
F8	1.27451	4.00000	1.72549	3.00000
F9	2.00000	4.00000	1.00000	3.00000
F10	1.00000	3.17647	3.82353	2.00000
F11	1.00000	4.00000	2.00000	3.00000
F12	1.00000	4.00000	2.80392	2.19608
F13	1.00000	4.00000	2.33333	2.66667
F14	1.03922	4.00000	1.96078	3.00000
F15	1.00000	4.00000	2.00000	3.00000
F16	1.00000	4.00000	3.00000	2.00000
F17	1.21569	4.00000	3.00000	1.78431
F18	1.00000	4.00000	3.00000	2.00000
F19	1.01961	4.00000	3.00000	1.98039
F20	1.33333	3.60784	3.39216	1.66667
F21	1.98039	4.00000	2.62745	1.39216
F22	1.01961	3.54902	3.45098	1.98039
F23	2.39216	4.00000	2.39216	1.21569
F24	2.19608	4.00000	2.41176	1.39216
F25	1.00000	4.00000	2.00000	3.00000
F26	2.19608	4.00000	1.21569	2.58824
F27	1.00000	4.00000	2.00000	3.00000
F28	1.00000	4.00000	3.00000	2.00000
F29	1.00000	4.00000	3.00000	2.00000
F30	1.00000	4.00000	3.00000	2.00000

Table A.25
Comparison of optimization algorithms on benchmark functions (10D).

Function	Metric	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	Mean	**1.444E−01**	6.322E+02	1.288E+03	2.980E+02	1.641E+03	1.071E+00
	Std	3.870E−03	1.604E−03	2.915E+03	6.024E−05	2.158E+04	5.895E−05
F3	Mean	4.347E−14	6.081E−14	5.239E−14	5.239E−14	**3.790E−14**	4.904E−14
	Std	4.663E−15	6.755E−15	5.629E−15	7.881E−15	6.107E−15	6.755E−15
F4	Mean	8.582E−14	7.022E−14	6.019E−14	6.019E−14	**3.121E−14**	7.022E−14
	Std	1.441E−14	1.238E−14	1.238E−14	9.007E−15	1.743E−01	1.190E−14
F5	Mean	**2.848E+01**	2.876E+01	3.718E+01	3.215E+01	2.940E+01	3.664E+01
	Std	1.126E−13	1.689E−08	8.331E−04	2.108E−04	1.017E+00	2.410E−04
F6	Mean	**3.325E−01**	2.527E+00	4.212E+00	4.823E+00	1.660E+00	2.732E+00
	Std	1.531E−12	1.103E−13	1.013E−13	1.090E−13	1.138E−04	1.080E−13
F7	Mean	**3.199E+01**	3.515E+01	3.910E+01	3.734E+01	3.252E+01	3.999E+01
	Std	8.106E−14	8.332E−14	9.234E−14	8.332E−14	2.419E−01	7.486E−14
F8	Mean	**2.468E+01**	2.565E+01	2.991E+01	2.811E+01	2.702E+01	2.903E+01
	Std	1.316E−07	4.549E−08	1.191E−06	2.939E−07	6.737E−01	2.368E−07
F9	Mean	5.293E−01	1.305E+00	3.162E+00	4.382E+00	**4.706E−01**	5.777E+00
	Std	5.855E−14	4.954E−14	4.504E−14	6.530E−14	5.275E−14	3.378E−14
F10	Mean	**7.096E+02**	7.381E+02	8.120E+02	8.516E+02	8.002E+02	8.451E+02
	Std	3.451E−04	7.779E−01	9.611E−03	3.764E−01	3.196E+01	9.215E−01
F11	Mean	**1.385E+01**	2.795E+01	3.873E+01	2.910E+01	2.219E+01	2.789E+01
	Std	3.639E−08	1.995E−13	7.978E−04	1.982E−13	1.568E+00	2.080E−03
F12	Mean	**5.804E+03**	6.867E+03	8.183E+03	8.232E+03	1.209E+04	7.576E+03
	Std	7.818E−01	3.014E+00	1.569E+01	2.095E−02	7.324E+04	2.956E−01
F13	Mean	**1.687E+02**	2.393E+02	2.079E+02	2.160E+02	2.220E+02	1.889E+02
	Std	4.353E−03	2.352E−05	5.132E−03	2.701E−06	9.213E+02	2.056E−02
F14	Mean	5.345E+01	6.225E+01	7.506E+01	7.386E+01	**4.782E+01**	6.955E+01
	Std	6.533E+01	1.249E−02	4.472E−03	3.781E−03	3.730E+01	4.600E−06
F15	Mean	**4.618E+01**	5.854E+01	6.264E+01	6.719E+01	5.274E+01	4.947E+01
	Std	1.233E+01	1.261E−02	4.855E−01	7.072E−07	3.805E+00	2.003E−02
F16	Mean	**8.583E+01**	1.126E+02	1.727E+02	1.402E+02	1.485E+02	1.457E+02
	Std	5.327E−03	2.206E−02	2.156E−02	3.201E−02	5.396E−01	2.559E−02
F17	Mean	**4.630E+01**	5.737E+01	6.821E+01	5.879E+01	6.678E+01	6.233E+01
	Std	3.085E−09	3.098E−13	7.764E−07	1.350E−05	3.064E+00	2.878E−07
F18	Mean	1.475E+02	1.353E+02	9.425E+01	7.157E+02	6.201E+02	**1.140E+02**
	Std	3.154E−02	3.366E−03	1.096E−04	7.748E+00	1.248E+04	3.704E−03
F19	Mean	**1.841E+01**	3.838E+01	3.167E+01	3.272E+01	3.993E+01	3.602E+01
	Std	1.240E−01	6.357E−07	1.236E−04	2.531E−04	1.051E+04	1.242E−02
F20	Mean	**8.025E+01**	8.231E+01	9.868E+01	8.058E+01	8.792E+01	9.212E+01
	Std	1.892E−02	4.128E−13	3.861E−13	4.388E−11	1.018E+01	3.614E−03
F21	Mean	**1.000E+02**	1.561E+02	1.692E+02	1.651E+02	1.762E+02	1.598E+02
	Std	1.584E−13	1.354E−04	1.747E−06	3.793E−05	4.754E−01	2.424E−05
F22	Mean	1.000E+02	1.022E+02	1.019E+02	1.093E+02	1.007E+02	**9.787E+01**
	Std	1.621E−13	4.594E−13	4.802E−13	4.432E−13	5.690E−02	4.684E−13
F23	Mean	**2.000E+02**	3.290E+02	3.273E+02	3.289E+02	3.249E+02	3.255E+02
	Std	1.801E−14	2.870E−02	1.773E−02	7.355E−03	6.124E−01	4.674E−03
F24	Mean	**1.941E+02**	3.230E+02	3.378E+02	3.309E+02	3.567E+02	3.500E+02
	Std	1.028E−14	1.159E−03	5.362E−04	3.948E−03	8.779E−01	2.390E−03
F25	Mean	**2.000E+02**	4.205E+02	4.238E+02	4.187E+02	4.246E+02	4.281E+02
	Std	0.000E+00	2.882E−13	1.797E−05	3.062E−13	5.984E−08	1.087E−03
F26	Mean	**1.922E+02**	3.276E+02	3.400E+02	3.597E+02	3.495E+02	3.598E+02
	Std	0.000E+00	2.452E−02	1.927E−02	8.903E−03	1.929E+00	7.991E−03
F27	Mean	**2.494E+02**	3.745E+02	3.737E+02	3.733E+02	3.737E+02	3.743E+02
	Std	5.657E+00	8.403E−01	6.699E−01	1.069E+00	3.746E+00	7.257E−01
F28	Mean	**2.000E+02**	3.870E+02	3.960E+02	3.978E+02	3.856E+02	3.751E+02
	Std	0.000E+00	1.313E+01	1.557E+01	2.194E+01	2.986E+01	1.803E+01
F29	Mean	**2.164E+02**	3.128E+02	3.284E+02	3.085E+02	2.873E+02	3.165E+02
	Std	5.916E−01	2.628E−01	2.007E−01	1.423E−01	1.135E+01	1.198E−01
F30	Mean	**7.435E+02**	2.864E+03	1.080E+03	2.558E+03	1.700E+03	1.445E+03
	Std	4.920E+03	1.376E+05	1.137E+05	1.607E+05	7.410E+05	7.891E+04

Table A.26
Comparison of optimization algorithms on benchmark functions (30D).

Function	Metric	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	Mean	**4.011E+00**	4.083E+01	1.344E+02	3.360E+02	3.820E+02	1.799E+02
	Std	1.100E−03	4.256E−05	4.498E−05	2.259E−06	1.163E+03	1.536E−05
F3	Mean	**3.109E−12**	4.772E−12	4.621E−12	3.227E−12	2.957E−06	3.548E−12
	Std	4.923E−14	5.340E−14	5.498E−14	4.684E−14	5.221E−13	5.008E−14
F4	Mean	4.797E+01	5.641E+01	5.221E+01	**4.570E+01**	4.895E+01	5.869E+01
	Std	4.056E−01	3.953E−01	2.186E−01	3.173E−01	2.786E+00	4.671E−01
F5	Mean	**1.628E+02**	1.680E+02	1.850E+02	1.896E+02	1.719E+02	1.968E+02
	Std	2.948E−03	1.031E−04	1.007E−04	2.154E−03	2.254E−01	7.234E−04
F6	Mean	3.322E+01	3.158E+01	3.679E+01	3.703E+01	**2.738E+01**	3.629E+01
	Std	5.825E−05	7.286E−05	1.156E−06	3.891E−09	1.460E−01	2.802E−06
F7	Mean	1.639E+02	2.180E+02	2.325E+02	2.464E+02	**1.487E+02**	2.502E+02
	Std	1.470E−13	1.501E−13	1.710E−13	1.855E−13	9.020E−01	1.521E−13
F8	Mean	**1.378E+02**	1.436E+02	1.420E+02	1.490E+02	1.520E+02	1.508E+02
	Std	2.414E−08	8.155E−06	1.023E−03	1.316E−06	1.675E−13	1.453E−04
F9	Mean	3.978E+02	1.519E+03	1.546E+03	1.621E+03	**3.415E+02**	1.674E+03
	Std	9.777E−12	1.260E−01	1.343E−12	5.935E−03	1.150E+01	1.669E−01
F10	Mean	**3.507E+03**	3.829E+03	3.615E+03	3.685E+03	3.843E+03	3.716E+03
	Std	8.296E+00	1.130E+01	9.465E+00	4.841E+00	6.522E+01	5.299E+00
F11	Mean	**1.174E+02**	1.365E+02	1.379E+02	1.296E+02	1.205E+02	1.367E+02
	Std	1.013E−01	1.441E−01	9.368E−02	2.620E−01	5.804E+00	8.054E−02
F12	Mean	**1.511E+05**	2.114E+05	2.078E+05	1.879E+05	6.277E+05	1.504E+05
	Std	8.603E+03	1.952E+03	3.889E+03	2.229E+03	1.014E+06	7.063E+03
F13	Mean	**5.306E+03**	9.184E+03	8.563E+03	9.055E+03	1.127E+04	8.993E+03
	Std	1.782E−03	2.147E−02	5.443E−02	3.295E−02	2.587E+04	3.575E−02
F14	Mean	2.859E+02	**2.241E+02**	2.444E+02	2.359E+02	5.377E+02	2.503E+02
	Std	4.856E−02	3.151E−02	4.491E−02	1.493E−03	3.570E+02	1.160E−01
F15	Mean	1.394E+03	1.694E+03	1.616E+03	2.116E+03	2.484E+03	**1.389E+03**
	Std	2.686E−01	1.385E−02	4.772E−02	4.054E−01	2.983E+02	3.490E−02
F16	Mean	1.022E+03	1.045E+03	1.039E+03	1.105E+03	**1.017E+03**	1.055E+03
	Std	3.510E−01	1.938E−01	1.075E+00	4.056E−01	7.383E+00	3.766E−01
F17	Mean	5.657E+02	**5.037E+02**	5.830E+02	5.708E+02	5.203E+02	6.025E+02
	Std	2.675E−03	5.682E−02	2.416E−01	9.815E−02	1.021E+00	3.130E−01
F18	Mean	5.348E+03	4.885E+03	6.429E+03	1.590E+04	2.951E+04	**3.494E+03**
	Std	2.479E−03	5.495E−04	3.975E−05	1.968E+01	3.652E+04	8.184E−04
F19	Mean	6.509E+02	**2.900E+02**	3.920E+02	7.909E+02	9.057E+02	5.781E+02
	Std	1.167E−01	1.067E−01	4.930E−03	7.415E−03	1.514E+03	1.557E−02
F20	Mean	**5.334E+02**	5.964E+02	6.096E+02	6.017E+02	5.593E+02	5.831E+02
	Std	2.698E−01	6.661E−02	2.214E−01	1.127E−01	1.959E+01	2.178E−02
F21	Mean	3.575E+02	**3.536E+02**	3.807E+02	3.586E+02	3.787E+02	3.724E+02
	Std	2.910E−02	2.209E−02	3.363E−02	3.956E−02	1.013E+00	4.838E−02
F22	Mean	1.734E+03	**1.558E+03**	2.055E+03	2.102E+03	2.196E+03	2.128E+03
	Std	1.297E+00	2.347E+00	6.299E+00	1.925E+00	4.941E+01	3.316E+00
F23	Mean	5.638E+02	**5.514E+02**	6.056E+02	5.730E+02	5.874E+02	5.783E+02
	Std	3.403E−01	2.236E−01	1.436E−01	1.382E−01	2.161E+00	1.143E−01
F24	Mean	**6.203E+02**	6.393E+02	6.362E+02	6.598E+02	6.536E+02	6.501E+02
	Std	3.719E−02	2.752E−02	3.999E−02	9.843E−02	2.925E+00	2.441E−01
F25	Mean	3.793E+02	3.789E+02	3.785E+02	3.796E+02	**3.777E+02**	3.790E+02
	Std	9.596E−02	1.070E−01	8.854E−02	1.376E−01	1.380E+00	1.696E−01
F26	Mean	**2.300E+03**	2.331E+03	2.803E+03	2.315E+03	2.427E+03	2.563E+03
	Std	9.578E−01	1.095E+00	1.228E+00	3.323E−01	1.900E+01	1.124E+00
F27	Mean	**4.973E+02**	5.000E+02	4.979E+02	5.000E+02	4.992E+02	4.981E+02
	Std	7.482E+01	7.032E+01	7.361E+01	7.056E+01	1.413E+02	7.927E+01
F28	Mean	**3.804E+02**	3.895E+02	3.912E+02	4.019E+02	3.899E+02	3.927E+02
	Std	4.518E+01	6.888E+01	3.737E+01	1.428E+02	2.442E+02	7.553E+01
F29	Mean	8.082E+02	7.535E+02	7.747E+02	7.445E+02	8.327E+02	**7.423E+02**
	Std	5.948E+00	7.605E+00	6.499E+00	3.848E+00	7.859E+01	3.231E+00
F30	Mean	**4.715E+03**	5.798E+03	7.841E+03	6.928E+03	9.228E+03	5.894E+03
	Std	2.530E+04	6.701E+04	2.584E+04	5.569E+04	1.074E+06	7.102E+04

Table A.27
Comparison of optimization algorithms on benchmark functions (50D).

Function	Metric	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	Mean	**2.045E+03**	3.696E+03	4.974E+03	3.904E+03	2.219E+03	3.179E+03
	Std	5.643E−03	3.850E+01	7.031E+03	3.305E−06	3.530E+01	1.497E−06
F3	Mean	4.343E−07	4.556E−08	4.475E−08	8.415E−08	3.344E−05	**2.473E−08**
	Std	3.785E−09	1.747E−12	7.605E−13	3.537E−13	3.199E−12	6.274E−12
F4	Mean	7.577E+01	**7.437E+01**	9.538E+01	1.058E+02	8.798E+01	9.780E+01
	Std	6.187E−01	5.176E−01	9.005E−01	8.151E−01	7.737E+00	7.516E−01
F5	Mean	3.120E+02	3.157E+02	3.263E+02	3.308E+02	**3.068E+02**	3.341E+02
	Std	1.379E−13	1.115E−13	6.133E−03	7.641E−05	2.304E−03	5.476E−05
F6	Mean	**4.385E+01**	4.626E+01	4.945E+01	4.815E+01	4.665E+01	4.859E+01
	Std	1.013E−06	3.849E−05	3.723E−07	1.207E−09	1.262E−01	1.007E−04
F7	Mean	3.266E+02	5.526E+02	5.579E+02	5.531E+02	**3.083E+02**	5.583E+02
	Std	1.651E−13	2.062E−13	2.555E−13	2.101E−13	2.521E−13	2.501E−13
F8	Mean	3.154E+02	**3.021E+02**	3.154E+02	3.276E+02	3.167E+02	3.206E+02
	Std	6.025E−06	5.041E−05	2.144E−05	3.069E−04	1.278E+00	5.229E−03
F9	Mean	2.971E+03	7.005E+03	6.563E+03	7.005E+03	**2.030E+03**	5.993E+03
	Std	4.319E−09	8.706E−01	1.156E−02	5.913E−02	1.509E+02	8.323E−01
F10	Mean	**6.325E+03**	6.455E+03	6.584E+03	6.459E+03	6.578E+03	6.439E+03
	Std	1.168E+01	8.013E+00	4.457E+00	9.711E+00	1.835E+02	8.449E+00
F11	Mean	**2.265E+02**	2.425E+02	2.354E+02	**2.265E+02**	2.300E+02	2.274E+02
	Std	1.027E−01	1.127E−01	7.843E−02	7.924E−02	1.741E+00	1.241E−01
F12	Mean	**8.049E+05**	1.330E+06	1.432E+06	1.414E+06	1.387E+06	1.829E+06
	Std	3.071E+03	3.948E+06	4.213E+05	2.297E+06	1.661E+06	1.081E+04
F13	Mean	8.567E+03	1.116E+04	9.688E+03	**7.873E+03**	4.133E+04	8.466E+03
	Std	2.076E−02	9.060E−02	3.177E−02	1.772E−01	3.190E+05	1.066E−01
F14	Mean	1.392E+04	**9.207E+03**	1.294E+04	9.309E+03	2.454E+04	1.636E+04
	Std	2.680E+00	1.942E+01	3.992E+01	2.046E+01	3.908E+04	1.005E+02
F15	Mean	6.019E+03	**5.460E+03**	5.759E+03	9.635E+03	6.344E+04	7.994E+03
	Std	9.476E−03	1.189E−02	1.224E−01	6.569E−02	3.537E+05	1.832E−02
F16	Mean	1.830E+03	1.858E+03	1.931E+03	**1.767E+03**	1.771E+03	1.824E+03
	Std	1.727E+00	1.850E+00	1.714E+00	2.240E+00	2.188E+01	1.521E+00
F17	Mean	1.418E+03	1.519E+03	1.631E+03	1.563E+03	**1.417E+03**	1.630E+03
	Std	1.108E−01	2.060E−01	3.388E−01	1.184E−02	1.519E+01	5.239E−01
F18	Mean	5.172E+04	7.276E+04	**4.956E+04**	1.134E+05	2.411E+05	6.397E+04
	Std	9.750E−01	7.307E+03	5.729E+03	1.067E+03	3.457E+05	9.568E+02
F19	Mean	2.781E+03	2.346E+03	**2.241E+03**	4.637E+03	1.440E+04	2.775E+03
	Std	1.804E−05	1.301E−03	6.019E−03	8.414E−02	6.183E+04	3.570E−03
F20	Mean	1.221E+03	1.165E+03	1.262E+03	**1.151E+03**	1.243E+03	1.244E+03
	Std	1.507E+00	6.895E−01	4.277E−01	6.086E−01	6.624E+01	5.377E−01
F21	Mean	**5.400E+02**	5.498E+02	5.931E+02	5.554E+02	5.527E+02	5.806E+02
	Std	2.578E−03	1.598E−03	4.902E−03	7.488E−03	1.327E+00	2.316E−02
F22	Mean	**6.669E+03**	6.902E+03	7.083E+03	6.892E+03	7.270E+03	7.137E+03
	Std	1.154E+01	1.148E+01	1.219E+01	7.911E+00	1.444E+02	5.371E+00
F23	Mean	9.039E+02	**8.998E+02**	9.488E+02	9.640E+02	9.131E+02	9.420E+02
	Std	1.839E−01	3.164E−01	1.586E−01	4.106E−01	8.796E+00	1.650E−01
F24	Mean	**9.367E+02**	9.445E+02	1.018E+03	9.650E+02	1.025E+03	9.924E+02
	Std	1.262E−01	2.035E−01	3.727E−01	9.386E−02	3.723E+00	1.354E−01
F25	Mean	4.698E+02	4.637E+02	**4.570E+02**	4.645E+02	4.681E+02	4.653E+02
	Std	1.082E+00	6.519E−01	9.025E−01	6.428E−01	7.151E+00	4.852E−01
F26	Mean	**4.071E+03**	5.366E+03	4.848E+03	4.797E+03	4.796E+03	4.612E+03
	Std	3.522E−01	5.412E−02	3.512E−02	4.485E−01	4.950E+01	5.256E−02
F27	Mean	**5.000E+02**	**5.000E+02**	**5.000E+02**	**5.000E+02**	**5.000E+02**	**5.000E+02**
	Std	9.916E+01	1.743E+02	1.543E+02	1.342E+02	2.353E+02	1.737E+02
F28	Mean	**4.891E+02**	4.923E+02	4.986E+02	4.948E+02	4.929E+02	4.970E+02
	Std	5.164E+02	6.644E+02	7.417E+02	6.728E+02	2.012E+03	4.268E+02
F29	Mean	1.218E+03	1.187E+03	**1.136E+03**	1.171E+03	1.500E+03	1.180E+03
	Std	1.816E+01	1.052E+01	1.230E+01	9.998E+00	2.591E+02	5.846E+00
F30	Mean	**3.591E+04**	4.558E+04	4.327E+04	5.066E+04	9.238E+04	4.471E+04
	Std	4.022E+05	9.161E+05	7.222E+05	1.125E+06	9.886E+06	1.478E+06

Table A.28
Comparison of optimization algorithms on benchmark functions (100D).

Function	Metric	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	Mean	6.888E+03	**6.372E+03**	8.518E+03	7.503E+03	6.384E+03	8.772E+03
	Std	1.469E−03	1.142E−06	1.321E−06	1.225E−04	6.184E+02	6.926E−06
F3	Mean	3.890E+00	5.875E+00	**7.484E−01**	2.079E+00	5.306E+02	8.019E−01
	Std	9.311E−04	7.500E−08	5.364E−09	3.358E−08	2.011E−10	2.055E−07
F4	Mean	**2.364E+02**	2.377E+02	2.429E+02	**2.364E+02**	2.391E+02	2.431E+02
	Std	1.032E+00	7.565E−01	3.160E−01	9.617E−01	6.642E+00	6.679E−01
F5	Mean	**7.653E+02**	7.773E+02	8.071E+02	8.068E+02	7.924E+02	8.087E+02
	Std	8.632E−04	8.619E−04	3.992E−03	6.308E−04	3.206E−03	1.364E−04
F6	Mean	**5.297E+01**	5.463E+01	5.577E+01	5.652E+01	5.685E+01	5.663E+01
	Std	1.186E−07	6.043E−07	3.288E−10	4.195E−04	2.199E−01	3.005E−09
F7	Mean	**1.018E+03**	1.653E+03	1.736E+03	1.768E+03	9.080E+02	1.730E+03
	Std	3.402E−13	4.315E−13	4.301E−13	4.479E−13	3.173E−13	4.837E−13
F8	Mean	**8.534E+02**	8.768E+02	9.153E+02	8.856E+02	8.842E+02	9.021E+02
	Std	5.605E−03	2.365E−03	7.995E−03	2.534E−03	4.516E−04	1.324E−03
F9	Mean	1.372E+04	1.915E+04	1.910E+04	1.930E+04	**1.224E+04**	1.971E+04
	Std	2.248E−08	2.513E−06	9.470E−01	6.457E−04	3.794E+03	6.554E−08
F10	Mean	1.438E+04	**1.432E+04**	1.474E+04	1.460E+04	1.490E+04	1.453E+04
	Std	1.066E+01	1.027E+01	7.906E+00	1.638E+01	2.165E+02	4.835E+00
F11	Mean	1.348E+03	1.371E+03	1.345E+03	1.300E+03	**1.255E+03**	1.292E+03
	Std	6.022E−01	2.257E−01	4.236E−01	8.427E−01	4.260E+04	4.566E−01
F12	Mean	**3.400E+06**	1.328E+07	9.500E+06	1.328E+07	1.633E+07	1.106E+07
	Std	2.030E+03	2.557E+05	1.035E+06	2.735E+05	8.891E+06	1.664E+06
F13	Mean	1.140E+04	1.149E+04	1.479E+04	1.333E+04	**1.085E+04**	1.258E+04
	Std	7.214E+03	1.782E−02	4.078E+02	1.802E−01	2.276E+01	4.200E−03
F14	Mean	5.096E+03	4.282E+03	**3.754E+03**	6.192E+03	1.423E+05	4.092E+03
	Std	1.093E−01	5.947E−02	5.374E−03	2.319E−01	1.105E+05	9.869E−02
F15	Mean	**3.732E+03**	4.638E+03	5.970E+03	4.032E+03	2.940E+04	6.429E+03
	Std	5.426E−01	2.677E−01	1.796E+00	5.839E−01	1.803E+05	3.809E+04
F16	Mean	**3.941E+03**	4.089E+03	4.019E+03	4.060E+03	4.311E+03	4.174E+03
	Std	2.469E+00	6.412E+00	4.737E+00	2.510E+00	5.556E+01	6.265E+00
F17	Mean	**3.399E+03**	3.425E+03	3.711E+03	3.674E+03	3.556E+03	3.515E+03
	Std	3.224E+00	1.831E+00	4.431E−01	2.695E+00	4.523E+01	1.061E+00
F18	Mean	**7.446E+03**	1.142E+04	3.416E+04	1.908E+04	1.611E+05	2.484E+04
	Std	2.627E−01	1.783E+00	6.449E+02	9.716E−03	1.250E+05	1.728E−04
F19	Mean	3.091E+03	4.712E+03	3.277E+03	3.847E+03	**3.003E+03**	3.314E+03
	Std	5.875E−04	2.937E+00	2.816E−03	6.575E−03	6.513E+00	1.380E−02
F20	Mean	3.140E+03	**3.080E+03**	3.279E+03	3.153E+03	3.465E+03	3.260E+03
	Std	1.681E+00	1.857E+00	9.487E−01	2.244E+00	1.386E+02	7.175E−01
F21	Mean	**1.217E+03**	1.234E+03	1.317E+03	1.300E+03	1.263E+03	1.279E+03
	Std	1.445E−01	3.613E−02	1.193E−01	3.992E−02	1.096E+00	9.758E−02
F22	Mean	**1.564E+04**	1.590E+04	1.596E+04	**1.564E+04**	1.613E+04	1.599E+04
	Std	1.504E+01	1.452E+01	1.892E+01	2.087E+01	1.894E+02	2.026E+01
F23	Mean	1.708E+03	1.615E+03	**1.574E+03**	1.676E+03	1.863E+03	1.658E+03
	Std	3.885E−01	3.299E−01	4.671E−01	7.308E−01	7.799E+00	1.649E−01
F24	Mean	1.908E+03	**1.826E+03**	1.868E+03	1.895E+03	2.016E+03	1.897E+03
	Std	4.180E−01	2.851E−01	3.295E−01	5.475E−01	1.831E+01	4.015E−01
F25	Mean	8.061E+02	8.012E+02	8.014E+02	8.076E+02	8.309E+02	**7.981E+02**
	Std	1.778E+00	1.104E+00	1.735E+00	2.329E+00	1.125E+01	1.047E+00
F26	Mean	**1.555E+04**	1.753E+04	1.582E+04	1.726E+04	1.660E+04	1.611E+04
	Std	6.194E−01	2.054E+00	7.828E−01	2.268E+00	1.651E+01	7.216E−01
F27	Mean	**5.000E+02**	**5.000E+02**	**5.000E+02**	**5.000E+02**	**5.000E+02**	**5.000E+02**
	Std	2.068E+02	5.146E+02	4.816E+02	4.821E+02	6.554E+02	4.541E+02
F28	Mean	**5.398E+02**	5.486E+02	5.513E+02	5.508E+02	5.906E+02	5.547E+02
	Std	1.696E+00	1.225E+00	1.243E+00	1.040E+00	4.075E+02	1.034E+00
F29	Mean	**3.384E+03**	3.602E+03	3.522E+03	3.460E+03	4.290E+03	3.496E+03
	Std	2.546E+01	2.454E+01	2.272E+01	2.295E+01	4.289E+02	3.473E+01
F30	Mean	6.129E+04	5.440E+04	5.247E+04	4.577E+04	1.941E+05	**3.184E+04**
	Std	1.495E+05	7.191E+05	4.362E+05	2.392E+05	5.300E+06	2.819E+05

Table A.29
Wilcoxon test results (10D).

Function	AEAS vs.				
	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	8.592E−04	1.881E−04	3.797E−04	2.208E−12	2.448E−04
F3	NAN	NAN	NAN	NAN	NAN
F4	3.567E−09	1.252E−09	1.252E−09	1.252E−09	3.567E−09
F5	6.016E−01	1.441E−03	3.202E−01	9.840E−01	4.884E−03
F6	3.190E−06	1.051E−08	3.648E−07	2.524E−01	3.584E−05
F7	6.870E−02	5.257E−03	1.430E−02	2.903E−01	5.962E−04
F8	3.953E−01	3.401E−01	5.162E−01	9.786E−01	5.314E−01
F9	9.184E−01	5.490E−01	9.627E−01	6.456E−01	7.423E−01
F10	4.822E−01	6.472E−02	2.815E−02	8.422E−02	2.208E−02
F11	3.844E−06	3.534E−10	1.479E−05	4.429E−03	2.163E−06
F12	3.991E−01	1.540E−01	6.972E−02	3.013E−02	2.812E−01
F13	1.950E−02	9.041E−02	1.851E−01	7.181E−02	7.735E−01
F14	2.208E−02	1.363E−04	1.478E−04	9.298E−02	1.316E−03
F15	9.429E−02	2.232E−01	2.287E−02	5.337E−01	7.531E−01
F16	6.539E−01	7.430E−05	8.422E−02	1.942E−01	3.864E−02
F17	2.035E−01	5.266E−04	6.186E−02	2.170E−02	6.570E−02
F18	4.189E−02	1.227E−03	5.822E−02	1.785E−01	2.021E−02
F19	1.096E−04	1.658E−03	3.341E−04	3.679E−02	2.036E−04
F20	9.147E−01	5.147E−02	6.442E−01	7.941E−01	2.309E−01
F21	8.323E−11	1.761E−12	6.411E−13	3.262E−11	8.327E−11
F22	1.391E−20	1.391E−20	4.029E−19	4.029E−19	2.279E−16
F23	1.391E−20	1.391E−20	1.391E−20	1.391E−20	1.391E−20
F24	1.521E−11	9.740E−14	1.293E−12	9.337E−19	3.845E−16
F25	3.970E−19	1.367E−20	3.987E−19	1.337E−20	1.376E−20
F26	1.035E−14	1.561E−15	1.508E−17	3.456E−16	6.392E−17
F27	2.944E−07	3.406E−07	3.797E−07	3.662E−07	2.944E−07
F28	7.788E−21	8.959E−21	9.517E−21	2.838E−19	2.358E−19
F29	7.409E−15	2.100E−16	2.929E−15	2.938E−13	1.991E−15
F30	1.850E−01	5.877E−01	9.893E−01	9.947E−01	9.626E−01

Table A.30
Wilcoxon test results (30D).

Function	AEAS vs.				
	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	4.328E−05	1.545E−07	7.858E−05	4.065E−08	6.275E−09
F3	3.286E−01	2.960E−01	6.752E−01	8.191E−07	6.716E−01
F4	1.072E−07	5.338E−06	6.263E−04	8.822E−06	1.723E−07
F5	8.543E−02	2.175E−03	3.608E−04	4.326E−02	8.995E−07
F6	1.254E−01	7.432E−04	3.093E−04	3.385E−01	9.975E−03
F7	3.679E−02	6.419E−04	8.543E−05	6.639E−05	2.147E−04
F8	4.538E−02	1.848E−02	3.599E−03	1.171E−03	5.961E−04
F9	2.298E−07	7.400E−10	5.293E−08	2.558E−07	3.018E−10
F10	2.415E−01	9.466E−01	7.078E−01	5.647E−02	5.559E−01
F11	8.304E−01	5.740E−01	9.520E−01	2.415E−01	4.991E−01
F12	2.842E−01	4.988E−02	1.237E−01	3.444E−02	5.831E−01
F13	6.782E−01	5.740E−01	7.992E−01	4.834E−02	6.929E−01
F14	1.321E−01	1.000E+00	9.520E−01	2.753E−01	6.635E−01
F15	7.430E−05	3.428E−04	5.363E−07	3.235E−12	6.829E−05
F16	9.573E−01	6.635E−01	4.066E−01	6.299E−01	8.724E−01
F17	5.911E−02	9.947E−01	7.531E−01	2.107E−01	9.200E−01
F18	2.157E−01	6.880E−01	1.783E−04	5.149E−03	7.889E−01
F19	5.924E−01	9.413E−01	9.169E−02	2.937E−04	7.633E−01
F20	1.011E−01	4.684E−02	9.298E−02	3.953E−01	2.207E−01
F21	4.726E−12	1.814E−14	2.430E−12	8.941E−14	1.041E−13
F22	2.046E−01	7.480E−03	5.156E−03	4.407E−03	4.005E−03
F23	1.873E−01	1.286E−03	1.919E−01	4.056E−02	8.067E−02
F24	4.219E−01	6.782E−01	3.991E−02	1.873E−01	7.181E−02
F25	6.880E−01	8.829E−01	3.154E−01	4.535E−01	6.157E−01
F26	4.140E−03	6.070E−05	1.682E−02	1.929E−03	1.538E−03
F27	6.299E−01	6.831E−01	1.409E−01	8.356E−01	6.782E−01
F28	9.652E−01	8.299E−01	8.221E−01	4.876E−01	8.904E−01
F29	7.480E−01	2.580E−01	8.777E−01	2.962E−02	7.480E−01
F30	9.786E−01	2.362E−01	5.381E−01	4.887E−04	6.251E−01

Table A.31
Wilcoxon test results (50D).

Function	AEAS vs.				
	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	2.431E−03	5.067E−02	7.253E−04	2.965E−01	1.919E−01
F3	5.860E−02	5.162E−01	2.772E−03	7.909E−18	8.882E−01
F4	8.724E−01	2.021E−02	1.400E−04	3.276E−02	2.715E−03
F5	4.142E−01	6.280E−02	2.720E−02	6.110E−01	8.533E−03
F6	1.559E−01	7.615E−04	1.255E−02	1.127E−01	3.756E−03
F7	5.981E−12	5.676E−13	3.015E−11	4.759E−02	1.365E−12
F8	7.329E−01	6.002E−02	7.427E−03	2.182E−01	2.674E−02
F9	2.841E−13	2.881E−11	1.124E−12	2.815E−02	1.051E−08
F10	2.469E−01	4.028E−01	1.680E−01	6.347E−01	1.502E−01
F11	1.000E+00	5.694E−01	2.753E−01	2.996E−01	1.851E−01
F12	4.266E−03	1.363E−04	6.583E−03	3.879E−01	1.944E−03
F13	5.119E−01	7.531E−01	1.579E−01	9.298E−02	7.379E−01
F14	5.831E−01	4.396E−02	2.541E−03	8.309E−05	2.724E−01
F15	8.461E−01	7.992E−01	3.163E−03	1.098E−02	4.466E−02
F16	8.200E−01	2.309E−01	5.469E−01	4.991E−01	8.671E−01
F17	4.781E−01	1.950E−02	1.722E−01	7.941E−01	2.248E−02
F18	3.027E−01	3.169E−02	8.882E−01	3.257E−04	2.469E−01
F19	2.812E−01	4.028E−01	1.464E−01	3.637E−05	6.782E−01
F20	4.864E−01	4.180E−01	2.873E−01	5.469E−01	6.110E−01
F21	4.180E−01	1.783E−04	2.035E−01	1.483E−01	1.163E−02
F22	8.388E−04	3.236E−05	3.093E−04	6.603E−07	6.639E−05
F23	3.806E−01	3.663E−01	2.903E−01	3.916E−01	5.924E−01
F24	4.297E−01	1.942E−01	9.573E−01	2.903E−01	6.299E−01
F25	1.919E−01	9.786E−01	1.373E−01	4.177E−03	1.446E−01
F26	1.774E−02	1.265E−01	1.534E−01	3.899E−01	2.458E−01
F27	6.587E−01	5.033E−01	1.082E−01	6.733E−01	4.657E−01
F28	1.185E−02	2.482E−05	9.975E−03	5.561E−02	9.919E−04
F29	4.822E−01	2.011E−01	2.637E−01	5.399E−04	4.739E−01
F30	5.649E−01	5.786E−01	4.781E−01	1.638E−09	2.934E−01

Table A.32
Wilcoxon test results (100D).

Function	AEAS vs.				
	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	9.253E−01	9.573E−01	9.253E−01	8.777E−01	9.893E−01
F3	1.659E−01	6.002E−02	3.453E−01	2.699E−15	3.718E−09
F4	6.299E−01	4.948E−01	3.592E−01	6.831E−01	8.200E−01
F5	4.180E−01	6.854E−03	3.232E−03	1.270E−01	4.838E−03
F6	2.283E−01	5.367E−03	1.286E−03	1.944E−03	8.388E−04
F7	5.652E−14	1.851E−15	1.661E−15	2.369E−02	5.280E−16
F8	1.267E−15	2.895E−14	1.638E−09	1.330E−08	2.571E−13
F9	5.931E−11	4.443E−10	1.378E−10	4.611E−02	3.558E−12
F10	4.739E−01	6.280E−02	9.041E−02	7.135E−03	2.415E−01
F11	2.021E−02	1.521E−01	9.893E−01	6.016E−01	7.992E−01
F12	7.992E−01	3.879E−01	5.649E−01	6.782E−01	9.094E−01
F13	8.252E−01	7.128E−01	2.934E−01	2.157E−01	3.734E−01
F14	5.425E−01	5.822E−02	6.063E−01	2.204E−04	5.734E−02
F15	3.806E−01	1.743E−01	3.698E−01	7.379E−01	1.579E−01
F16	4.199E−07	9.694E−06	2.801E−06	1.681E−08	1.336E−07
F17	6.782E−01	2.021E−02	2.328E−02	1.873E−01	3.770E−01
F18	5.831E−01	9.466E−01	7.844E−07	1.243E−16	8.461E−01
F19	4.450E−03	5.425E−01	1.540E−01	3.058E−01	3.453E−01
F20	5.142E−05	2.558E−07	4.002E−06	2.205E−08	7.844E−07
F21	9.041E−01	4.056E−02	6.472E−02	7.684E−01	3.318E−01
F22	6.016E−01	9.360E−01	2.415E−01	8.096E−01	5.337E−01
F23	6.539E−01	1.807E−01	3.027E−01	9.111E−06	4.822E−01
F24	3.501E−02	2.107E−01	6.684E−01	3.927E−02	5.381E−01
F25	4.657E−01	2.753E−01	2.257E−01	1.944E−03	6.880E−01
F26	8.988E−01	1.427E−01	8.513E−01	3.662E−01	2.283E−01
F27	7.078E−01	7.582E−01	6.587E−01	3.122E−01	1.158E−01
F28	4.838E−03	1.411E−03	3.232E−03	8.817E−13	2.648E−04
F29	7.028E−01	1.000E+00	6.831E−01	1.351E−06	9.413E−01
F30	5.970E−01	3.627E−01	6.251E−01	7.092E−11	3.953E−01

Table A.33
Friedman test results (10D).

Function	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	2.25490	3.47059	3.50980	3.35294	5.03922	3.37255
F3	3.50000	3.50000	3.50000	3.50000	3.50000	3.50000
F4	4.87255	3.25490	3.20588	3.20588	3.20588	3.25490
F5	3.01961	3.21569	4.20588	3.35294	3.17647	4.02941
F6	2.52941	3.72549	4.31373	4.23529	2.25490	3.94118
F7	2.86275	3.45098	3.84314	3.64706	3.09804	4.09804
F8	3.60784	3.16667	3.60784	3.61765	3.26471	3.73529
F9	3.52941	3.56863	3.61765	3.48039	3.38235	3.42157
F10	2.98039	3.19608	3.60784	3.68627	3.66667	3.86275
F11	2.15686	3.80392	4.59804	3.77451	2.90196	3.76471
F12	2.98039	3.50980	3.64706	3.56863	3.88235	3.41176
F13	3.01961	3.96078	3.66667	3.49020	3.72549	3.13725
F14	2.64706	3.62745	4.19608	4.21569	2.45098	3.86275
F15	3.09804	3.66667	3.72549	3.94118	3.33333	3.23529
F16	3.00000	2.96078	4.21569	3.60784	3.49020	3.72549
F17	3.03922	3.54902	4.19608	3.66667	2.64706	3.90196
F18	3.98039	3.31373	2.98039	3.11765	4.45098	3.15686
F19	2.78431	4.01961	3.90196	4.05882	2.27451	3.96078
F20	3.35294	3.23529	3.96078	3.43137	3.43137	3.58824
F21	1.88235	3.57843	4.03922	3.77451	4.14706	3.57843
F22	1.09804	4.19608	4.00000	3.74510	3.82353	4.13725
F23	1.00000	4.31373	3.96078	4.17647	3.86275	3.68627
F24	1.40196	3.53922	3.88235	3.86275	4.00000	4.31373
F25	1.03922	4.06863	3.96078	3.74510	3.77451	4.41176
F26	1.33333	3.75490	3.74510	3.94118	4.00000	4.22549
F27	2.05882	3.39216	3.98039	3.62745	3.94118	4.00000
F28	1.03922	3.71569	4.20588	4.10784	4.11765	3.81373
F29	1.25490	4.03922	4.37255	3.92157	3.17647	4.23529
F30	3.41176	3.05882	3.72549	3.78431	3.64706	3.37255

Table A.34
Friedman test results (30D).

Function	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	2.01961	3.43137	3.78431	3.60784	4.00000	4.15686
F3	3.10784	3.38235	3.36275	3.27451	4.89216	2.98039
F4	2.25490	4.01961	3.76471	3.33333	3.62745	4.00000
F5	2.54902	3.25490	3.80392	3.94118	3.37255	4.07843
F6	2.80392	3.39216	4.09804	4.01961	2.78431	3.90196
F7	3.03922	3.45098	4.17647	4.47059	1.72549	4.13725
F8	2.45098	3.39216	3.56863	3.62745	3.98039	3.98039
F9	2.47059	4.03922	4.39216	4.25490	1.56863	4.27451
F10	3.17647	3.74510	3.25490	3.45098	3.88235	3.49020
F11	3.56863	3.35294	3.68627	3.72549	3.03922	3.62745
F12	2.94118	3.74510	3.64706	3.47059	3.96078	3.23529
F13	3.66667	3.43137	3.60784	3.90196	2.90196	3.49020
F14	3.68627	3.25490	3.54902	3.64706	3.37255	3.49020
F15	2.05882	3.49020	3.43137	3.98039	4.66667	3.37255
F16	3.45098	3.37255	3.50980	3.80392	3.43137	3.43137
F17	3.68627	3.07843	3.68627	3.49020	3.31373	3.74510
F18	3.03922	3.47059	3.13725	4.33333	3.90196	3.11765
F19	3.21569	2.98039	3.27451	3.70588	4.52941	3.29412
F20	3.11765	3.74510	3.66667	3.66667	3.21569	3.58824
F21	1.33333	3.45098	4.35294	3.56863	4.25490	4.03922
F22	2.68627	3.31373	3.77451	3.67647	3.80392	3.74510
F23	3.11765	3.50980	4.11765	3.33333	3.54902	3.37255
F24	3.21569	3.37255	3.29412	3.94118	3.70588	3.47059
F25	3.23529	3.47059	3.31373	3.86275	3.60784	3.50980
F26	2.57843	3.64706	4.04902	3.47059	3.44118	3.81373
F27	3.64706	3.33333	3.19608	4.09804	3.47059	3.25490
F28	3.52941	3.45098	3.63725	3.64706	3.28431	3.45098
F29	3.17647	3.41176	3.54902	3.31373	4.09804	3.45098
F30	3.23529	3.19608	3.49020	3.49020	4.39216	3.19608

Table A.35
Friedman test results (50D).

Function	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	2.68627	4.00000	3.64706	3.98039	3.25490	3.43137
F3	2.71569	3.24510	2.82353	3.47059	5.92157	2.82353
F4	2.88235	3.15686	3.72549	4.09804	3.50980	3.62745
F5	3.07843	3.37255	3.72549	3.94118	2.94118	3.94118
F6	2.78431	3.41176	3.80392	3.66667	3.47059	3.86275
F7	2.00000	4.23529	4.50980	4.31373	1.64706	4.29412
F8	2.96078	3.27451	3.62745	3.96078	3.33333	3.84314
F9	2.15686	4.49020	4.27451	4.49020	1.76471	3.82353
F10	3.96078	3.47059	3.49020	3.29412	3.52941	3.25490
F11	3.68627	3.62745	3.58824	3.37255	3.49020	3.23529
F12	2.86275	3.54902	3.80392	3.72549	3.15686	3.90196
F13	3.68627	3.84314	3.76471	3.01961	3.11765	3.56863
F14	2.98039	2.86275	3.62745	3.94118	4.33333	3.25490
F15	3.13725	2.96078	3.01961	4.13725	3.98039	3.76471
F16	3.39216	3.56863	4.01961	3.31373	3.23529	3.47059
F17	3.23529	3.43137	3.92157	3.56863	2.92157	3.92157
F18	3.64706	3.35294	2.78431	3.56863	4.58824	3.05882
F19	3.37255	2.92157	3.03922	3.86275	4.70588	3.09804
F20	3.43137	3.27451	3.80392	3.05882	3.78431	3.64706
F21	3.03922	3.07843	4.07843	3.41176	3.50980	3.88235
F22	2.31373	3.66667	3.62745	3.66667	3.82353	3.90196
F23	3.43137	3.27451	3.82353	3.86275	3.05882	3.54902
F24	3.49020	3.13725	3.90196	3.25490	3.64706	3.56863
F25	2.94118	3.56863	3.05882	3.70588	4.05882	3.66667
F26	2.98039	4.05882	3.56863	3.52941	3.42157	3.44118
F27	3.27451	3.19608	3.60784	3.92157	3.47059	3.52941
F28	2.62745	3.54902	4.23529	3.60784	3.39216	3.58824
F29	3.54902	3.41176	3.03922	3.15686	4.66667	3.17647
F30	3.03922	3.15686	3.11765	3.25490	5.09804	3.33333

Table A.36
Friedman test results (100D).

Function	AEAS	AEAS-1	AEAS-2	AEAS-3	AEAS-4	AEAS-5
F1	3.60784	3.56863	3.56863	3.31373	3.37255	3.56863
F3	3.50980	3.23529	3.13725	3.68627	5.78431	1.64706
F4	3.78431	3.41176	3.37255	3.47059	3.49020	3.47059
F5	2.88235	3.21569	3.92157	3.86275	3.31373	3.80392
F6	2.78431	3.11765	3.54902	4.00000	3.94118	3.60784
F7	1.84314	4.17647	4.41176	4.58824	1.50980	4.47059
F8	1.49020	4.31373	4.09804	3.54902	3.58824	3.96078
F9	1.98039	4.27451	4.13725	4.27451	1.98039	4.35294
F10	2.98039	3.27451	3.74510	3.62745	4.01961	3.35294
F11	3.35294	4.03922	3.78431	3.33333	3.11765	3.37255
F12	3.72549	3.56863	3.11765	3.62745	3.45098	3.50980
F13	3.45098	3.37255	3.37255	3.86275	3.09804	3.84314
F14	3.43137	3.31373	2.86275	3.78431	4.62745	2.98039
F15	3.19608	3.45098	3.74510	3.60784	3.33333	3.66667
F16	1.98039	3.90196	3.49020	3.66667	4.15686	3.80392
F17	3.11765	3.23529	3.74510	3.96078	3.54902	3.39216
F18	2.68627	2.92157	2.64706	4.17647	5.68627	2.88235
F19	3.25490	4.21569	3.43137	3.60784	3.00000	3.49020
F20	1.90196	3.52941	3.92157	3.50980	4.31373	3.82353
F21	3.17647	3.07843	3.88235	3.88235	3.50980	3.47059
F22	3.43137	3.50980	3.62745	3.01961	3.78431	3.62745
F23	3.35294	3.21569	2.72549	3.37255	4.70588	3.62745
F24	3.64706	2.86275	3.29412	3.54902	4.17647	3.47059
F25	3.09804	3.29412	3.54902	3.41176	4.43137	3.21569
F26	3.64706	3.66667	3.23529	3.70588	3.47059	3.27451
F27	3.27451	3.39216	3.39216	3.49020	3.78431	3.66667
F28	2.27451	3.49020	3.37255	3.33333	4.96078	3.56863
F29	3.09804	3.49020	3.19608	3.15686	4.98039	3.07843
F30	3.33333	3.09804	2.90196	3.17647	5.50980	2.98039

Table A.37
Comparative results for the compression/tension spring design.

Algorithms	d	D	N
TSRL-CMM	0.051698	0.356992	11.27030
AVOA	0.052784	0.383690	9.864500
WOA	0.051469	0.351488	11.59960
MPA	0.050000	0.317415	14.03010
PSO	0.050000	0.315845	14.28170
GWOCS	0.053880	0.411820	8.66280
CSCAHHO	0.050000	0.317222	14.05310
PSOGSA	0.050000	0.311080	15.00000
PSOGWO	0.051196	0.345032	12.00620
QLPSO	0.050000	0.347232	13.02510
QLSOPSO	0.050000	0.362548	13.01561

Table A.38
Comparative results for the welded beam design case.

Algorithms	d	l	t	b
TSRL-CMM	0.205730	3.253100	9.036600	0.205730
AVOA	0.205630	3.255000	9.036600	0.205730
WOA	0.187960	3.606200	9.022900	0.206350
MPA	0.200590	3.383600	8.948700	0.209830
PSO	0.205300	3.271100	9.052500	0.206010
GWOCS	0.194520	3.472400	9.036400	0.205740
CSCAHHO	0.208420	3.196100	9.073300	0.209360
PSOGSA	0.205188	3.114960	10.00000	0.205188
PSOGWO	0.204680	3.279400	9.033700	0.206070
QLPSO	0.205151	3.245100	9.036600	0.205730
QLSOPSO	0.217342	3.263100	9.025400	0.205730

Table A.39
Comparative results for the tubular column design.

Algorithms	d	t
TSRL-CMM	5.452200	0.291620
AVOA	5.452200	0.291620
WOA	5.450800	0.291850
MPA	5.452200	0.291640
PSO	5.473900	0.290950
GWOCS	5.452200	0.291620
CSCAHHO	5.450200	0.291940
PSOGSA	5.291800	0.330470
PSOGWO	5.452300	0.291620
QLPSO	5.452200	0.291620
QLSOPSO	5.452200	0.291620

Table A.40
Comparative results for the pressure vessel design.

Algorithms	T_s	T_h	R	L
TSRL-CMM	0.735414	0.365131	38.10062	233.4127
AVOA	0.780101	0.385608	40.41975	198.6108
WOA	0.876953	0.425927	44.36754	150.4111
MPA	0.788024	0.389397	40.50036	200.0000
PSO	0.781377	0.405110	40.33289	200.0000
GWOCS	0.850925	0.449590	44.08841	153.4464
CSCAHHO	0.832168	0.446002	43.11577	169.1112
PSOGSA	0.817664	0.445793	40.86497	200.0000
PSOGWO	0.778738	0.385042	40.34080	199.7214
QLPSO	0.763318	0.380196	39.54565	211.1401
QLSOPSO	0.778171	0.384651	40.31976	199.9981

Table A.41
Comparative results for the piston lever design.

Algorithms	H	B	x	D
TSRL-CMM	0.050000	2.041495	4.083028	120.0000
AVOA	0.050000	2.041495	4.083028	120.0000
WOA	0.052349	2.066047	4.091259	119.4775
MPA	0.050793	2.061573	4.087300	120.0000
PSO	0.050000	2.042093	4.084460	120.0000
GWOCS	15.62643	16.93581	3.390885	119.1788
CSCAHHO	0.050000	2.171922	4.155761	115.6238
PSOGSA	0.050000	4.493779	4.042080	120.0000
PSOGWO	0.050000	2.041898	4.083132	120.0000
QLPSO	0.052616	2.041442	4.082922	120.0000
QLSOPSO	0.052033	2.270749	4.079865	119.9470

Table A.42
Comparative results for the three-bar truss design.

Algorithms	X_{A1}	X_{A2}
TSRL-CMM	0.786100	0.406910
AVOA	0.786100	0.406910
WOA	0.770770	0.452110
MPA	0.787150	0.404360
PSO	0.786100	0.406910
GWOCS	0.786110	0.406890
CSCAHHO	0.786030	0.407120
PSOGSA	0.781560	0.419440
PSOGWO	0.786100	0.406910
QLPSO	0.786100	0.406910
QLSOPSO	0.786100	0.406910

Table A.43
Comparative results for the gear train design.

Algorithms	A	B	C	D
TSRL-CMM	66.54750	29.41390	22.95990	69.45300
AVOA	49.36910	15.68940	18.91630	43.09870
WOA	48.87190	15.76900	19.25820	43.15560
MPA	48.60280	18.73070	16.38120	42.94710
PSO	49.33310	22.25480	17.99220	56.08040
GWOCS	48.57770	15.92860	19.22830	42.69720
CSCAHHO	48.78240	16.07070	18.92980	42.64030
PSOGSA	53.02750	30.11960	13.20000	51.21490
PSOGWO	43.46650	16.47430	18.89340	48.78540
QLSPO	53.18450	25.56992	15.11563	51.26185
QLSOPSO	48.01122	21.56828	16.77818	53.64815

Table A.44
Comparative results for the speed reducer design.

Algorithms	x_1	x_2	x_3	x_4	x_5	x_6	x_7
TSRL-CMM	9.226300	0.779790	8.440100	7.215800	7.734400	3.366000	5.301400
AVOA	3.500000	0.700000	17.00000	7.300000	7.802500	3.352000	5.281200
WOA	3.512560	0.700000	17.00000	7.300000	7.802500	3.352000	5.293500
MPA	3.536700	0.700000	17.00000	7.300000	7.802500	3.384000	5.303700
PSO	3.600000	0.700000	17.00000	7.300000	7.832700	3.373000	5.293500
GWOCS	3.523900	0.704333	17.29010	7.909180	7.893200	3.425000	5.281200
CSCAHHO	3.500020	0.700000	17.00000	7.300000	7.802500	3.364000	5.281200
PSOGSA	3.600000	0.700000	17.00000	7.300000	8.005000	3.364000	5.403200
PSOGWO	3.500250	0.700000	17.00000	7.300000	7.802500	3.352000	5.281200
QLPSO	3.156740	0.700000	17.00000	7.300000	7.562400	3.354000	5.234500
QLSOPSO	3.521450	0.700000	17.00000	7.300000	7.151500	3.254000	5.123500

Table A.45
Comparative results for the SDM.

Algorithms	I_{ph} (A)	I_{sd} (A)	R_s (Ω)	R_{sh} (Ω)	n
TSRL-CMM	0.760790	3.09E−07	0.036581	52.75240	1.516350
AVOA	0.760732	4.55E−07	0.034758	60.97120	1.557370
WOA	0.761039	6.26E−07	0.034094	68.70090	1.592810
MPA	0.762095	4.68E−07	0.038318	100.0000	1.557661
PSO	0.760737	3.21E−07	0.036403	53.96400	1.520290
GWOCS	0.759590	2.07E−07	0.038341	75.74100	1.475600
CSCAHHO	0.758522	1.00E−06	0.040175	100.0000	1.653362
PSOGSA	0.760556	1.00E−06	0.033125	100.0000	1.646558
PSOGWO	0.763452	9.74E−07	0.030974	38.78680	1.645660
QLPSO	0.854561	8.25E−08	0.042152	100.0000	1.254622
QLSOPSO	0.856154	1.55E−07	0.025451	90.2356	1.754812

Table A.46
Comparative results for the DDM.

Algorithms	I_{ph} (A)	I_{sd1} (A)	I_{sd2} (A)	R_s (Ω)	R_{sh} (Ω)	n_1	n_2
TSRL-CMM	1.519170	1.10E−06	1.63E−07	0.034193	48.0774	1.577610	2.224290
AVOA	1.000000	2.05E−09	1.90E−10	0.022164	100.0000	1.999992	1.000000
WOA	1.000000	4.12E−09	1.90E−10	0.022133	100.0000	1.948030	1.000000
MPA	1.000000	4.12E−09	1.90E−10	0.022133	100.0000	1.948030	1.000000
PSO	1.000000	4.12E−09	1.90E−10	0.022175	100.0000	2.000000	1.000000
GWOCS	0.999107	9.99E−07	7.61E−07	0.016277	44.5422	1.889240	1.621310
CSCAHHO	1.000000	1.00E−06	1.00E−06	0.025002	43.4793	1.632970	2.000000
PSOGSA	1.519170	1.10E−06	1.63E−07	0.034193	48.0774	1.577610	2.224290
PSOGWO	1.000000	1.57E−07	7.64E−09	0.022133	100.0000	2.000000	1.201481
QLPSO	1.000000	2.01E−08	5.24E−09	0.015624	95.2451	2.000000	1.558741
QLSOPSO	1.020525	8.24E−09	7.23E−09	0.021458	100.0000	1.942355	1.000000

Table A.47
Comparative results for the TDM.

Algorithms	I_{ph} (A)	I_{sd1} (A)	I_{sd2} (A)	I_{sd3} (A)	R_s (Ω)	R_{sh} (Ω)	n_1	n_2	n_3
TSRL-CMM	1.522890	4.93E−07	3.89E−07	6.66E−07	0.036549	23.3693	1.498340	2.011540	2.005280
AVOA	1.000000	1.03E−08	7.86E−08	3.46E−10	5.27E−08	100.0000	1.220374	1.000349	1.000040
WOA	1.000000	3.20E−09	3.31E−07	8.26E−08	0.006688	100.0000	1.148177	2.000000	2.000000
MPA	1.000000	2.14E−07	1.24E−07	1.88E−10	0.022595	100.0000	2.000000	2.000000	1.000000
PSO	1.000000	4.31E−09	1.90E−10	7.22E−09	0.021443	100.0000	1.451355	1.000000	1.682005
GWOCS	0.999438	3.75E−07	7.75E−08	9.99E−07	0.050651	99.8582	1.676810	1.607300	1.711160
CSCAHHO	0.983391	1.00E−06	1.00E−06	1.00E−06	0.048132	78.7907	1.739580	1.913240	1.815520
PSOGSA	1.000000	1.00E−06	3.42E−07	4.54E−07	1.61E−07	100.0000	1.642430	2.000000	2.000000
PSOGWO	1.000000	9.10E−08	9.91E−08	7.83E−09	6.85E−09	100.0000	2.000000	2.000000	1.204144
QLPSO	1.000000	2.25E−09	5.24E−10	2.54E−09	3.65E−08	100.0000	1.895600	2.000000	2.001452
QLSOPSO	1.000000	9.25E−10	5.68E−10	5.24E−10	5.88E−09	100.0000	2.000000	1.985555	2.000012

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